

Thesis draft

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Chapter 1

Introduction

The standard model of cosmology describes a universe that emerged from a hot, dense initial state and has been expanding and cooling ever since. As it cooled, the fundamental forces of nature underwent a series of symmetry breaking transitions, each leaving behind observable imprints on the structure of the universe we see today. Understanding these transitions from first principles requires a synthesis of quantum field theory, statistical mechanics, and cosmology, and the central tool that makes this synthesis possible is the effective action.

This thesis develops that tool systematically, beginning with its abstract foundations and ending with concrete physical predictions for gravitational wave signals detectable by next-generation experiments. The thread connecting all five parts is a single question: how does the vacuum structure of a quantum field theory change as a function of temperature, and what are the cosmological consequences of those changes?

The Effective Action and the Quantum Vacuum

In classical field theory the vacuum is simply the minimum of the classical potential. In quantum field theory this picture is corrected by loop diagrams, which shift the location of the minimum, change the shape of the potential, and can even generate new minima that have no classical counterpart. The object that encodes all of these quantum corrections is the effective action $\Gamma[\varphi_c]$, the generating functional for one-particle-irreducible Green's functions. Its equation of motion, $\delta\Gamma/\delta\varphi_c = 0$, governs the full quantum dynamics of the field with all loop corrections included, and its restriction to constant field configurations defines the effective potential $V_{\text{eff}}(\varphi_c)$, whose minima are the true vacua of the quantum theory.

The construction of the effective action from the path integral, through the generating functionals $Z[J]$ and $W[J]$ and the Legendre transform, is developed in Part I. The physical content of this object is richer than its definition suggests. The effective potential is simultaneously the generating functional for zero-momentum 1PI diagrams, the minimum energy density over all states with a given field expectation value, and the quantum-corrected potential that appears in the equation of motion for the classical field. All three interpretations are developed and used.

The explicit computation of the effective potential at one loop, first in scalar field theory and then in gauge theories, reveals one of the most striking consequences of quantum corrections: dimensional transmutation. In massless scalar electrodynamics, a theory with no classical mass scale and a classically scale-invariant action, quantum corrections generate

a nontrivial minimum of the effective potential at a nonzero field value, dynamically producing a mass scale from a dimensionless coupling constant. This mechanism, due to Coleman and Weinberg, is the prototype for a broad class of phenomena in which the vacuum structure of a theory is qualitatively different from what classical analysis would suggest. It also provides the first example of a theme that recurs throughout the thesis: quantum corrections do not merely shift numbers, they can change the qualitative character of the vacuum.

Finite Temperature and Phase Transitions

The effective potential computed in Part I describes the vacuum of the quantum field theory at zero temperature. In the early universe, matter was in thermal equilibrium at high temperatures, and the relevant object is not the ground state energy but the free energy. Part II extends the effective action formalism to finite temperature using the imaginary-time formalism of thermal field theory, in which the inverse temperature $\beta = 1/T$ plays the role of a compactified Euclidean time direction and quantum fields satisfy periodic or antiperiodic boundary conditions.

The key result of this extension is that finite-temperature corrections generate a positive mass term proportional to $T^2\varphi_c^2$, which competes with any negative mass term in the zero-temperature potential. This competition drives symmetry restoration: even if the zero-temperature theory has a spontaneously broken ground state, above a critical temperature T_c the symmetric phase becomes the global minimum of the finite-temperature effective potential. As the universe cooled through T_c , the symmetry was broken and a phase transition occurred. The order and character of that transition, whether it proceeds smoothly or via bubble nucleation, depends on the detailed structure of the effective potential and has profound observational consequences.

Part III applies this framework to the cosmological setting. The thermal history of the early universe is punctuated by a sequence of such symmetry breaking transitions: a possible GUT transition at $T \sim 10^{16}$ GeV, the electroweak transition at $T \sim 100$ GeV, and the QCD confinement transition at $T \sim 200$ MeV. For each of these, the effective potential, computed using the tools developed in Parts I and II, determines whether the transition is first or second order and what the thermodynamic parameters of the transition are.

False Vacuum Decay and Bubble Nucleation

When the effective potential has two minima separated by a barrier, the universe can find itself temporarily trapped in the metastable false vacuum. The transition to the true vacuum does not proceed by the field rolling classically over the barrier but by quantum and thermal fluctuations nucleating bubbles of the true vacuum within the surrounding false vacuum. Each bubble expands, driven by the pressure difference between the two phases, and eventually the bubbles percolate, completing the transition.

The quantitative description of this process, developed in Part IV, is one of the most elegant applications of the path integral formalism in quantum field theory. The nucleation rate is controlled by the bounce solution, a saddle-point of the Euclidean action that interpolates between the false and true vacua. At zero temperature the relevant bounce has $O(4)$ symmetry, while at finite temperature it has $O(3)$ symmetry, reflecting the

compactification of Euclidean time. In both cases the nucleation rate takes the form $\Gamma \sim e^{-B}$, where B is the Euclidean action of the bounce, and the dynamics of the resulting bubbles are determined by analytic continuation of the bounce solution from Euclidean to Minkowski space.

The phase transition is characterized by three dimensionless parameters: α , the ratio of the latent heat to the radiation energy density; $\tilde{\beta}/H$, the ratio of the transition rate to the Hubble rate; and v_w , the bubble wall velocity. These parameters determine the gravitational wave spectrum produced during the transition through the three mechanisms of bubble wall collisions, sound waves in the primordial plasma, and magnetohydrodynamic turbulence. For an electroweak-scale transition, the predicted spectrum peaks at millihertz frequencies, precisely the sensitivity band of the LISA space interferometer, providing a direct observational window into the thermal history of the early universe at energies far beyond the reach of terrestrial accelerators. The key references for this gravitational wave signal are Caprini et al. (JCAP 2016, arXiv:1512.06239) and Caprini et al. (JCAP 2020, arXiv:1910.13125).

Quantum Fields in Curved Spacetime and Semiclassical Gravity

All of the above is developed in flat spacetime, or at most in a classical FRW background treated as fixed. In the early universe, however, the curvature of spacetime is not negligible, and a more complete treatment is required. The appropriate framework is semiclassical gravity, in which the spacetime geometry is treated classically via the Einstein equations but the matter fields are promoted to quantum operators. The backreaction of quantum matter on the geometry is captured by the semiclassical Einstein equation,

$$G_{\mu\nu} = 8\pi G \langle \hat{T}_{\mu\nu} \rangle,$$

where the right-hand side is the expectation value of the quantum stress-energy tensor in some state. This is the natural next step beyond the flat-spacetime effective action formalism developed in the earlier parts of the thesis, and it is valid as long as the curvature remains well below the Planck scale so that quantum gravitational effects themselves can be ignored.

The necessity of this framework is already visible in the effective action. In a curved background the field equation acquires a non-minimal coupling to the curvature, $(\square - m^2 + \xi R)\phi = 0$, which shifts the effective mass and can qualitatively change the phase structure of the theory. More fundamentally, semiclassical gravity reveals that the vacuum itself is observer-dependent. In flat spacetime, the Poincaré invariance of the Minkowski metric singles out a unique vacuum state agreed upon by all inertial observers, and the effective potential is a well-defined object. In a general curved spacetime there is no such preferred frame. Different observers define inequivalent mode decompositions of the quantum field, related by Bogoliubov transformations, and the vacuum of one observer contains particles as seen by another.

Part V develops this formalism systematically. We begin with the causal structure of the Schwarzschild geometry, introducing the Kruskal and Penrose diagrams that make the global structure of spacetime transparent. Black hole thermodynamics follows naturally: the area of the horizon never decreases, in direct analogy with entropy, and a gedanken experiment involving a qubit lowered into a black hole fixes the Bekenstein-Hawking

entropy $S_{BH} = A/4$ and the generalized second law $\delta S_{\text{tot}} \geq 0$. The general Bogoliubov transformation formalism is then developed, leading to the key result that the vacuum of one observer can contain a thermal distribution of particles as seen by another whenever the β coefficients are nonzero.

The physical consequences are explored through two concrete examples that build toward Hawking radiation. The moving mirror shows that even in flat spacetime a time-dependent boundary condition produces a thermal flux of particles when the mirror asymptotically approaches a null ray, and the logarithmic relation between the in and out coordinates near that asymptote is the essential mathematical structure. The Unruh effect shows that a uniformly accelerating observer in the Minkowski vacuum measures a thermal bath at temperature $T = a/2\pi$, where a is the proper acceleration, because the causal horizon of the accelerating observer plays precisely the same role as the horizon of a black hole.

These two results are not analogies. They are the same physics as Hawking radiation, seen in simpler settings. When a black hole forms by gravitational collapse, the logarithmic relation between the coordinates at past and future null infinity produces, in the semiclassical gravity framework, a steady thermal flux of radiation at the Hawking temperature $T_H = \kappa/2\pi$, where κ is the surface gravity of the horizon. The black hole emits as if it were a black body at this temperature, and the resulting evaporation has deep consequences for the consistency of quantum mechanics and general relativity that remain incompletely understood.

The connection back to the rest of the thesis is direct. The de Sitter spacetime produced during inflation also possesses a cosmological horizon, and quantum fields in that background experience a Gibbons-Hawking temperature $T = H/2\pi$ by the same mechanism. Thermal fluctuations in this background seed the primordial density perturbations that grow into the large-scale structure of the universe, completing the circle from the microscopic effective action of Part I to the observable universe of today.

Outline

The thesis is organized as follows. Part I develops the effective action formalism at zero temperature, computing the one-loop effective potential in scalar theory, massless scalar electrodynamics, the Glashow-Salam-Weinberg electroweak model, and non-Abelian gauge theories, with renormalization carried out both by hard cutoff and by dimensional regularization in the $\overline{\text{MS}}$ scheme. Part II develops thermal field theory using the imaginary-time formalism and computes finite-temperature corrections to the effective potential for scalar fields, the GSW model, and minimal SU(5). Part III places these results in their cosmological context, covering the thermal history of the early universe, the characterization of first-order phase transitions, and the gravitational wave signatures they produce. Part IV analyzes false vacuum decay and bubble nucleation via the bounce solution, at both zero and finite temperature. Part V treats quantum field theory in curved spacetime, from the causal structure of the Schwarzschild geometry through black hole thermodynamics, the Bogoliubov transformation formalism, and the Unruh effect, culminating in the derivation of Hawking radiation and its physical interpretation. Technical derivations are collected in the appendices.

Part I

Effective Action and Effective Potential

Chapter 2

General Formalism of the Effective Action

The central object of this thesis is the *effective action* $\Gamma[\varphi_c]$, a functional that encodes the full quantum dynamics of a field theory in a single object. Before defining it, we build up to it systematically: starting from the generating functional $Z[J]$ for all Green functions, passing through the connected generating functional $W[J]$, and finally arriving at $\Gamma[\varphi_c]$ via a Legendre transform. The payoff is that the equation of motion derived from Γ governs the quantum-corrected vacuum structure of the theory, making it the natural tool for studying spontaneous symmetry breaking and phase transitions.

2.1 Generating Functionals

Rather than computing Green functions diagram by diagram, all of them, full, connected, and one-particle-irreducible, can be extracted from a small number of functionals by simple functional differentiation.

Coupling the field $\varphi(x)$ linearly to an arbitrary c -number source $J(x)$,

$$\mathcal{L}(\varphi, \partial_\mu \varphi) \longrightarrow \mathcal{L} + J(x) \varphi(x), \quad (2.1)$$

the generating functional for the full Green functions is defined as the vacuum-to-vacuum transition amplitude in the presence of this source,

$$Z[J] = \langle 0 | T \exp \left(i \int d^4x J(x) \varphi(x) \right) | 0 \rangle, \quad (2.2)$$

where $|0\rangle$ is the physical vacuum and $\varphi(x)$ is the Heisenberg-picture field, both defined at $J = 0$. Expanding in powers of J makes the content of this definition explicit,

$$Z[J] = \sum_{n=0}^{\infty} \frac{i^n}{n!} \int d^4x_1 \cdots d^4x_n J(x_1) \cdots J(x_n) G_n(x_1, \dots, x_n), \quad (2.3)$$

where $G_n(x_1, \dots, x_n) \equiv \langle 0 | T[\varphi(x_1) \cdots \varphi(x_n)] | 0 \rangle$ is the full n -point Green function. Each G_n is recovered by n functional derivatives of $Z[J]$ with respect to J , evaluated at $J = 0$. Equivalently, $Z[J]$ admits a path-integral representation,

$$Z[J] = N \int [\mathcal{D}\varphi] \exp \left(i \int d^4x [\mathcal{L}[\varphi(x)] + J(x) \varphi(x)] \right), \quad (2.4)$$

where N is fixed by $Z[0] = 1$. The two representations (2.2) and (2.4) describe the same object and will be used interchangeably throughout.

The full Green functions G_n contain disconnected contributions, products of lower-point functions, that carry no new dynamical information. The *connected* Green functions G_n^c , which isolate only the irreducible correlations, are generated by passing to the logarithm of $Z[J]$,

$$iW[J] = \ln Z[J], \quad (2.5)$$

the proof that this logarithm precisely removes all disconnected pieces is given in Appendix A. Since $Z[J]$ is the vacuum-to-vacuum amplitude, Eq. (2.5) means

$$e^{iW[J]} = \langle 0^+ | 0^- \rangle_J, \quad (2.6)$$

where $|0^\pm\rangle$ are the vacuum states in the far future and far past, so $W[J]$ is the *vacuum persistence amplitude* in the presence of the external source. This interpretation will be central when we discuss the metastability of the false vacuum in Chapter 14.

The first functional derivative of $W[J]$ defines the *classical field*,

$$\varphi_c(x) \equiv \frac{\delta W}{\delta J(x)} = \left[\frac{\langle 0^+ | \varphi(x) | 0^- \rangle}{\langle 0^+ | 0^- \rangle} \right]_J, \quad (2.7)$$

the source-dependent vacuum expectation value of $\varphi(x)$, which will serve as the argument of the effective action.

2.2 The Effective Action and the Effective Potential

Given $W[J]$ and the classical field φ_c via (2.7), we trade the source J for φ_c as the fundamental variable via the Legendre transform,

$$\Gamma[\varphi_c] = W[J] - \int d^4x J(x) \varphi_c(x), \quad (2.8)$$

where J on the right-hand side is understood as a functional of φ_c obtained by inverting (2.7). Differentiating with respect to φ_c and using (2.7) gives

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = -J(x). \quad (2.9)$$

In the physical situation $J = 0$, this becomes the quantum equation of motion,

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = 0, \quad (2.10)$$

the exact analogue of $\delta S / \delta \varphi = 0$ for the classical action, but now with all quantum corrections included. The vacua of the quantum theory are precisely the solutions to (2.10).

Expanding $\Gamma[\varphi_c]$ in powers of φ_c ,

$$\Gamma[\varphi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \Gamma^{(n)}(x_1, \dots, x_n) \varphi_c(x_1) \cdots \varphi_c(x_n), \quad (2.11)$$

the coefficients $\Gamma^{(n)}$ are the one-particle-irreducible (1PI) Green functions, or proper vertices, the sum of all Feynman diagrams with n external lines that cannot be disconnected by cutting a single internal propagator, with no external propagators attached.

For a translationally invariant configuration $\varphi_c = \text{const}$, the effective action takes the form $\Gamma = -\Omega V(\varphi_c)$, where Ω is the four-volume. More generally, for slowly-varying fields, Γ admits a derivative expansion,

$$\Gamma[\varphi_c] = \int d^4x \left[-V(\varphi_c) + \frac{1}{2}(\partial_\mu \varphi_c)^2 Z(\varphi_c) + \cdots \right], \quad (2.12)$$

where $V(\varphi_c)$ is the *effective potential*, an ordinary function rather than a functional. Comparing with (2.11), the n -th derivative of V equals the sum of all 1PI diagrams with n external legs at zero external momenta, so the effective potential encodes the full quantum-corrected vacuum structure of the theory. The physical mass and coupling are read off from it via

$$\mu^2 = \left. \frac{d^2 V}{d\varphi_c^2} \right|_{\varphi_c=0}, \quad \lambda = \left. \frac{d^4 V}{d\varphi_c^4} \right|_{\varphi_c=0}. \quad (2.13)$$

2.3 Spontaneous Symmetry Breaking

The effective potential is the natural language for spontaneous symmetry breaking. Suppose $\mathcal{L}$ has a discrete internal symmetry, e.g. $\varphi \rightarrow -\varphi$. This symmetry is spontaneously broken when the quantum equation of motion (2.10) admits a solution with a nonzero, spatially uniform vacuum expectation value,

$$\left. \frac{dV}{d\varphi_c} \right|_{\varphi_c=\langle\varphi\rangle} = 0, \quad \langle\varphi\rangle \neq 0, \quad (2.14)$$

even in the absence of any external source. Stability requires this stationary point to be a minimum of V .

Two phases are possible. In the symmetric phase, $\langle\varphi\rangle = 0$ and the $\varphi \rightarrow -\varphi$ symmetry is intact. In the broken phase, $\langle\varphi\rangle = v \neq 0$ and the symmetry is hidden. The physical spectrum in the broken phase is described by the shifted field $\varphi' = \varphi - v$, and the renormalization conditions (2.13) are displaced to the broken minimum,

$$\mu^2 = \left. \frac{d^2 V}{d\varphi_c^2} \right|_{\varphi_c=v}, \quad \lambda = \left. \frac{d^4 V}{d\varphi_c^4} \right|_{\varphi_c=v}. \quad (2.15)$$

With the formalism in place, we turn in the next chapter to an explicit one-loop computation of the effective potential, first in a simple scalar theory and then in the physically relevant case of scalar electrodynamics.

Chapter 3

Physical Meaning of the Effective Action and Effective Potential

We have defined $\Gamma[\varphi_c]$ and $V_{\text{eff}}(\bar{\varphi})$ formally in the previous chapter. Here we give them concrete physical meaning.

3.1 The Effective Potential as a Generator of 1PI Graphs

For a constant classical field $\bar{\varphi}$, inserting the Fourier representation of $\Gamma^{(n)}$ into the expansion (2.11) gives

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int \prod_{i=1}^n d^4 x_i \Gamma^{(n)}(x_1, \dots, x_n) \quad (3.1)$$

$$= \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int \prod_{i=1}^n \left(d^4 x_i \frac{d^4 k_i}{(2\pi)^4} e^{ik_i x_i} \right) \tilde{\Gamma}^{(n)}(k_1, \dots, k_n) \cdot (2\pi)^4 \delta^{(4)}\left(\sum_{i=1}^n k_i\right), \quad (3.2)$$

where $\int d^4 x e^{ikx} = (2\pi)^4 \delta^{(4)}(k)$ was used. Setting $k = 0$ gives $\int d^4 x = (2\pi)^4 \delta^{(4)}(0)$, so that

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int d^4 x \tilde{\Gamma}^{(n)}(0, \dots, 0). \quad (3.3)$$

Comparing this with $\Gamma(\bar{\varphi}) = \int d^4 x [-V_{\text{eff}}(\bar{\varphi})]$ gives

$$V_{\text{eff}}(\bar{\varphi}) = - \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \tilde{\Gamma}^{(n)}(0, \dots, 0). \quad (3.4)$$

Thus $V_{\text{eff}}(\bar{\varphi})$ is the generating functional for 1PI graphs at vanishing external momenta, computed by summing all such graphs and inserting a factor $\bar{\varphi}$ for each external line.

To lowest order in $\hbar$, only tree-level graphs contribute. For a Lagrangian of the form

$$\mathcal{L}(\varphi, \partial_\mu \varphi) = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - U(\varphi), \quad (3.5)$$

each term $g_i \varphi^i / i!$ in $U(\varphi)$ contributes exactly one tree-level 1PI graph, since all tree graphs with more than one vertex are not 1PI. Each term gives a contribution $-g_i \bar{\varphi}^i$, where $-g_i$

is the vertex factor and $\bar{\varphi}^i$ comes from the i external lines. The minus signs cancel and one recovers

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + \mathcal{O}(\hbar), \quad (3.6)$$

confirming that the classical potential is the leading-order approximation to the effective potential.

3.2 The Effective Potential as a Minimum Energy Density

A deeper interpretation follows from the variational principle.

Theorem 1. $V_{\text{eff}}(\bar{\varphi})$ is the minimum of the energy density expectation value over all normalized states $|a\rangle$ satisfying $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle, \quad (3.7)$$

for $|a\rangle$ which obeys $\delta\langle a|\mathcal{H}|a\rangle = 0$ subject to the constraints $\langle a|a\rangle = 1$, $\langle a|\varphi|a\rangle = \bar{\varphi}$.

Proof. We first solve the analogous problem in quantum mechanics, then translate to field theory. In QM the problem is to find a state $|a\rangle$ such that $\langle a|H|a\rangle$ is stationary subject to $\langle a|a\rangle = 1$ and $\langle a|A|a\rangle = A_c$ for some self-adjoint operator A . Introducing Lagrange multipliers E and J for each constraint, the problem becomes

$$\delta\langle a|(H - E - JA)|a\rangle = 0 \quad (\text{unconstrained}) \quad (3.8)$$

$$\langle a|a\rangle = 1 \quad (3.9)$$

$$\langle a|A|a\rangle = A_c. \quad (3.10)$$

Since H is self-adjoint, this yields

$$(H - E - JA)|a\rangle = 0, \quad (3.11)$$

with solution $|a\rangle = |a(E, J)\rangle$. Using $\langle a|a\rangle = 1$, E is solved as a function of J , i.e. $E = E(J)$, and $\langle a|A|a\rangle = A_c$ gives $J = J(A_c)$. Thus

$$(H - JA)|a(J)\rangle = E(J)|a(J)\rangle, \quad (3.12)$$

where $|a(J)\rangle$ is a normalized eigenstate of $(H - JA)$. Taking the expectation value with $\langle a(J)|$ and differentiating with respect to J ,

$$\begin{aligned} \left(\frac{\delta}{\delta J} \langle a(J)| \right) \underbrace{(H - E(J) - JA)|a(J)\rangle}_0 + \langle a(J)| \frac{\delta}{\delta J} (H - E(J) - JA)|a(J)\rangle \\ + \langle a(J)| (H - E(J) - JA) \underbrace{\frac{\delta}{\delta J} |a(J)\rangle}_0 = 0, \end{aligned} \quad (3.13)$$

which gives

$$-\frac{\delta E}{\delta J} - \underbrace{\langle a(J)|A|a(J)\rangle}_{A_c} = 0 \quad \Rightarrow \quad \frac{\delta E}{\delta J} = -A_c. \quad (3.14)$$

Substituting back into the expectation value of the eigenvalue equation,

$$\langle a(J)|H|a(J)\rangle = E(J) - J \frac{\delta E}{\delta J}, \quad (3.15)$$

which is the energy of the ground state of H in the space of states satisfying $\langle a(J)|A|a(J)\rangle = A_c$.

We now translate this to field theory. In field theory the only normalizable eigenstate of a Hamiltonian is the vacuum, so $|a(J)\rangle$ is identified with the vacuum of the sourced theory. Adding a source $J(x)\varphi(x)$ to the Hamiltonian density, adiabatically switched on at $t = 0$ and off at $t = T$ over spatial volume V , the adiabatic theorem gives

$$e^{iW(J)} = \langle 0^+ | 0^- \rangle = \exp[-iVT \varepsilon(J)], \quad (3.16)$$

so $-W(J)$ is the ground state action of the sourced theory. The Legendre transform structure,

$$\Gamma(\bar{\varphi}) = W(J) - J \frac{\delta W}{\delta J}, \quad \frac{\delta W}{\delta J} = \bar{\varphi}, \quad (3.17)$$

is formally identical to the quantum-mechanical result above, identifying $-\Gamma(\bar{\varphi})$ as the ground state action of the sourceless theory in the sector with $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$-\Gamma(\bar{\varphi}) = \langle a|H|a\rangle, \quad (3.18)$$

$$-\frac{\Gamma(\bar{\varphi})}{VT} = \frac{\langle a|H|a\rangle}{VT}, \quad (3.19)$$

and therefore

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle \quad (3.20)$$

for the lowest-energy state satisfying $\langle a|a\rangle = 1$ and $\langle a|\varphi|a\rangle = \bar{\varphi}$. $\square$

This result has an important consequence: since $V_{\text{eff}}(\bar{\varphi})$ is defined as a minimum of energy over a convex set of states, it is convex. Its minima therefore correspond to the stable equilibrium configurations of the quantum field theory, a fact that will be central to the analysis of phase transitions in later chapters.

3.3 Variational Definition of the Full Effective Action

The variational interpretation of V_{eff} established above can be generalized to the full effective action.

Theorem 2. $\Gamma(\varphi)$ is the stationary value of

$$\Gamma(\varphi) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle \quad (3.21)$$

subject to the constraints

$$\langle \psi_-, t | \Phi(\mathbf{x}) | \psi_+, t \rangle = \varphi(t, \mathbf{x}), \quad \langle \psi_-, t | \psi_+, t \rangle = 1, \quad \lim_{t \rightarrow \pm\infty} |\psi_{\pm}, t\rangle = |0\rangle. \quad (3.22)$$

Proof. The proof follows the same structure as Theorem 2. Introducing Lagrange multipliers $J(t, \mathbf{x})$ and $w(t)$ for the two constraints,

$$\Gamma_m(\varphi) = \int dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle + \int d^4x J(t, \mathbf{x}) (\langle \psi_-, t | \Phi | \psi_+, t \rangle - \varphi(t, \mathbf{x})) - w(t) (\langle \psi_-, t | \psi_+, t \rangle - 1). \quad (3.23)$$

Varying with respect to $\langle \psi_-, t |$, the last term drops out and since $\langle \delta\psi_-, t |$ is arbitrary,

$$\left(i\partial_t - H + \int d^3x J(t, \mathbf{x}) \Phi(\mathbf{x}) \right) | \psi_+, t \rangle = w(t) | \psi_+, t \rangle. \quad (3.24)$$

Varying with respect to $| \psi_+, t \rangle$ and integrating by parts,

$$\int dt \langle \psi_-, t | i\partial_t | \delta\psi_+, t \rangle = [\langle \psi_-, t | \delta\psi_+, t \rangle]_{-\infty}^{\infty} - \int dt i (\partial_t \langle \psi_-, t |) | \delta\psi_+, t \rangle, \quad (3.25)$$

where the boundary term vanishes because $\lim_{t \rightarrow \pm\infty} | \psi_{\pm}, t \rangle = | 0 \rangle$. Hence

$$\int dt \langle \psi_-, t | i\partial_t | \delta\psi_+, t \rangle = - \int dt i (\partial_t \langle \psi_-, t |) | \delta\psi_+, t \rangle. \quad (3.26)$$

Since $| \delta\psi_+, t \rangle$ is arbitrary,

$$\delta_{| \psi_+, t \rangle} \Gamma_m(\varphi) = \int dt \langle \psi_-, t | \left(-i\overleftarrow{\partial}_t + H + \int d^3x J(\mathbf{x}) \Phi(\mathbf{x}) + w(t) \right) | \delta\psi_+, t \rangle = 0, \quad (3.27)$$

and taking the Hermitian conjugate,

$$\left[i\partial_t - H + \int d^3x J(\mathbf{x}) \Phi(\mathbf{x}) \right] | \psi_-, t \rangle = w^*(t) | \psi_-, t \rangle. \quad (3.28)$$

Defining the rescaled states

$$| +, t \rangle = e^{i \int_{-\infty}^t dt' w(t')} | \psi_+, t \rangle, \quad (3.29)$$

$$| -, t \rangle = e^{-i \int_t^{\infty} dt' w^*(t')} | \psi_-, t \rangle, \quad (3.30)$$

these are asymptotic ground states satisfying the sourced Schrödinger equation with no inhomogeneous term. Their overlap gives

$$\langle -, t | +, t \rangle = e^{iW(J)} = \exp \left[i \int_{-\infty}^{\infty} dt w(t) \right], \quad (3.31)$$

from which

$$W(J) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H + \int d^3x J \cdot \Phi | \psi_+, t \rangle. \quad (3.32)$$

As shown in Appendix D, $\delta W(J)/\delta J = \varphi$, so the Legendre transform gives

$$W(J) - \int d^4x J\varphi = \Gamma(\varphi) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle, \quad (3.33)$$

which shows that $\Gamma(\varphi)$ is indeed the stationary value. $\square$

The equation (2.9) has a further physical interpretation. When the source vanishes, $\delta\Gamma/\delta\bar{\varphi} = 0$ becomes the exact quantum equation of motion for $\varphi(\mathbf{x}, t)$. Expanding Γ in powers of momentum about vanishing external momenta,

$$\Gamma(\varphi) = \int d^4x \left[-V_{\text{eff}}(\varphi) + \frac{1}{2} (\partial_\mu \varphi \partial^\mu \varphi) Z(\varphi) + \cdots \right], \quad (3.34)$$

and renormalizing the field so that $Z(\varphi) = 1$, the equation of motion becomes

$$\square \varphi = -V'_{\text{eff}}(\varphi). \quad (3.35)$$

In the static limit this is precisely the classical Klein-Gordon equation, but with the classical potential replaced by the effective potential. Thus quantum corrections to the dynamics are fully captured by V_{eff} , which is yet another reason it is the central object of study.

Chapter 4

Toy Model: One-Loop Effective Potential and Renormalization

We illustrate the general formalism of the previous chapter with an explicit computation in the simplest setting: a single real scalar field with Lagrangian

$$\mathcal{L}(x) = \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - V(\phi(x)), \quad (4.1)$$

and classical potential

$$V(\phi) = \frac{\alpha_0}{4!} \phi^4 - \frac{m_0^2}{2} \phi^2. \quad (4.2)$$

We work in units $\hbar = c = 1$ throughout.

4.1 The Classical Field and the Effective Action

The classical minimum of $V(\phi)$ sits at some ϕ_0 , but the vacuum expectation value $\langle \phi \rangle$ of the quantum field need not coincide with it. It is defined by

$$\langle \phi \rangle \equiv \lim_{J \rightarrow 0} \frac{\langle 0^+ | \phi_{\text{op}} | 0^- \rangle_J}{\langle 0^+ | 0^- \rangle_J}, \quad (4.3)$$

where the generating functional $W[J]$ encodes the full quantum theory via

$$\exp \frac{i}{\hbar} W[J] = \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{S[\phi] + (J, \phi)\}, \quad (4.4)$$

with $S[\phi] = \int d^4x \mathcal{L}(x)$ and $(J, \phi) = \int d^4x J(x) \phi(x)$. The classical field, the VEV in the presence of an external source $J(x)$, is

$$\phi_c(x) \equiv \frac{\langle 0^+ | \phi_{\text{op}}(x) | 0^- \rangle_J}{\langle 0^+ | 0^- \rangle_J} = \frac{\delta W[J]}{\delta J(x)}, \quad (4.5)$$

and $\langle \phi \rangle$ is recovered as $\lim_{J \rightarrow 0} \phi_c(x)$. The loop expansion is an expansion in powers of $\hbar c \alpha$, but it performs an infinite resummation of Feynman graphs rather than a perturbative truncation.

The natural question is: given a desired profile ϕ_c , what source J produces it? To answer this, we trade J for ϕ_c as the independent variable via the Legendre transform,

$$\Gamma[\phi_c] \equiv W[J] - (J, \phi_c), \quad (4.6)$$

where J is eliminated in favour of ϕ_c by inverting (4.5). The functional $\Gamma[\phi_c]$ is the effective action, so named because it is a functional of the classical field ϕ_c and thus plays the same role for the quantum theory as $S[\phi]$ does classically. Differentiating (4.6) and using (4.5),

$$\begin{aligned}\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} &= \frac{\delta W[J]}{\delta\phi_c(x)} - J(x) - \int d^4y \frac{\delta J(y)}{\delta\phi_c(x)} \phi_c(y), \\ \frac{\delta W[J]}{\delta\phi_c(x)} &= \int d^4y \frac{\delta W[J]}{\delta J(y)} \frac{\delta J(y)}{\delta\phi_c(x)} = \int d^4y \frac{\delta J(y)}{\delta\phi_c(x)} \phi_c(y),\end{aligned}\quad (4.7)$$

so that

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x). \quad (4.8)$$

At $J = 0$, translational invariance forces ϕ_c to be a constant, so $\langle\phi\rangle$ is the root of

$$\left. \frac{d\Gamma[\phi_c]}{d\phi_c} \right|_{\phi_c=\langle\phi\rangle} = 0. \quad (4.9)$$

4.2 The Effective Action as a Generator of 1PI Green's Functions

Expanding $\Gamma[\phi_c]$ in powers of ϕ_c ,

$$\Gamma[\phi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \Gamma_n(x_1, \dots, x_n) \phi_c(x_1) \cdots \phi_c(x_n), \quad (4.10)$$

we claim that Γ_n is the n -point 1PI Green function, the sum of all connected 1PI Feynman graphs with n external legs. To see this, consider the functional integral

$$\mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \}. \quad (4.11)$$

In the limit $\hbar \rightarrow 0$ the integrand is dominated by its saddle point, which occurs at $\phi = \phi_c$ since (4.8) is precisely the saddle-point condition. Thus

$$\mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \} \xrightarrow{\hbar \rightarrow 0} \mathcal{N} \exp \frac{i}{\hbar} \{ \Gamma[\phi_c] + (J, \phi_c) \}, \quad (4.12)$$

and by (4.6) the right-hand side equals $\exp \frac{i}{\hbar} W[J]$, so

$$\exp \frac{i}{\hbar} W[J] \xrightarrow{\hbar \rightarrow 0} \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \}. \quad (4.13)$$

In this limit the right-hand side is the sum of all tree graphs of a theory whose action is $\Gamma[\phi]$, with vertices given by the Γ_n of (4.10). Since $W[J]$ generates all connected Green functions of the original theory, and any connected graph decomposes into 1PI components, it follows that Γ_n is a 1PI Green function of the original theory. $\square$

4.3 The Effective Potential

Alternatively, $\Gamma[\phi_c]$ may be expanded in derivatives of ϕ_c ,

$$\Gamma[\phi_c] = \int d^4x \left[-U(\phi_c(x)) + \frac{1}{2} \partial_\mu \phi_c(x) \partial^\mu \phi_c(x) Z(\phi_c(x)) + \dots \right], \quad (4.14)$$

where $U(\phi_c)$ is the effective potential. For a constant configuration $\phi_c(x) = \rho$ this reduces to

$$\Gamma[\rho] = -\Omega U(\rho), \quad (4.15)$$

where Ω is the total spacetime volume. To connect $U(\rho)$ to the 1PI Green functions, we pass to momentum space. The Fourier transforms of ϕ_c and Γ_n are defined by

$$\tilde{\phi}_c(k) = \int d^4x e^{ik \cdot x} \phi_c(x), \quad (4.16)$$

$$\phi_c(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \tilde{\phi}_c(k), \quad (4.17)$$

$$\tilde{\Gamma}_n(k_1, \dots, k_n) (2\pi)^4 \delta^4(k_1 + \dots + k_n) = \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) e^{i(k_1 x_1 + \dots + k_n x_n)}. \quad (4.18)$$

Substituting these into (4.10),

$$\begin{aligned} \Gamma[\phi_c] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) \phi_c(x_1) \dots \phi_c(x_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) e^{-i(k_1 x_1 + \dots + k_n x_n)} \tilde{\phi}(k_1) \dots \tilde{\phi}(k_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} (2\pi)^4 \delta^4(k_1 + \dots + k_n) \tilde{\Gamma}_n(k_1, \dots, k_n) \tilde{\phi}(k_1) \dots \tilde{\phi}(k_n). \end{aligned} \quad (4.19)$$

Setting $\phi_c(x) = \rho$ gives $\tilde{\phi}_c(k) = (2\pi)^4 \delta^4(k) \rho$, so each momentum integral collapses,

$$\begin{aligned} \Gamma[\rho] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} (2\pi)^4 \delta^4(k_1 + \dots + k_n) \tilde{\Gamma}_n(k_1, \dots, k_n) \\ &\quad \cdot (2\pi)^4 \delta^4(k_1) \rho \dots (2\pi)^4 \delta^4(k_n) \rho \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \underbrace{(2\pi)^4 \delta^4(0)}_{\Omega} \rho^n \tilde{\Gamma}_n(0, \dots, 0), \end{aligned} \quad (4.20)$$

and therefore

$$\Gamma[\rho] = \Omega \sum_{n=0}^{\infty} \frac{\rho^n}{n!} \tilde{\Gamma}_n(0), \quad (4.21)$$

where $\tilde{\Gamma}_n(0) \equiv \tilde{\Gamma}_n(0, \dots, 0)$ and $\Omega = (2\pi)^4 \delta^4(0)$ is the total spacetime volume. Comparing (4.21) with $\Gamma[\rho] = -\Omega U(\rho)$,

$$U(\rho) = - \sum_{n=0}^{\infty} \frac{\rho^n}{n!} \tilde{\Gamma}_n(0), \quad (4.22)$$

which is simply the statement that the Taylor coefficients of U are the zero-momentum 1PI Green functions,

$$-\left.\frac{d^n U(\rho)}{d\rho^n}\right|_{\rho=0} = \tilde{\Gamma}_n(0). \quad (4.23)$$

The effective potential is thus entirely determined by the 1PI Green functions of the original theory evaluated at vanishing external momenta. In the next section we compute it explicitly at one loop.

4.4 Spontaneous Symmetry Breaking and the Shifted Field

When spontaneous symmetry breaking occurs, it is more natural to work with the shifted field

$$\eta(x) = \phi(x) - \langle\phi\rangle, \quad (4.24)$$

so that the effective potential in terms of the shifted variable becomes

$$U(\rho - \langle\phi\rangle) = - \sum_{n=0}^{\infty} \frac{[\rho - \langle\phi\rangle]^n}{n!} \tilde{\Gamma}_n^{(s)}(0), \quad (4.25)$$

where $\tilde{\Gamma}_n^{(s)}(0)$ are the 1PI Green functions defined with respect to $\eta(x)$. The physically relevant quantities extracted from $U(\rho)$ at the minimum $\rho = \langle\phi\rangle$ are

$$\left.\frac{dU(\rho)}{d\rho}\right|_{\rho=\langle\phi\rangle} = 0, \quad (4.26)$$

which determines $\langle\phi\rangle$, together with the renormalized mass and coupling constant,

$$\left.\frac{d^2 U(\rho)}{d\rho^2}\right|_{\rho=\langle\phi\rangle} = m^2, \quad \left.\frac{d^4 U(\rho)}{d\rho^4}\right|_{\rho=\langle\phi\rangle} = \alpha. \quad (4.27)$$

Since $m^2 \geq 0$, these conditions mean that $\langle\phi\rangle$ sits at a minimum of $U(\rho)$, in direct analogy with ρ_0 being a minimum of the classical potential $V(\rho)$. The parameters m^2 and α are the free parameters of the renormalized theory.

4.5 One-Loop Effective Potential

We now compute $U(\rho)$ to one-loop order via saddle-point integration. Starting from

$$\exp \frac{i}{\hbar} W[J] = \mathcal{N} \int \mathcal{D}\phi \exp \frac{i}{\hbar} \{S[\phi] + (J, \phi)\}, \quad (4.28)$$

the saddle point occurs at $\phi = \phi_0$, defined by

$$\left.\frac{\delta S[\phi]}{\delta\phi(x)}\right|_{\phi=\phi_0} = -J(x). \quad (4.29)$$

Here $\phi_0(x)$ is a functional of J , and as $J \rightarrow 0$ it satisfies the classical equation of motion. Expanding $S[\phi]$ about ϕ_0 ,

$$\begin{aligned} S[\phi_0 + \phi_c] &= S[\phi_0] + \int d^4x \phi_c(x) \left[\frac{\delta S[\phi]}{\delta \phi} \right]_{\phi=\phi_0} \\ &\quad + \frac{1}{2} \int d^4x d^4y \phi_c(x) \phi_c(y) \left[\frac{\delta^2 S[\phi]}{\delta \phi(x) \delta \phi(y)} \right]_{\phi=\phi_0} \\ &\quad + S_2[\phi_c, \phi_0], \end{aligned} \quad (4.30)$$

where S_2 collects all higher-order terms. Introducing the propagator function

$$\langle x | i\Delta^{-1}[\phi_0] | y \rangle \equiv \frac{\delta^2 S[\phi]}{\delta \phi(x) \delta \phi(y)} \Big|_{\phi=\phi_0}, \quad (4.31)$$

the expansion becomes

$$S[\phi_0 + \phi_c] = S[\phi_0] - (J, \phi_c) + \frac{1}{2} (\phi_c, i\Delta^{-1}[\phi_0] \phi_c) + S_2[\phi_c, \phi_0]. \quad (4.32)$$

Substituting into the path integral and shifting $\phi \rightarrow \phi + \phi_0$,

$$\begin{aligned} \exp \frac{i}{\hbar} W[J] &= \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ S[\phi + \phi_0] + (\phi + \phi_0, J) \} \\ &= \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \left\{ S[\phi_0] + \frac{1}{2} (\phi, i\Delta^{-1}[\phi_0] \phi) + S_2[\phi, \phi_0] + (\phi_0, J) \right\}, \end{aligned} \quad (4.33)$$

which factors as

$$= \exp \frac{i}{\hbar} \{ S[\phi_0] + (\phi_0, J) \} \cdot \left\{ \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{2\hbar} (\phi, i\Delta^{-1}[\phi_0] \phi) \right\} \cdot \left\langle \exp \frac{i}{\hbar} S_2 \right\rangle, \quad (4.34)$$

where the Gaussian integral evaluates to $\{\det i\Delta^{-1}[\phi_0]\}^{-1/2}$ and $\langle \cdot \rangle$ denotes the functional average with respect to the Gaussian weight $\exp \frac{i}{2\hbar} (\phi, i\Delta^{-1}[\phi_0] \phi)$. Taking the logarithm,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2} \ln \det \{ i\Delta^{-1}[\phi_0] \} + W_2[J], \quad (4.35)$$

where $W_2[J] = -i\hbar \ln \langle \exp \frac{i}{\hbar} S_2 \rangle$.

To determine the order of $W_2[J]$, rescale $\phi = \hbar^{1/2} \tilde{\phi}$,

$$\begin{aligned} \hbar^{-1} S_2[\phi, \phi_0] &= \hbar^{-1} \phi \phi \phi \left[\frac{\delta^3 S}{\delta \phi^3} \right]_{\phi=\phi_0} + \dots \\ &= \hbar^{-1/2} \tilde{\phi} \tilde{\phi} \tilde{\phi} \left[\frac{\delta^3 S}{\delta \phi^3} \right]_{\phi=\phi_0} + \dots, \end{aligned} \quad (4.36)$$

so that

$$W_2[J] = -i\hbar \ln [1 + \mathcal{O}(\hbar^{1/2})] \sim \mathcal{O}(\hbar^{3/2}). \quad (4.37)$$

Since the loop expansion of $W[J]$ contains only integer powers of $\hbar$, the fractional contribution must actually vanish and $W_2[J] \sim \mathcal{O}(\hbar^2)$. Thus the first two terms in (4.35) are the tree-level approximation and the third term is the complete one-loop correction,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2} \ln \det \{ i\Delta^{-1}[\phi_0] \} + \mathcal{O}(\hbar^2). \quad (4.38)$$

To obtain $\Gamma[\phi_c] = W[J] - (J, \phi_c)$, we express $S[\phi_0]$ as a functional of ϕ_c . At tree level $\phi_c = \phi_0$, so $\phi_c - \phi_0 \equiv \phi_1 \sim \mathcal{O}(\hbar)$,

$$\phi_c(x) \equiv \frac{\delta W[J]}{\delta J(x)} = \phi_0(x) + \phi_1(x). \quad (4.39)$$

Expanding $S[\phi_0] = S[\phi_c - \phi_1]$ about ϕ_c ,

$$\begin{aligned} S[\phi_0] &= S[\phi_c] - \int d^4x \phi_1(x) \left\{ \frac{\delta S[\phi_0]}{\delta \phi} \right\}_{\phi=\phi_c} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_1, J) + \mathcal{O}(\hbar^2). \end{aligned} \quad (4.40)$$

Substituting into $\Gamma[\phi_c]$,

$$\begin{aligned} \Gamma[\phi_c] &= S[\phi_c] + \underbrace{(\phi_1, J)}_{\mathcal{O}(\hbar)} + (\phi_c, J) + \frac{i\hbar}{2} \ln \det \{i\Delta^{-1}[\phi_c]\} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_c, J) + \frac{i\hbar}{2} \ln \det \{i\Delta^{-1}[\phi_c]\} + \mathcal{O}(\hbar^2), \end{aligned} \quad (4.41)$$

where the (ϕ_1, J) term is absorbed into $\mathcal{O}(\hbar^2)$. Setting $J = 0$ and $\phi_c(x) = \rho$ (const), and using $\Gamma[\rho] = -\Omega U(\rho)$ and $S[\rho] = -\Omega V(\rho)$,

$$-\Omega U(\rho) = -\Omega V(\rho) + \frac{i\hbar}{2} \ln \det \{i\Delta^{-1}[\rho]\} + \mathcal{O}(\hbar^2), \quad (4.42)$$

$$\boxed{U(\rho) = V(\rho) - \frac{i\hbar}{2\Omega} \ln \det \{i\Delta^{-1}[\rho]\} + \mathcal{O}(\hbar^2).} \quad (4.43)$$

The determinant is evaluated using $\ln \det = \text{Tr} \ln$,

$$\begin{aligned} \ln \det \{i\Delta^{-1}[\rho]\} &= \text{Tr} \ln \{i\Delta^{-1}[\rho]\} \\ &= \int \frac{d^4k}{(2\pi)^4} \ln \langle k | i\Delta^{-1}[\rho] | k \rangle, \end{aligned} \quad (4.44)$$

so that

$$U(\rho) = V(\rho) - \frac{i\hbar}{2} \int \frac{d^4k}{(2\pi)^4} \Omega^{-1} \ln \langle k | i\Delta^{-1}[\rho] | k \rangle + \mathcal{O}(\hbar^2). \quad (4.45)$$

To evaluate $\Omega^{-1} \langle k | i\Delta^{-1}[\rho] | k \rangle$, expand $\phi(x) = \rho + \phi_1(x)$ with ρ constant and $\partial V / \partial \phi|_{\phi=\rho} = 0$,

$$\begin{aligned} S[\phi] &= \int d^4x \left[\frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 - V(\phi_1 + \rho) \right] \\ &= -\frac{1}{2} \int d^4x \phi_1 [\square + V''(\rho)] \phi_1 + \dots \end{aligned} \quad (4.46)$$

By (4.31),

$$\begin{aligned} \langle x | i\Delta^{-1}[\rho] | y \rangle &= \frac{\delta^2 S[\phi]}{\delta \phi_1(x) \delta \phi_1(y)} \Big|_{\phi_1=0} \\ &= -[\square + V''(\rho)] \delta^4(x - y). \end{aligned} \quad (4.47)$$

Taking the matrix element between momentum eigenstates

$$\Omega^{-1} \langle k | i\Delta^{-1}[\rho] | k \rangle = k^2 - V''(\rho). \quad (4.48)$$

One way to see how we get this is the following: We evaluate $\langle k' | i\Delta^{-1}[\rho] | k \rangle$ using $\langle x | k \rangle = e^{ikx}$,

$$\begin{aligned} \langle k' | i\Delta^{-1}[\rho] | k \rangle &= \int d^4x d^4y \langle k' | x \rangle \langle x | i\Delta^{-1}[\rho] | y \rangle \langle y | k \rangle \\ &= \int d^4x d^4y e^{ik'x} [-(\square + V''(\rho))] \delta^4(x - y) e^{iky} \\ &= \int d^4x e^{-i(k-k')x} [-(-k^2 + V''(\rho))] \\ &= [k^2 - V''(\rho)] (2\pi)^4 \delta^4(k - k'). \end{aligned} \quad (4.49)$$

Setting $k = k'$ and dividing by $\Omega = (2\pi)^4 \delta^4(0)$,

$$\langle k | i\Delta^{-1}(\rho) | k \rangle = [k^2 - V''(\rho)] \Omega, \quad (4.50)$$

$$\Rightarrow \Omega^{-1} \langle k | i\Delta^{-1}(\rho) | k \rangle = k^2 - V''(\rho). \quad (4.51)$$

Substituting into (4.45) and rotating to Euclidean momentum space via $k^0 \rightarrow ik_E^0$, $k^2 \rightarrow -k_E^2$, $d^4k = i d^4k_E$,

$$U(\rho) = V(\rho) + \frac{\hbar}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln [k_E^2 + V''(\rho)] + \mathcal{O}(\hbar^2), \quad (4.52)$$

where the $i\pi$ contribution from $\ln(-k_E^2 - V'')$ is a field-independent constant and has been dropped. The momentum integral is evaluated in Appendix E, with the result

$$\frac{\hbar}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln(k_E^2 + V'') = \hbar \left[\frac{\Lambda^2}{32\pi^2} V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2} \left(-\frac{1}{2} + \ln \frac{V''(\rho)}{\Lambda^2} \right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right) \right]. \quad (4.53)$$

We can therefore write

$$U(\rho) = V(\rho) + \hbar V_1(\rho) + \mathcal{O}(\hbar^2), \quad (4.54)$$

where the one-loop correction is

$$V_1(\rho) = \frac{\Lambda^2}{32\pi^2} V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2} \left[-\frac{1}{2} + \ln \frac{V''(\rho)}{\Lambda^2} \right]. \quad (4.55)$$

To separate the divergent from the convergent part, introduce an arbitrary renormalization scale μ via

$$\ln \frac{V''}{\Lambda^2} = \ln \frac{V''}{\mu^2} + \ln \frac{\mu^2}{\Lambda^2}, \quad (4.56)$$

so that

$$\begin{aligned} V_1(\rho) &= \frac{\Lambda^2}{32\pi^2} V''(\rho) + \frac{(V''(\rho))^2}{64\pi^2} \left[-\frac{1}{2} + \ln \frac{V''}{\mu^2} + \ln \frac{\mu^2}{\Lambda^2} \right] \\ &= \underbrace{\frac{1}{32\pi^2} \left[\Lambda^2 V''(\rho) - \frac{1}{2} [V''(\rho)]^2 \ln \frac{\Lambda^2}{\mu^2} \right]}_{\text{divergent}} + \underbrace{\frac{(V'')^2}{64\pi^2} \left[-\frac{1}{2} + \ln \frac{V''}{\mu^2} \right]}_{\text{convergent}}. \end{aligned} \quad (4.57)$$

The divergent part must be removed by renormalization, which we turn to next.

4.6 Renormalization

The theory is renormalizable if the counterterms needed to cancel the divergences are of the same form as terms already present in the Lagrangian. For a classical potential $V(\rho)$ that is a polynomial of degree n , the second derivative $V''(\rho)$ is of degree $n - 2$, and the one-loop correction $V_1(\rho)$ in (4.55) is of degree $2(n - 2) = 2n - 4$ from the $(V'')^2$ term. Renormalizability then requires

$$2n - 4 \leq n \quad \Rightarrow \quad n \leq 4, \quad (4.58)$$

a condition satisfied by our choice of $V(\rho)$.

The bare parameters are split into renormalized parameters and counterterms,

$$\alpha_0 = \alpha_1 + \delta\alpha, \quad (4.59)$$

$$m_0^2 = m_1^2 + \delta m^2, \quad (4.60)$$

where $\delta\alpha$ and δm^2 are divergent in perturbation theory but α_1 and m_1^2 are required to be finite. Since the counterterms only arise at the quantum level, $\delta\alpha, \delta m^2 \sim \mathcal{O}(\hbar)$. The Lagrangian becomes

$$\mathcal{L}(x) = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) + \underbrace{\frac{\delta m^2}{2} \phi^2 - \frac{\delta\alpha}{4!} \phi^4}_{\text{counterterms}}, \quad (4.61)$$

with the renormalized classical potential

$$V(\phi) = \frac{\alpha_1}{4!} \phi^4 - \frac{m_1^2}{4} \phi^2. \quad (4.62)$$

The classical vacuum value ρ_0 is found from

$$\begin{aligned} \frac{dV(\phi)}{d\phi} &= \frac{\alpha_1}{3!} \phi^3 - \frac{m_1^2}{2} \phi = 0 \\ \Rightarrow \phi \left(\frac{\alpha_1}{3} \phi^2 - m_1^2 \right) &= 0, \end{aligned} \quad (4.63)$$

giving $\phi = 0$ or

$$\rho_0 = m_1 \left(\frac{3}{\alpha_1} \right)^{\frac{1}{2}}. \quad (4.64)$$

Since the counterterms are of order $\hbar$, they contribute at the same order as the one-loop correction and the full effective potential is

$$U(\rho) = V(\rho) + \hbar V_1(\rho) + \frac{\delta\alpha}{4!} \rho^4 - \frac{\delta m^2}{2} \rho^2. \quad (4.65)$$

Choosing $\delta\alpha$ and δm^2 to precisely cancel the divergent part of $V_1(\rho)$,

$$\delta\alpha = \frac{3\alpha_1^2 \hbar}{32\pi^2} \ln \frac{\Lambda^2}{\mu^2}, \quad (4.66)$$

$$\delta m^2 = \frac{\alpha_1 \hbar}{32\pi^2} \left(\Lambda^2 + \frac{m_1^2}{2} \ln \frac{\Lambda^2}{\mu^2} \right), \quad (4.67)$$

where the right-hand sides are ambiguous up to finite additive terms, the effective potential becomes

$$U(\rho) = V(\rho) + \frac{\hbar}{64\pi^2} [V''(\rho)]^2 \left[-\frac{1}{2} + \ln \frac{V''(\rho)}{\mu^2} \right]. \quad (4.68)$$

4.7 Physical Parameters from the Effective Potential

We now extract the VEV, the renormalized mass, and the renormalized coupling constant from (4.68). Setting $k \equiv \hbar/(32\pi^2)$ for brevity, the successive derivatives of $U(\rho)$ are computed using $V = \frac{\alpha_1}{4!}\rho^4 - \frac{m_1^2}{4}\rho^2$, so $V' = \frac{\alpha_1}{3!}\rho^3 - \frac{m_1^2}{2}\rho$, $V'' = \frac{\alpha_1}{2}\rho^2 - \frac{m_1^2}{2}$, $V''' = \alpha_1\rho$, $V'''' = \alpha_1$, $V''''' = 0$.

Differentiating (4.68),

$$\begin{aligned} U' &= V' + \frac{\hbar}{64\pi^2} \left[2V''V''' \left(-\frac{1}{2} + \ln \frac{V''}{\mu^2} \right) + (V'')^2 \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2 \right] \\ &= V' + \frac{\hbar}{64\pi^2} \left[-V''V''' + 2V''V''' \ln \frac{V''}{\mu^2} + V''V''' \right], \quad (4.69) \end{aligned}$$

$$\Rightarrow U' = V' + k V''V''' \ln \frac{V''}{\mu^2}. \quad (4.70)$$

Differentiating again,

$$U'' = V'' + k \left[(V''')^2 \ln \frac{V''}{\mu^2} + V''V'''' \ln \frac{V''}{\mu^2} + V''V''' \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2 \right], \quad (4.71)$$

$$\Rightarrow U'' = V'' + k \left[\{(V''')^2 + \alpha_1 V''\} \ln \frac{V''}{\mu^2} + (V''')^2 \right]. \quad (4.72)$$

The third derivative,

$$\begin{aligned} U''' &= V''' + k \left[\{(V''')^2 + \alpha_1 V''\}' \ln \frac{V''}{\mu^2} \right. \\ &\quad \left. + \{(V''')^2 + \alpha_1 V''\} \left(\ln \frac{V''}{\mu^2} \right)' + 2V''''V'''' \right] \\ &= V''' + k \left[(2V''''V'''' + \alpha_1 V''') \ln \frac{V''}{\mu^2} + \{(V''')^2 + \alpha_1 V''\} \frac{V'''}{V''} + 2\alpha_1 V''' \right], \quad (4.73) \end{aligned}$$

$$\Rightarrow U''' = V''' + k \left[3\alpha_1 V''' \ln \frac{V''}{\mu^2} + \frac{(V''')^3}{V''} + 3\alpha_1 V''' \right]. \quad (4.74)$$

The fourth derivative,

$$\begin{aligned} U'''' &= V'''' + k \left[3\alpha_1 V'''' \ln \frac{V''}{\mu^2} + 3\alpha_1 V'''' \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2 \right. \\ &\quad \left. + \frac{3(V''')^2 V''''}{V''} - \frac{(V''')^3}{(V'')^2} \cdot V'' + 3\alpha_1 V'''' \right] \\ &= \alpha_1 + k \left[3\alpha_1^2 \ln \frac{V''}{\mu^2} + 3\alpha_1 \frac{(V''')^2}{V''} + \frac{3\alpha_1 (V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2 \right] \\ &= \alpha_1 + k \left[3\alpha_1^2 \ln \frac{V''}{\mu^2} + \frac{6\alpha_1 (V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2 \right], \quad (4.75) \end{aligned}$$

$$\Rightarrow U'''' = \alpha_1 + k\alpha_1^2 \left[3 \ln \frac{V''}{\mu^2} + \frac{6\alpha_1 \rho^2}{V''} - \frac{\alpha_1^2 \rho^4}{(V'')^2} + 3 \right]. \quad (4.76)$$

The VEV

The minimum $\langle\phi\rangle = \rho_0 + \rho_1$ is found by expanding $U'(\rho_0 + \rho_1) = 0$ to first order,

$$U'(\rho_0) + \rho_1 U''(\rho_0) = 0 \quad \Rightarrow \quad \rho_1 = -\frac{U'(\rho_0)}{U''(\rho_0)}. \quad (4.77)$$

We first evaluate $V''(\rho_0)$ and $V'''(\rho_0)$ at the classical minimum $\rho_0^2 = 3m_1^2/\alpha_1$,

$$V''(\rho_0) = \frac{\alpha_1}{2}\rho_0^2 - \frac{m_1^2}{2} = \frac{\alpha_1}{2} \cdot \frac{3m_1^2}{\alpha_1} - \frac{m_1^2}{2} = \frac{3m_1^2}{2} - \frac{m_1^2}{2} = m_1^2, \quad (4.78)$$

$$\Rightarrow V''(\rho_0) = m_1^2. \quad (4.79)$$

Also, $V'(\rho_0) = 0$ by definition of ρ_0 , so from (4.70),

$$U'(\rho_0) = k m_1^2 \alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2}. \quad (4.80)$$

For $U''(\rho_0)$, the $\mathcal{O}(\alpha_1^2)$ terms are dropped,

$$\begin{aligned} U''(\rho_0) &= m_1^2 + k \left[(\alpha_1^2 \rho_0^2 + \alpha_1 m_1^2) \ln \frac{m_1^2}{\mu^2} + \alpha_1^2 \rho_0^2 \right] \\ &= m_1^2 + k \alpha_1 m_1^2 \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \quad (4.81)$$

$$\Rightarrow U''(\rho_0) = m_1^2 \left(1 + \alpha_1 k \ln \frac{m_1^2}{\mu^2} \right) + \mathcal{O}(\alpha_1^2). \quad (4.82)$$

Hence,

$$\begin{aligned} \rho_1 &= -\frac{k m_1^2 \alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2}}{m_1^2 \left(1 + \alpha_1 k \ln \frac{m_1^2}{\mu^2} \right)} \\ &= -k \alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2} \left(1 + \alpha_1 k \ln \frac{m_1^2}{\mu^2} \right)^{-1} \\ &= -k \alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2} \left[1 - \alpha_1 k \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2) \right] \\ &= -k \alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \quad (4.83)$$

$$\frac{\rho_1}{\rho_0} = -\frac{\hbar \alpha_1}{32\pi^2} \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2). \quad (4.84)$$

Setting $\hbar = 1$, the full VEV is

$$\boxed{\langle\phi\rangle = m_1 \left(\frac{3}{\alpha_1} \right)^{\frac{1}{2}} \left[1 - \frac{\alpha_1}{32\pi^2} \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2) \right]}. \quad (4.85)$$

The Renormalized Mass

The renormalized mass is $m^2 \equiv U''(\rho_0 + \rho_1)$. Expanding,

$$m^2 = U''(\rho_0) + \rho_1 U'''(\rho_0). \quad (\text{I})$$

From (4.74), dropping $\mathcal{O}(\alpha_1^2)$ terms,

$$\begin{aligned} U'''(\rho_0) &= \alpha_1 \rho_0 + k \left[3\alpha_1^2 \rho_0 \ln \frac{m_1^2}{\mu^2} + \frac{(\alpha_1 \rho_0)^3}{m_1^2} + 3\alpha_1^2 \rho_0 \right] \\ &= \alpha_1 \rho_0 + \mathcal{O}(\alpha_1^2), \end{aligned} \quad (\text{II})$$

$$U'''(\rho_0) = \alpha_1 \rho_0 + k \left[3\alpha_1^2 \rho_0 \ln \frac{m_1^2}{\mu^2} \right]. \quad (\text{III})$$

Also,

$$U''(\rho_0) = m_1^2 + k \left[(\alpha_1^2 \rho_0^2 + \alpha_1 m_1^2) \ln \frac{m_1^2}{\mu^2} + \alpha_1^2 \rho_0^2 \right]. \quad (4.86)$$

Substituting (II) and (III) into (I), using $\rho_1 = -k\alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2}$ and $\rho_1 U'''(\rho_0) = -k\alpha_1 \rho_0 \ln \frac{m_1^2}{\mu^2} \cdot 3m_1^2 + \mathcal{O}(\alpha_1^2)$,

$$\begin{aligned} m^2 &= m_1^2 \left(1 + 4\alpha_1 k \ln \frac{m_1^2}{\mu^2} + 3\alpha_1 k \right) - 3m_1^2 \alpha_1 k \ln \frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2) \\ &= m_1^2 \left[1 + 4\alpha_1 k \ln \frac{m_1^2}{\mu^2} + 3\alpha_1 k - 3\alpha_1 k \ln \frac{m_1^2}{\mu^2} \right] + \mathcal{O}(\alpha_1^2) \\ &= m_1^2 \left[1 + \alpha_1 k \ln \frac{m_1^2}{\mu^2} + 3\alpha_1 k \right] + \mathcal{O}(\alpha_1^2) \\ &= m_1^2 \left[1 + 3\alpha_1 k \left(1 + \frac{1}{3} \ln \frac{m_1^2}{\mu^2} \right) \right] + \mathcal{O}(\alpha_1^2), \end{aligned} \quad (4.87)$$

$$\Rightarrow \boxed{m^2 = m_1^2 \left[1 + \frac{3\alpha_1}{32\pi^2} \left(1 + \frac{1}{3} \ln \frac{m_1^2}{\mu^2} \right) + \mathcal{O}(\alpha_1^2) \right]}. \quad (4.88)$$

The Renormalized Coupling Constant

The renormalized coupling is $\alpha \equiv U''''(\rho_0 + \rho_1)$. Evaluating (4.76) at $\rho = \rho_0$ and keeping terms to $\mathcal{O}(\alpha_1)$,

$$\begin{aligned} \alpha &= \alpha_1 + k\alpha_1^2 \left[3 \ln \frac{V''}{\mu^2} + \frac{6\alpha_1 \rho^2}{V''} - \frac{\alpha_1^2 \rho^4}{(V'')^2} + 3 \right] \\ &= \alpha_1 \left[1 + 3\alpha_1 k \left(1 + \ln \frac{V''}{\mu^2} \right) + \mathcal{O}(\alpha_1^2) \right], \end{aligned} \quad (4.89)$$

$$\Rightarrow \boxed{\alpha = \alpha_1 \left[1 + \frac{3\alpha_1}{32\pi^2} \left(1 + \ln \frac{m_1^2}{\mu^2} \right) + \mathcal{O}(\alpha_1^2) \right]}. \quad (4.90)$$

4.8 The Massless Case and the Running Coupling Constant

We now consider the massless limit $m_1 = 0$, in which

$$V' = \frac{\alpha_1}{6} \rho^3, \quad V'' = \frac{\alpha_1 \rho^2}{2}. \quad (4.91)$$

From (4.76),

$$\begin{aligned} U''''(\rho) &= \alpha_1 + k\alpha_1^2 \left[3 \ln \frac{\alpha_1 \rho^2}{2\mu^2} + \frac{6\alpha_1 \rho^2}{\frac{\alpha_1 \rho^2}{2}} - \frac{\alpha_1^2 \rho^4}{\frac{\alpha_1^2 \rho^4}{4}} + 3 \right] \\ &= \alpha_1 + k\alpha_1^2 \left[3 \ln \frac{\alpha_1 \rho^2}{2\mu^2} + 12 - 4 + 3 \right], \end{aligned} \quad (4.92)$$

$$\Rightarrow U''''(\rho) = \alpha_1 + k\alpha_1^2 \left[11 + 3 \ln \frac{\alpha_1 \rho^2}{2\mu^2} \right]. \quad (4.93)$$

Setting $\alpha_1 \equiv \alpha$ and defining the running coupling constant $\alpha(\rho) \equiv U''''(\rho)$,

$$\alpha(\rho) = \alpha + \frac{k\alpha^2}{1} \left[11 + 3 \ln \frac{\alpha \rho^2}{2\mu^2} \right]. \quad (4.94)$$

We introduce a new scale μ_0 defined by the renormalization condition $U''''(\mu_0) = \alpha$, which gives

$$\begin{aligned} \alpha &= \alpha + k\alpha^2 \left[11 + 3 \ln \frac{\alpha \mu_0^2}{2\mu^2} \right] \\ \Rightarrow \ln \frac{\alpha \mu_0^2}{2\mu^2} &= -\frac{11}{3}. \end{aligned} \quad (4.95)$$

Using this to eliminate the explicit $\ln(\alpha/2\mu^2)$ dependence,

$$\begin{aligned} \alpha(\rho) &= \alpha + k\alpha^2 \left[3 \cdot \frac{11}{3} + 3 \ln \frac{\alpha \rho^2}{2\mu^2} \right] \\ &= \alpha + k\alpha^2 \left[-3 \ln \frac{\alpha \mu_0^2}{2\mu^2} + 3 \ln \frac{\alpha \rho^2}{2\mu^2} \right] \\ &= \alpha + 3k\alpha^2 \ln \left(\frac{\frac{\alpha \rho^2}{2\mu^2}}{\frac{\alpha \mu_0^2}{2\mu^2}} \right), \end{aligned} \quad (4.96)$$

$$\Rightarrow \boxed{\alpha(\rho) = \alpha + \frac{3k\alpha^2}{32\pi^2} \ln \frac{\rho^2}{\mu_0^2}}, \quad (4.97)$$

where $\alpha = \alpha(\mu_0)$ is the value of the running coupling at $\rho = \mu_0$. Inverting,

$$\frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0) + \frac{3k\alpha^2(\mu_0)}{32\pi^2} \ln \frac{\rho^2}{\mu_0^2}} = \frac{1}{\alpha(\mu_0)} \left[1 + \frac{3k\alpha(\mu_0)}{32\pi^2} \ln \frac{\rho^2}{\mu_0^2} \right]^{-1}, \quad (4.98)$$

$$\Rightarrow \frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0)} \left(1 - \frac{3k\alpha(\mu_0)}{32\pi^2} \ln \frac{\rho^2}{\mu_0^2} \right) + \text{higher order}. \quad (4.99)$$

4.9 Spontaneous Symmetry Breaking by Radiative Corrections?

A natural question arises: can SSB occur in the massless theory purely due to radiative corrections, generating a nonzero $\langle\phi\rangle$ dynamically? From (4.70) with $m_1 = 0$,

$$\begin{aligned} U'(\rho) &= V' + k V'' V''' \ln \frac{V''}{\mu^2} \\ &= \frac{\alpha}{6} \rho^3 + k \frac{\alpha \rho^2}{2} \cdot \alpha \rho \cdot \ln \frac{\alpha \rho^2}{2\mu^2} \\ &= \frac{\rho^3}{6} \left[\alpha + 3k\alpha^2 \ln \frac{\alpha \rho^2}{2\mu^2} \right]. \end{aligned} \quad (4.100)$$

Writing $\ln \frac{\alpha \rho^2}{2\mu^2} = \ln \frac{\rho^2}{\mu^2} + \ln \frac{\alpha}{2}$, where the last term is a field-independent constant,

$$U'(\rho) = \frac{\rho^3}{6} \left[\alpha + \frac{3\hbar\alpha^2}{32\pi^2} \left(\ln \frac{\rho^2}{\mu^2} + \text{const} \right) \right]. \quad (4.101)$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root gives

$$\begin{aligned} \alpha + \frac{3k\alpha^2}{32\pi^2} \ln \frac{\langle\phi\rangle^2}{\mu^2} &= 0 \\ \Rightarrow \alpha \left(1 + \frac{3k\alpha}{32\pi^2} \ln \frac{\langle\phi\rangle^2}{\mu^2} \right) &= 0 \\ \Rightarrow \hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} &= -\frac{32\pi^2}{3} + \mathcal{O}(\hbar). \end{aligned} \quad (4.102)$$

This root must be rejected. The loop expansion is valid only when $\hbar\alpha$ is small, but (4.102) requires

$$\hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} \sim \mathcal{O}(1), \quad (4.103)$$

meaning higher-order terms are not small and cannot be neglected. The minimum lies outside the domain of validity of the one-loop approximation. Let us explain this in slightly more details. The loop expansion is an expansion in powers of $\hbar$, with the L -loop contribution carrying a factor of $\hbar^L$. The one-loop approximation is therefore valid when the one-loop correction is small compared to the tree-level term,

$$\frac{\hbar V_1(\rho)}{V(\rho)} \ll 1, \quad (4.104)$$

or more precisely in this case, when

$$\hbar\alpha \ln \frac{\langle\phi\rangle^2}{\mu^2} \ll 1. \quad (4.105)$$

The spurious root requires precisely the opposite, $\hbar\alpha \ln(\langle\phi\rangle^2/\mu^2) \sim \mathcal{O}(1)$, so that the logarithm is of order $1/(\hbar\alpha)$ and $\langle\phi\rangle$ sits exponentially far from μ . At such field values the two-loop contribution goes as

$$(\hbar\alpha)^2 \cdot \left(\ln \frac{\langle\phi\rangle^2}{\mu^2} \right)^2 \sim \mathcal{O}(1), \quad (4.106)$$

and similarly for all higher loops, so every order contributes equally and truncating at one loop is unjustified. The root is therefore rejected, not of validity of the approximation.

The technical reason for this failure is that the two terms in (4.101) are of order α and α^2 respectively, and cannot cancel each other within the range of validity of perturbation theory: an $\mathcal{O}(\alpha)$ quantity cannot cancel an $\mathcal{O}(\alpha^2)$ quantity. However, if the model were modified so that α and α^2 were replaced by two independent coupling constants, the cancellation could occur and one would obtain a genuine theory of SSB by radiative corrections. The simplest such model is massless scalar electrodynamics, to which we turn in the next section.

Chapter 5

Massless Scalar Electrodynamics

In renormalized massless scalar electrodynamics, the scalar field acquires an arbitrary vacuum expectation value $\langle\phi\rangle$ even though classically it vanishes. This means $\langle\phi\rangle$ emerges as a spontaneously generated physical mass scale, with both scalar and vector particles developing dynamical masses proportional to it. This phenomenon is called *dimensional transmutation*: a theory with no classical mass scale develops a preferred mass scale through radiative corrections.

5.1 The Lagrangian and Unitary Gauge

The Lagrangian of massless scalar electrodynamics is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + [(\partial_\mu - ieA_\mu)\phi^*][(\partial^\mu + ieA^\mu)\phi] - \frac{\lambda_0}{6}(\phi^*\phi)^2, \quad (5.1)$$

where e is the renormalized coupling constant and λ_0 is the bare scalar coupling, necessary to renormalize the scalar-scalar scattering amplitude. Transforming away the phase of ϕ by going to unitary gauge and rescaling $\phi \rightarrow \phi/\sqrt{2}$, the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}e^2A_\mu A^\mu\phi^2 + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{\lambda_0}{4!}\phi^4, \quad (5.2)$$

subject to the Lorenz condition $\partial_\mu A^\mu = 0$. The corresponding action is

$$S[A, \phi] = \int d^4x \left[\frac{1}{2}A^\mu(\partial^2 + e^2\phi^2)A_\mu - \frac{1}{2}\phi\partial^2\phi - \frac{\lambda_0}{4!}\phi^4 \right]. \quad (5.3)$$

5.2 Integrating Out the Vector Field

To obtain an effective action for ϕ alone, we integrate out the vector field,

$$\exp iS[\phi] = \mathcal{N} \int (\mathcal{D}A) \delta[\partial_\mu A^\mu] \exp iS[A, \phi], \quad (5.4)$$

where $\delta[\partial_\mu A^\mu]$ enforces the Lorenz constraint. Each transverse component of A_μ contributes a Gaussian determinant,

$$\begin{aligned} [\det(\partial^2 + e^2\phi^2)]^{-\frac{1}{2}} &= \exp \left[-\frac{1}{2} \ln \det(\partial^2 + e^2\phi^2) \right] \\ &= \exp \left[-\frac{1}{2} \text{Tr} \ln(\partial^2 + e^2\phi^2) \right]. \end{aligned} \quad (5.5)$$

There are three such transverse components, so the effective action becomes

$$S[\phi] = - \int d^4x \left(\frac{1}{2} \phi \partial^2 \phi + \frac{\lambda_0}{4!} \phi^4 \right) - \frac{3}{2} \text{Tr} \ln(\partial^2 + e^2 \phi^2). \quad (5.6)$$

The one-loop effective potential for ϕ , to lowest order in λ_0 , is then

$$U(\rho) = \frac{\lambda_0}{4!} \rho^4 + \frac{3}{2} \int \frac{d^4 k_E}{(2\pi)^4} \ln(k_E^2 + e^2 \rho^2), \quad (5.7)$$

where the second term sums all one-loop graphs with external ϕ lines and vector boson running in the loop, since we integrated out A_μ . Scalar loops contribute only at higher order.

5.3 The Renormalized Effective Potential

Using the machinery developed in the previous chapter, specifically Eq.(4.68), the renormalized effective potential in this case takes the form

$$U(\rho) = \rho^4 \left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2} \left(-\frac{1}{2} + \ln \frac{e^2 \rho^2}{\mu^2} \right) \right], \quad (5.8)$$

where λ is the running coupling constant at the renormalization point μ . Note that e^4 is a one-loop result and λ is a tree-level result, but they can be comparable since they are independent coupling constants. This is what allows the cancellation that was impossible in the pure scalar case.

Differentiating (5.8),

$$\begin{aligned} U'(\rho) &= 4\rho^3 [\dots] + \rho^4 \cdot \frac{3e^4}{64\pi^2} \cdot \frac{1}{\frac{e^2 \rho^2}{\mu^2}} \cdot \frac{2\rho e^2}{\mu^2} \\ &= 4\rho^3 [\dots] + \frac{3e^4}{64\pi^2} \rho^3 \cdot 2 \\ &= \rho^3 \left[4 \cdot \frac{\lambda}{4!} + 4 \cdot \frac{3e^4}{64\pi^2} \left(-\frac{1}{2} \right) + 4 \cdot \frac{3e^4}{64\pi^2} \ln \frac{e^2 \rho^2}{\mu^2} + \frac{3e^4}{64\pi^2} \cdot 2 \right], \end{aligned} \quad (5.9)$$

$$\Rightarrow U'(\rho) = 4\rho^3 \left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2} \ln \frac{e^2 \rho^2}{\mu^2} \right]. \quad (5.10)$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root,

$$\begin{aligned} \frac{\lambda}{4!} + \frac{3e^4}{64\pi^2} \ln \frac{e^2 \langle\phi\rangle^2}{\mu^2} &= 0 \\ \Rightarrow \frac{3e^4}{64\pi^2} \ln \frac{e^2 \langle\phi\rangle^2}{\mu^2} &= -\frac{\lambda}{4!}. \end{aligned} \quad (5.11)$$

Unlike the pure scalar case, this root is within the domain of validity of the approximation since λ and e^4 are independent and can be chosen to make the two terms comparable without requiring the logarithm to be large. Using this condition to eliminate μ from (5.8),

$$\begin{aligned} U(\rho) &= \rho^4 \left[-\frac{3e^4}{64\pi^2} \ln \frac{e^2 \langle\phi\rangle^2}{\mu^2} + \frac{3e^4}{64\pi^2} \left(-\frac{1}{2} \right) + \frac{3e^4}{64\pi^2} \ln \frac{e^2 \rho^2}{\mu^2} \right] \\ &= \frac{3e^4 \rho^4}{64\pi^2} \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right]. \end{aligned} \quad (5.12)$$

The scalar particle mass follows from $m_S^2 \equiv U''(\langle\phi\rangle)$. Setting $a \equiv 3e^4/(64\pi^2)$ for brevity, we differentiate $U(\rho) = a\rho^4 \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right]$,

$$\begin{aligned} U'(\rho) &= a \cdot 4\rho^3 \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right] + a\rho^4 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 2a\rho^3 \left[2 \left(-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right) + 1 \right] \\ &= 4a\rho^3 \ln \frac{\rho^2}{\langle\phi\rangle^2}. \end{aligned} \tag{5.13}$$

Differentiating once more,

$$\begin{aligned} U''(\rho) &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 4a\rho^3 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 8a\rho^2. \end{aligned} \tag{5.14}$$

Evaluating at $\rho = \langle\phi\rangle$,

$$\begin{aligned} U''(\langle\phi\rangle) &= 12a\langle\phi\rangle^2 \ln \frac{\langle\phi\rangle^2}{\langle\phi\rangle^2} + 8a\langle\phi\rangle^2 \\ &= 8 \cdot \frac{3e^4}{64\pi^2} \langle\phi\rangle^2, \end{aligned} \tag{5.15}$$

$$\Rightarrow \boxed{m_S^2 = \frac{3e^4}{8\pi^2} \langle\phi\rangle^2}. \tag{5.16}$$

The vector field acquires mass through the Higgs mechanism,

$$m_V^2 = e^2 \langle\phi\rangle^2. \tag{5.17}$$

Both masses are proportional to $\langle\phi\rangle$, confirming that the dynamically generated VEV sets the mass scale of the theory.

5.4 Dimensional Transmutation

To renormalize the theory we introduce a momentum cutoff, which represents a hidden scale. After renormalization this hidden scale is replaced by an arbitrary finite renormalization point μ . In a classically massless theory one might expect all choices of μ to be equivalent, but what we have found is that there is a preferred renormalization point: the one at which the particles develop mass dynamically. Dimensional transmutation refers precisely to this emergence of a preferred scale in a theory with no intrinsic mass, whereby the infinite cutoff momentum becomes transmuted into a finite physical mass. This mechanism is widely believed to operate in QCD with massless quarks, where no intrinsic mass scale exists in the Lagrangian yet hadrons acquire mass dynamically.

Chapter 6

One-Loop Effective Potential in Non-Abelian Gauge Theories

The mechanism of spontaneous symmetry breaking via radiative corrections, which we saw explicitly in massless scalar electrodynamics, works equally well for non-Abelian gauge theories. The computation of the one-loop effective potential is conceptually the same as in the scalar case, though the technical details are more involved due to the richer field content. We follow the treatment of Brandenberger closely.

6.1 The Theory

Consider a non-Abelian gauge theory with gauge group G and generators τ_a , a set of Dirac bispinor fields ψ^a , and a set of scalar fields φ_i , each forming a representation space for G . The Lagrangian is

$$\mathcal{L}(A, \psi, \varphi) = -\frac{1}{4}\text{Tr}(F_{\mu\nu}F^{\mu\nu}) + i\text{Tr}(\bar{\psi}\not{D}\psi) + \text{Tr}(\bar{\psi}M\psi) - \text{Tr}(\bar{\psi}\Gamma\varphi\psi) + \frac{1}{2}\text{Tr}(D_\mu\varphi)^\dagger D^\mu\varphi - U(\varphi), \quad (6.1)$$

where the gauge field, field strength, and covariant derivative are

$$A_\mu = A_\mu^a \tau_a, \quad (6.2)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu], \quad (6.3)$$

$$D_\mu = \partial_\mu + igA_\mu. \quad (6.4)$$

In D_μ , the τ_a are the matrices representing the generators in the appropriate representation space. The trace refers to the appropriate representation space: if the Higgs and Fermi fields are represented as column vectors, the trace reduces to the scalar product, e.g. $\text{Tr}(D_\mu\varphi)^\dagger D^\mu\varphi = (D_\mu\varphi)^\dagger D^\mu\varphi$. The matrix Γ denotes the Yukawa coupling constants.

The Feynman rules in general involve extra fields, the ghost fields ω and $\bar{\omega}$, scalar fields obeying Fermi statistics (Faddeev and Popov, 1967). They arise as a consequence of the field-dependent functional determinant in the Faddeev-Popov ansatz for the generating functional. In covariant gauges, the ghost term in the effective Lagrangian is $\text{Tr} \bar{\omega} \partial_\mu A^\mu \omega$, and contains no Higgs couplings to ghost fields. Hence to one-loop order no ghosts arise in our calculation of the effective potential. The one-loop effective potential therefore takes the form

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + V_{\text{c.t.}}^{(1)} + V_{1\text{-loop}}, \quad (6.5)$$

where $V_{\text{c.t.}}^{(1)}$ contains all counterterms to order $\hbar$ and $V_{1\text{-loop}}$ is the sum of all one-loop graphs with external φ lines at zero momenta. By charge conservation, the particle type cannot change within a single loop, so the contributing one-loop diagrams are those with Higgs bosons, fermions, or gauge bosons running in the loop. The corresponding contributions are denoted $V_S^{(1)}$, $V_f^{(1)}$, and $V_g^{(1)}$.

6.2 Diagonalization of the Scalar Mass Matrix

The scalar sector of the theory is described by the Hessian of the potential, evaluated at the background field $\bar{\varphi}$,

$$W^{ij}(\bar{\varphi}) = \left. \frac{\partial^2 U(\varphi)}{\partial \varphi_i \partial \varphi_j} \right|_{\varphi=\bar{\varphi}}. \quad (6.6)$$

This matrix controls the quadratic fluctuations around the background. To see this explicitly, expand around the background $\varphi_i(x) = \bar{\varphi}_i + \eta_i(x)$, where η_i are small fluctuations. Since $\bar{\varphi}$ satisfies the classical equations of motion, the linear term vanishes and the quadratic Lagrangian is

$$\mathcal{L}_{\text{quad}} = \frac{1}{2}(\partial_\mu \eta_i)^2 - \frac{1}{2}\eta_i W^{ij}(\bar{\varphi})\eta_j. \quad (6.7)$$

If W^{ij} is not diagonal, the fields η_i are mixed and do not correspond to independent physical particles. To decouple them, we diagonalize W .

For an even quartic potential, the most general form is

$$U(\varphi) = \frac{1}{2}m_{ij}\varphi_i\varphi_j + \frac{1}{4!}\lambda_{ijkl}\varphi_i\varphi_j\varphi_k\varphi_l, \quad (6.8)$$

with $m_{ij} = m_{ji}$ and $\lambda_{ijkl} = \lambda_{(ijkl)}$. Computing the second derivative,

$$W^{ab}(\bar{\varphi}) = m_{ab} + \frac{1}{2}\lambda_{abkl}\bar{\varphi}_k\bar{\varphi}_l. \quad (6.9)$$

Since m_{ab} is symmetric and λ_{abkl} is symmetric in a, b , the matrix W^{ab} is real and symmetric. By the spectral theorem, any real symmetric matrix can be diagonalized by an orthogonal transformation: there exists an orthogonal matrix O such that

$$O^T W O = \text{diag}(w_1, \dots, w_N). \quad (6.10)$$

Defining the rotated fields $\chi_i = O_{ij}\eta_j$, and using the fact that orthogonal transformations preserve the kinetic term, the quadratic Lagrangian becomes

$$\mathcal{L}_{\text{quad}} = \sum_i \left[\frac{1}{2}(\partial_\mu \chi_i)^2 - \frac{1}{2}w_i \chi_i^2 \right]. \quad (6.11)$$

Each mode χ_i is now an independent scalar with mass squared w_i , propagating freely without mixing. These are the physical Higgs modes.

As a concrete example, for the gauge-invariant quartic potential $U(\varphi) = \frac{1}{2}m^2\varphi_i\varphi_i + \frac{\lambda}{4!}(\varphi_i\varphi_i)^2$, choosing the background along a single direction $\bar{\varphi}_i = v\delta_{i1}$ gives

$$W^{11} = m^2 + \frac{\lambda}{2}v^2 \quad (\text{radial/Higgs mode}), \quad (6.12)$$

$$W^{aa} = m^2 + \frac{\lambda}{6}v^2 \quad (N-1 \text{ degenerate transverse modes, } a \neq 1), \quad (6.13)$$

$$W^{ab} = 0 \quad \text{for } a \neq b, \quad (6.14)$$

so the matrix is already diagonal in this basis.

6.3 The Higgs Boson Contribution $V_S^{(1)}$

For $V_S^{(1)}$, the only change relative to the scalar field theory example is the presence of more than one scalar particle. The relevant diagrams are those of the single scalar case, now with one internal line associated with φ_i and the other with φ_j .

The effective coupling constant, which includes the $\bar{\varphi}$ factors for internal links and the Bose combinatorial factors, is

$$W^{ij} = \frac{\partial^2 U(\varphi)}{\partial \varphi_i \partial \varphi_j} \Big|_{\varphi=\bar{\varphi}}. \quad (6.15)$$

In the single scalar field theory, $W = \frac{1}{2}\lambda\bar{\varphi}^2$. If $U(\varphi)$ is an even quartic polynomial, $W(\bar{\varphi})$ is a quadratic form and can be diagonalized as shown above. In the diagonal basis, the different Higgs fields decouple and $V_S^{(1)}$ is obtained by adding the contributions of the new basis fields. Each individual term is given by the scalar Coleman-Weinberg integral. Neglecting the λ^2 corrections to terms already contained in $V_{\text{eff}}^{(0)}$, the cutoff-independent part of the one-loop integral for a single mode with $m^2 = \frac{\lambda}{2}\bar{\varphi}^2$ is

$$V_{1\text{-loop}}|_{\text{c.i.}} = \frac{1}{64\pi^2} \left(\frac{\lambda\bar{\varphi}^2}{2} \right)^2 \ln \frac{\lambda\bar{\varphi}^2}{2}, \quad (6.16)$$

where the divergent terms proportional to Λ^2 and $\ln \Lambda^2$ have been absorbed into renormalization of the mass and coupling constant respectively. Generalizing to the diagonal basis,

$$\begin{aligned} V_S^{(1)}(\bar{\varphi}) &= \frac{1}{64\pi^2} \sum_i (W^{ii})^2 \ln W^{ii} \\ &= \frac{1}{64\pi^2} \text{Tr}(W^2 \ln W), \end{aligned} \quad (6.17)$$

which is basis-independent since the trace is invariant under orthogonal transformations.

6.4 The Fermion Contribution $V_f^{(1)}$

The fermion contribution $V_f^{(1)}$ is given by the diagrams in Fig. 3 of Brandenberger, for the case $M = 0$. Since the trace of an odd number of γ matrices vanishes, diagrams with an odd number of vertices are zero. With all indices inserted explicitly, the Yukawa coupling term from (6.1) is

$$\sum_{a,b,i} \bar{\varphi}_i \Gamma_{ab}^i \psi^a \psi^b. \quad (6.18)$$

The diagram with two external scalar field insertions $\bar{\varphi}_i$ and $\bar{\varphi}_j$ consists of a closed fermion loop with two internal massless propagators $\not{k}/k^2$ and two Yukawa vertices contributing Γ_{ab}^i and Γ_{ba}^j . Assembling all factors, the overall sign (-1) from the fermion loop and the symmetry factor $\frac{1}{2}$ combine to give

$$-\frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} \sum_{i,j,a,b} \text{Tr}_S \bar{\varphi}_i \Gamma_{ab}^i \frac{\not{k}}{k^4} \Gamma_{ba}^j \bar{\varphi}_j. \quad (6.19)$$

Using $(\not{k})(\not{k}) = k^2$, the propagator product simplifies to $\not{k}^2/k^4 = 1/k^2$. The spinor trace then acts only on the identity in spinor space, giving $\text{Tr}_S(\mathbf{1}) = 4$ in four dimensions. Defining the matrix $(\bar{\varphi}\Gamma)_{ab} \equiv \sum_i \bar{\varphi}_i \Gamma_{ab}^i$, the sum over indices becomes $\sum_{a,b} (\bar{\varphi}\Gamma)_{ab} (\bar{\varphi}\Gamma)_{ba} = \text{Tr}_f(\bar{\varphi}\Gamma)^2$, and (6.19) reduces to

$$-\frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{4}{k^2} \text{Tr}_f(\bar{\varphi}\Gamma)^2. \quad (6.20)$$

The graph with $2n$ vertices yields analogously

$$\frac{1}{2n} \int \frac{d^4 k}{(2\pi)^4} \frac{1}{(k^2)^n} \text{Tr}_f(\bar{\varphi}\Gamma)^{2n}, \quad (6.21)$$

so that summing over all diagrams,

$$V_f^{(1)}(\bar{\varphi}) = -4i \text{Tr}_f \sum_{n=1}^{\infty} \int \frac{d^4 k}{(2\pi)^4} \frac{1}{2n} \left[\frac{(\bar{\varphi}\Gamma)^2}{k^2} \right]^n. \quad (6.22)$$

This result is analogous to (6.16), and its evaluation follows the same procedure as in the scalar case. The cutoff-independent part gives

$$V_f^{(1)}(\bar{\varphi}) = -\frac{1}{64\pi^2} 4 \text{Tr}_f [(\bar{\varphi}\Gamma)^4 \ln(\bar{\varphi}\Gamma)^2]. \quad (6.23)$$

6.5 The Gauge Boson Contribution $V_g^{(1)}$

Finally we compute $V_g^{(1)}$, the gauge boson loop contribution. The only vertex contributing to one-loop order is the $A^2\varphi^2$ vertex, which stems from the following term in the Lagrangian density,

$$\frac{1}{2} \text{Tr}(igA_\mu\varphi)^\dagger(igA^\mu\varphi) = \frac{1}{2} g_{\alpha\beta} \text{Tr}[(\tau_\alpha A_\mu^\alpha\varphi)^\dagger(\tau_\beta A^{\beta\mu}\varphi)] = \frac{1}{2} A_\mu^\alpha M_{\alpha\beta}^2 A^{\beta\mu}, \quad (6.24)$$

with the field-dependent gauge boson mass matrix

$$M_{\alpha\beta}^2 = g_{\alpha\beta} \text{Tr}[(\tau_\alpha\varphi)^\dagger\tau_\beta\varphi]. \quad (6.25)$$

A few comments on (6.25) are in order. First, if G is not a simple group, there will in general be a different coupling constant associated with each field A_μ^a . Second, the trace is in the Higgs representation space. Third, $M_{\alpha\beta}^2|_{\varphi=\bar{\varphi}}$ is the gauge boson mass matrix in the broken phase with vacuum expectation value $\bar{\varphi}$, and is a constant matrix in the sense of (6.6). Since in Landau gauge the gauge fields have three degrees of freedom (they have a Lie algebra trace), the calculation of $V_g^{(1)}$ is analogous to that of $V_S^{(1)}$, and

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64\pi^2} \text{Tr}(M^4 \ln M^2)|_{\varphi=\bar{\varphi}}. \quad (6.26)$$

6.6 The Full One-Loop Effective Potential

Combining the three contributions (6.17), (6.23), and (6.26), the full one-loop effective potential is

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + \frac{1}{64\pi^2} \text{Tr}(W^2 \ln W) - \frac{4}{64\pi^2} \text{Tr}_f [(\bar{\varphi}\Gamma)^4 \ln(\bar{\varphi}\Gamma)^2] + \frac{3}{64\pi^2} \text{Tr}(M^4 \ln M^2)|_{\varphi=\bar{\varphi}}, \quad (6.27)$$

where all traces are evaluated at $\varphi = \bar{\varphi}$. This is the master formula for the one-loop effective potential in a non-Abelian gauge theory. The scalar, fermion, and gauge boson loops each contribute a term of the Coleman-Weinberg form $m^4 \ln m^2$, with the mass matrix appropriate to each field type. In the next chapter we apply this formula to GUT models, where the gauge group is larger and the symmetry breaking pattern is richer.

Chapter 7

The Glashow-Salam-Weinberg Model

The Glashow-Salam-Weinberg (GSW) model is the quantum field theory of electroweak interactions, unifying the electromagnetic and weak forces into a single gauge theory with gauge group $SU(2)_L \times U(1)_Y$. It is one of the cornerstones of the Standard Model and provides the first physical example of spontaneous symmetry breaking in a gauge theory, the Higgs mechanism, which we studied abstractly in the previous chapter. Here we apply the effective potential formalism to this model, computing the field-dependent gauge boson masses and the resulting one-loop correction to the effective potential.

7.1 The Electroweak Lagrangian

The gauge fields of the theory are the $SU(2)_L$ triplet W_μ^i ($i = 1, 2, 3$) and the $U(1)_Y$ singlet V_μ , with corresponding field strengths $W_{\mu\nu}^i$ and $V_{\mu\nu}$. The matter content includes left-handed fermion doublets and right-handed singlets, but for the purposes of computing the effective potential it is the scalar sector that is relevant. The Higgs field φ is a complex doublet under $SU(2)_L$ with hypercharge $Y = 1$. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}W_{\mu\nu}^i W^{i\mu\nu} - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + (D_\mu\varphi)^\dagger(D^\mu\varphi) - V(\varphi), \quad (7.1)$$

where the covariant derivative is

$$D_\mu = \partial_\mu - ig\frac{\sigma^i}{2}W_\mu^i - ig'\frac{Y}{2}V_\mu, \quad (7.2)$$

with g and g' the $SU(2)_L$ and $U(1)_Y$ coupling constants respectively, and σ^i the Pauli matrices. The classical potential $V(\varphi) = -\mu^2\varphi^\dagger\varphi + \lambda(\varphi^\dagger\varphi)^2$ has a minimum at a nonzero field value when $\mu^2 > 0$, triggering spontaneous symmetry breaking.

7.2 Spontaneous Symmetry Breaking and Mass Generation

The Higgs field acquires a vacuum expectation value

$$\langle\varphi\rangle = \begin{pmatrix} 0 \\ a \end{pmatrix}, \quad (7.3)$$

which breaks $SU(2)_L \times U(1)_Y$ down to the electromagnetic $U(1)_{\text{em}}$. The gauge boson masses are read off from the kinetic term evaluated at the VEV. Since the VEV is constant, the ∂_μ term vanishes and

$$D_\mu \langle \varphi \rangle = \left(-ig \frac{\sigma^i}{2} W_\mu^i - ig' \frac{Y}{2} V_\mu \right) \begin{pmatrix} 0 \\ a \end{pmatrix}. \quad (7.4)$$

Expanding the $SU(2)$ part using

$$\sigma^i W_\mu^i = \begin{pmatrix} W_\mu^3 & W_\mu^1 - iW_\mu^2 \\ W_\mu^1 + iW_\mu^2 & -W_\mu^3 \end{pmatrix}, \quad (7.5)$$

acting on the VEV gives

$$\frac{\sigma^i}{2} W_\mu^i \begin{pmatrix} 0 \\ a \end{pmatrix} = \frac{a}{2} \begin{pmatrix} W_\mu^1 - iW_\mu^2 \\ -W_\mu^3 \end{pmatrix}. \quad (7.6)$$

For the $U(1)$ part, with hypercharge $Y = 1$,

$$\frac{Y}{2} V_\mu \begin{pmatrix} 0 \\ a \end{pmatrix} = \frac{a}{2} \begin{pmatrix} 0 \\ V_\mu \end{pmatrix}. \quad (7.7)$$

Combining,

$$D_\mu \langle \varphi \rangle = -ig \cdot \frac{a}{2} \begin{pmatrix} W_\mu^1 - iW_\mu^2 \\ -W_\mu^3 \end{pmatrix} - ig' \cdot \frac{a}{2} \begin{pmatrix} 0 \\ V_\mu \end{pmatrix} = \frac{ia}{2} \begin{pmatrix} gW_\mu^1 - igW_\mu^2 \\ -gW_\mu^3 + g'V_\mu \end{pmatrix}. \quad (7.8)$$

Taking the conjugate,

$$\overline{D_\mu \langle \varphi \rangle} = -\frac{ia}{2} (gW_\mu^1 + igW_\mu^2, -gW_\mu^3 + g'V_\mu), \quad (7.9)$$

so that

$$\overline{D_\mu \langle \varphi \rangle} D^\mu \langle \varphi \rangle = \frac{a^2}{4} (gW_\mu^1 + igW_\mu^2, -gW_\mu^3 + g'V_\mu) \begin{pmatrix} gW^{1\mu} - igW^{2\mu} \\ -gW^{3\mu} + g'V^\mu \end{pmatrix}. \quad (7.10)$$

Expanding the matrix product,

$$= \frac{a^2}{4} [g^2(W_\mu^{12} + W_\mu^{22}) + g^2W_\mu^{32} + g'^2V_\mu^2 - 2gg'W_\mu^3V_\mu]. \quad (7.11)$$

The fields W_μ^3 and V_μ mix, and to identify the physical mass eigenstates we introduce the Weinberg rotation,

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W V_\mu, \quad (7.12)$$

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W V_\mu, \quad (7.13)$$

where $\sin \theta_W = g' / \sqrt{g^2 + g'^2}$ and $\cos \theta_W = g / \sqrt{g^2 + g'^2}$. Inverting,

$$W_\mu^3 = \sin \theta_W A_\mu + \cos \theta_W Z_\mu, \quad (7.14)$$

$$V_\mu = \cos \theta_W A_\mu - \sin \theta_W Z_\mu. \quad (7.15)$$

Substituting into each term,

$$\begin{aligned}
g^2 W_\mu^{32} &= g^2 \sin^2 \theta_W A_\mu^2 + g^2 \cos^2 \theta_W Z_\mu^2 + 2g^2 \sin \theta_W \cos \theta_W A_\mu Z_\mu, \\
g'^2 V_\mu^2 &= g'^2 \cos^2 \theta_W A_\mu^2 + g'^2 \sin^2 \theta_W Z_\mu^2 - 2g'^2 \sin \theta_W \cos \theta_W A_\mu Z_\mu, \\
-2gg' W_\mu^3 V_\mu &= -2gg' (\sin \theta_W A_\mu + \cos \theta_W Z_\mu) (\cos \theta_W A_\mu - \sin \theta_W Z_\mu) \\
&= -2gg' \sin \theta_W \cos \theta_W A_\mu^2 + 2gg' \sin^2 \theta_W A_\mu Z_\mu - 2gg' \cos^2 \theta_W Z_\mu A_\mu + 2gg' \cos \theta_W \sin \theta_W Z_\mu^2.
\end{aligned} \tag{7.16}$$

Collecting by field bilinear,

$$\begin{aligned}
g^2 W_\mu^{32} + g'^2 V_\mu^2 - 2gg' W_\mu^3 V_\mu &= A_\mu^2 (g \sin \theta_W - g' \cos \theta_W)^2 \\
&\quad + Z_\mu^2 (g \cos \theta_W + g' \sin \theta_W)^2 \\
&\quad + 2 \sin \theta_W \cos \theta_W A_\mu Z_\mu \left(g^2 - g'^2 + gg' \left(\frac{\sin \theta_W}{\cos \theta_W} - \frac{\cos \theta_W}{\sin \theta_W} \right) \right).
\end{aligned} \tag{7.17}$$

The coefficient of A_μ^2 vanishes identically,

$$g \sin \theta_W - g' \cos \theta_W = \frac{gg'}{\sqrt{g^2 + g'^2}} - \frac{g'g}{\sqrt{g^2 + g'^2}} = 0, \tag{7.18}$$

and the $A_\mu Z_\mu$ cross term also vanishes. The coefficient of Z_μ^2 evaluates to

$$(g \cos \theta_W + g' \sin \theta_W)^2 = \left(\frac{g^2}{\sqrt{g^2 + g'^2}} + \frac{g'^2}{\sqrt{g^2 + g'^2}} \right)^2 = g^2 + g'^2. \tag{7.19}$$

The photon therefore remains massless, as required by the unbroken $U(1)_{\text{em}}$, and the kinetic term becomes

$$\boxed{\overline{D_\mu \langle \varphi \rangle} D^\mu \langle \varphi \rangle = \frac{a^2}{4} [g^2 (W_\mu^{12} + W_\mu^{22}) + (g^2 + g'^2) Z_\mu^2]}. \tag{7.20}$$

Reading off the field-dependent masses,

$$m^2(W_{1,2}) = \frac{1}{4} g^2 a^2, \tag{7.21}$$

$$m^2(Z) = \frac{1}{4} (g^2 + g'^2) a^2, \tag{7.22}$$

while the photon remains massless. These are the standard results of the Higgs mechanism in the electroweak theory.

7.3 One-Loop Gauge Field Contribution to the Effective Potential

With the field-dependent gauge boson masses identified, we can compute their contribution to the one-loop effective potential. By the same logic as in the scalar case, each massive

gauge boson with field-dependent mass matrix M^2 contributes a Coleman-Weinberg type term at one loop. The gauge field contribution is

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64\pi^2} \text{Tr}(M^4 \ln M^2) \Big|_{\rho=\bar{\varphi}}, \quad M_{\alpha\beta}^2 = \lambda_\alpha \lambda_\beta \text{Tr}[(t_\alpha \varphi)^\dagger \tau_\beta \varphi], \quad (7.23)$$

where the factor of 3 accounts for the three physical polarizations of each massive gauge boson. Substituting the masses of the two W bosons and the Z boson,

$$\begin{aligned} V_g^{(1)}(\bar{\varphi}) &= \frac{3}{64\pi^2} \left(\frac{1}{16}g^4 + \frac{1}{16}g'^4 + \frac{1}{16}(g^2 + g'^2)^2 \right) \bar{\varphi}^4 \ln \bar{\varphi}^2 \\ &= \frac{3}{64\pi^2} \cdot \frac{1}{16} [2g^4 + (g^2 + g'^2)^2] \bar{\varphi}^4 \ln \bar{\varphi}^2, \end{aligned} \quad (7.24)$$

$$\Rightarrow V_g^{(1)}(\bar{\varphi}) = \frac{3}{1024\pi^2} [2g^4 + (g^2 + g'^2)^2] \bar{\varphi}^4 \ln \bar{\varphi}^2. \quad (7.25)$$

The coefficient is determined entirely by the gauge couplings g and g' , and the structure $\bar{\varphi}^4 \ln \bar{\varphi}^2$ is the characteristic signature of the Coleman-Weinberg mechanism in a gauge theory.

Introduce $g = e/\sin \theta_W$ and $g' = e/\cos \theta_W$ to express the result in terms of measurable quantities. Computing each term,

$$g^4 = \frac{e^4}{\sin^4 \theta_W}, \quad (7.26)$$

$$(g^2 + g'^2)^2 = \frac{e^4}{\sin^4 \theta_W \cos^4 \theta_W}. \quad (7.27)$$

Combining,

$$2g^4 + (g^2 + g'^2)^2 = \frac{e^4}{\sin^4 \theta_W} \left(2 + \frac{1}{\cos^4 \theta_W} \right) = \frac{e^4 (2 + \cos^{-4} \theta_W)}{\sin^4 \theta_W}. \quad (7.28)$$

The prefactor rewrites as

$$\frac{3}{1024\pi^2} = \frac{3}{64} \cdot \frac{1}{16\pi^2} = \frac{3}{64} \left(\frac{1}{4\pi} \right)^2. \quad (7.29)$$

$$\frac{3}{1024\pi^2} [2g^4 + (g^2 + g'^2)^2] = \frac{3}{64} \left(\frac{e^2}{4\pi} \right)^2 \frac{2 + \cos^{-4} \theta_W}{\sin^4 \theta_W}, \quad (7.30)$$

and therefore

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64} \left(\frac{e^2}{4\pi} \right)^2 \left[\frac{2 + \cos^{-4} \theta_W}{\sin^4 \theta_W} \right] \bar{\varphi}^4 \ln \bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4), \quad (7.31)$$

which expresses the gauge boson contribution entirely in terms of the fine structure constant $\alpha_{\text{em}} = e^2/4\pi$ and the Weinberg angle θ_W .

We now derive the renormalized form of the gauge boson contribution. For the GSW model, the field-dependent masses are $m^2(W_{1,2}) = \frac{1}{4}g^2\bar{\varphi}^2$ for the two W bosons and $m^2(Z) = \frac{1}{4}(g^2 + g'^2)\bar{\varphi}^2$ for the Z boson. Each massive gauge boson with mass $m^2 = c\bar{\varphi}^2$ contributes to the one-loop integral, with three polarizations, a term

$$\frac{3c^2}{64\pi^2} \bar{\varphi}^4 \left[\ln \frac{c\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2} \right]. \quad (7.32)$$

Summing over the two W bosons and the Z boson and defining

$$A \equiv \frac{3}{1024\pi^2} [2g^4 + (g^2 + g'^2)^2], \quad (7.33)$$

the one-loop gauge boson potential, after the mass renormalization condition has been imposed to cancel the $\Lambda^2 \bar{\varphi}^2$ divergence, takes the form

$$V_g^{(1)}(\bar{\varphi}) = C^{(1)} \bar{\varphi}^4 + A \bar{\varphi}^4 \left[\ln \frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2} \right]. \quad (7.34)$$

We now compute four derivatives to apply the coupling constant renormalization condition. Differentiating once,

$$\begin{aligned} \frac{d}{d\bar{\varphi}} \left[A \bar{\varphi}^4 \left(\ln \frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2} \right) \right] &= 4A \bar{\varphi}^3 \left(\ln \frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2} \right) + A \bar{\varphi}^4 \cdot \frac{2}{\bar{\varphi}} \\ &= 4A \bar{\varphi}^3 \ln \frac{\bar{\varphi}^2}{\Lambda^2} - 2A \bar{\varphi}^3 + 2A \bar{\varphi}^3 \\ &= 4A \bar{\varphi}^3 \ln \frac{\bar{\varphi}^2}{\Lambda^2}, \end{aligned} \quad (7.35)$$

so that

$$\frac{dV_g^{(1)}}{d\bar{\varphi}} = 4C^{(1)} \bar{\varphi}^3 + 4A \bar{\varphi}^3 \ln \frac{\bar{\varphi}^2}{\Lambda^2}. \quad (7.36)$$

Differentiating again,

$$\begin{aligned} \frac{d}{d\bar{\varphi}} \left[4A \bar{\varphi}^3 \ln \frac{\bar{\varphi}^2}{\Lambda^2} \right] &= 12A \bar{\varphi}^2 \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 4A \bar{\varphi}^3 \cdot \frac{2}{\bar{\varphi}} \\ &= 12A \bar{\varphi}^2 \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 8A \bar{\varphi}^2, \end{aligned} \quad (7.37)$$

$$\frac{d^2 V_g^{(1)}}{d\bar{\varphi}^2} = 12C^{(1)} \bar{\varphi}^2 + 12A \bar{\varphi}^2 \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 8A \bar{\varphi}^2. \quad (7.38)$$

The third derivative,

$$\begin{aligned} \frac{d}{d\bar{\varphi}} \left[12A \bar{\varphi}^2 \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 8A \bar{\varphi}^2 \right] &= 24A \bar{\varphi} \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 12A \bar{\varphi}^2 \cdot \frac{2}{\bar{\varphi}} + 16A \bar{\varphi} \\ &= 24A \bar{\varphi} \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 24A \bar{\varphi} + 16A \bar{\varphi} \\ &= 24A \bar{\varphi} \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 40A \bar{\varphi}, \end{aligned} \quad (7.39)$$

$$\frac{d^3 V_g^{(1)}}{d\bar{\varphi}^3} = 24C^{(1)} \bar{\varphi} + 24A \bar{\varphi} \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 40A \bar{\varphi}. \quad (7.40)$$

The fourth derivative,

$$\begin{aligned} \frac{d}{d\bar{\varphi}} \left[24A \bar{\varphi} \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 40A \bar{\varphi} \right] &= 24A \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 24A \bar{\varphi} \cdot \frac{2}{\bar{\varphi}} + 40A \\ &= 24A \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 48A + 40A \\ &= 24A \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 88A, \end{aligned} \quad (7.41)$$

$$\frac{d^4 V_g^{(1)}}{d\bar{\varphi}^4} = 24C^{(1)} + 24A \ln \frac{\bar{\varphi}^2}{\Lambda^2} + 88A. \quad (7.42)$$

Imposing the coupling constant renormalization condition $\left. \frac{d^4 V_g^{(1)}}{d\bar{\varphi}^4} \right|_{\bar{\varphi}=M} = \lambda$,

$$24C^{(1)} + 24A \ln \frac{M^2}{\Lambda^2} + 88A = \lambda, \quad (7.43)$$

and solving for $C^{(1)}$,

$$C^{(1)} = \frac{\lambda - 88A}{24} - A \ln \frac{M^2}{\Lambda^2}. \quad (7.44)$$

Substituting back,

$$\begin{aligned} V_g^{(1)}(\bar{\varphi}) &= \left[\frac{\lambda - 88A}{24} - A \ln \frac{M^2}{\Lambda^2} \right] \bar{\varphi}^4 + A\bar{\varphi}^4 \left[\ln \frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2} \right] \\ &= \frac{\lambda}{24} \bar{\varphi}^4 - \frac{88A}{24} \bar{\varphi}^4 - A\bar{\varphi}^4 \ln \frac{M^2}{\Lambda^2} + A\bar{\varphi}^4 \ln \frac{\bar{\varphi}^2}{\Lambda^2} - \frac{A}{2} \bar{\varphi}^4. \end{aligned} \quad (7.45)$$

The two logarithms combine as

$$A\bar{\varphi}^4 \ln \frac{\bar{\varphi}^2}{\Lambda^2} - A\bar{\varphi}^4 \ln \frac{M^2}{\Lambda^2} = A\bar{\varphi}^4 \ln \frac{\bar{\varphi}^2}{M^2}, \quad (7.46)$$

and the remaining constant terms collect as

$$-\frac{88A}{24} - \frac{A}{2} = -\frac{88A}{24} - \frac{12A}{24} = -\frac{100A}{24} = -\frac{25A}{6}. \quad (7.47)$$

Therefore,

$$V_g^{(1)}(\bar{\varphi}) = \frac{\lambda}{4!} \bar{\varphi}^4 + A\bar{\varphi}^4 \left[\ln \frac{\bar{\varphi}^2}{M^2} - \frac{25}{6} \right], \quad (7.48)$$

and substituting back the definition of A and expressing in terms of the measurable quantities e and θ_W via (6.25),

$$\boxed{V_g^{(1)}(\bar{\varphi}) = \frac{\lambda}{4!} \bar{\varphi}^4 + \frac{3}{64} \left(\frac{e^2}{4\pi} \right)^2 \left[\frac{2 + \cos^{-4} \theta_W}{\sin^4 \theta_W} \right] \bar{\varphi}^4 \left[\ln \frac{\bar{\varphi}^2}{M^2} - \frac{25}{6} \right]}. \quad (7.49)$$

The $-25/6$ arises from collecting the finite constants after renormalization, specifically $-88/24 - 1/2 = -25/6$, and is a universal consequence of the renormalization condition imposed at the scale M , independent of the specific values of the gauge couplings.

7.4 Georgy-Glashow minimal SU(5) model

Chapter 8

Symmetry Breaking in Grand Unified Models

Part II

Finite Temperature Field Theory

Chapter 9

Finite Temperature Field Theory: General formalism

Conventional quantum field theory is formulated in empty space. The LSZ formalism expresses scattering matrix elements in terms of zero-temperature Green's functions, that is, vacuum expectation values of time-ordered strings of Heisenberg field operators. This is perfectly justified for particle accelerator experiments, where the background state is essentially the vacuum. In early universe cosmology, however, the background state is a thermal bath at temperature T , and a different formalism is required.

9.1 Finite-Temperature Green's Functions

The finite-temperature Green's function for a scalar field φ is defined by replacing the vacuum expectation value with a thermal average over the grand canonical ensemble,

$$G_n^\beta(x_1, \dots, x_n) = \frac{\text{Tr} \{ e^{-\beta H} T [\varphi(x_1) \cdots \varphi(x_n)] \}}{\text{Tr} e^{-\beta H}}, \quad (9.1)$$

where $\beta = 1/T$ is the inverse temperature and the trace is taken over all states of the system. This is the average value of the time-ordered product of n field operators in the grand canonical ensemble with no chemical potential.

The finite-temperature generating functional is defined analogously,

$$Z^\beta(J) = \frac{\text{Tr} [e^{-\beta H} T \exp(i \int J(x) \varphi(x) d^4x)]}{\text{Tr} e^{-\beta H}}. \quad (9.2)$$

The essential difference between the zero-temperature and finite-temperature formalisms lies in the boundary conditions on the set of paths over which the functional integral is taken. At zero temperature the transition amplitude is

$${}_H \langle \psi_1, t_1 | \psi_2, t_2 \rangle_H = \int [d\varphi(s, \mathbf{x})] \psi_1^*[\varphi(t_2, \mathbf{x})] \psi_2[\varphi(t_1, \mathbf{x})] \exp \left[i \int_{t_i}^{t_f} ds \int d^3x \mathcal{L}(\varphi, \partial\varphi) \right], \quad (9.3)$$

with $t_i \rightarrow -\infty$ and $t_f \rightarrow +\infty$. At finite temperature the boundary wavefunctionals $\psi^*[\varphi(t_f, \mathbf{x}_f)] \psi[\varphi(t_i, \mathbf{x}_i)]$ are replaced by

$$\sum e^{-\beta E(\psi)} \psi^*[\varphi(t_f, \mathbf{x}_f)] \psi[\varphi(t_i, \mathbf{x}_i)], \quad (9.4)$$

which implements the thermal weighting of states. It is also worth noting that the finite-temperature effective potential is not gauge invariant: only in physical gauges does the trace correspond to a sum over all physical states of the system.

9.2 Periodicity and the KMS Condition

A fundamental consequence of the thermal boundary conditions is that the finite-temperature Green's functions satisfy periodicity conditions in Euclidean time. Working with Euclidean times $t \in [0, -i\beta]$, the time-ordering is

$$T[\varphi(x)\varphi(y)] = \begin{cases} \varphi(x)\varphi(y) & \text{if } ix^0 > iy^0, \\ (-1)^s \varphi(y)\varphi(x) & \text{if } ix^0 < iy^0, \end{cases} \quad (9.5)$$

where $s = 0$ for bosonic fields and $s = 1$ for fermionic fields. To derive the periodicity condition, we evaluate the two-point function at $x^0 = 0$ with $iy^0 > ix^0$,

$$\begin{aligned} (\text{Tr } e^{-\beta H}) G_2^\beta(x-y)|_{x^0=0} &= (-1)^s \text{Tr} [e^{-\beta H} \varphi(y)\varphi(x)] \\ &= (-1)^s \text{Tr} [e^{-\beta H} \varphi(y^0; \mathbf{y}) \varphi(0, \mathbf{x})] \\ &= (-1)^s \text{Tr} [\varphi(0, \mathbf{x}) e^{-\beta H} \varphi(y^0; \mathbf{y})] \\ &= (-1)^s \text{Tr} \left[e^{-\beta H} \underbrace{e^{\beta H} \varphi(0, \mathbf{x}) e^{-\beta H}}_{\varphi(-i\beta, \mathbf{x})} \varphi(y^0; \mathbf{y}) \right] \\ &= (-1)^s \text{Tr} [e^{-\beta H} \varphi(-i\beta, \mathbf{x}) \varphi(y^0; \mathbf{y})] \\ &= (-1)^s (\text{Tr } e^{-\beta H}) G_2^\beta(x-y)|_{x^0=-i\beta}, \end{aligned} \quad (9.6)$$

where in the third line we used the cyclicity of the trace, and in the fourth line we used the Heisenberg equation of motion to identify $e^{\beta H} \varphi(0, \mathbf{x}) e^{-\beta H} = \varphi(-i\beta, \mathbf{x})$. Dividing through by $\text{Tr } e^{-\beta H}$,

$$\boxed{G_2^\beta(x-y)|_{x^0=0} = (-1)^s G_2^\beta(x-y)|_{x^0=-i\beta}.} \quad (9.7)$$

This is the Kubo-Martin-Schwinger (KMS) condition: bosonic Green's functions are periodic in Euclidean time with period β , while fermionic Green's functions are antiperiodic.

9.3 Matsubara Frequencies and Feynman Rules

The periodicity in Euclidean time has a direct consequence for the momentum-space structure of the theory. Vertex factors are unaffected by the change in boundary conditions, and the momentum-space propagator remains of the same form. However, the periodicity forces the energy component of the loop momentum to be discretized. The allowed frequencies, known as Matsubara frequencies, are

$$\omega_n = \frac{2n\pi}{\beta} \quad (\text{Bose fields}), \quad (9.8)$$

$$\omega_n = \frac{(2n+1)\pi}{\beta} \quad (\text{Fermi fields}), \quad (9.9)$$

for integer n . The Feynman rules are modified accordingly. The correspondence between zero-temperature and finite-temperature rules is summarized in the following table.

	Zero temperature	Finite temperature
Loop integral	$\int \frac{d^4 k}{(2\pi)^4}$	$\frac{1}{-i\beta} \sum_{n=-\infty}^{\infty} \int \frac{d^3 k}{(2\pi)^3}$
Vertex δ -function	$(2\pi)^4 \delta^{(4)}(\sum k_i)$	$\frac{\beta}{i} (2\pi)^3 \delta_{\sum \omega_n} \delta^{(3)}\left(\sum \mathbf{k}_i\right)$

The replacement of the continuous energy integral by a discrete sum over Matsubara frequencies is the defining feature of the imaginary-time formalism. In the next section we use these rules to compute the finite-temperature corrections to the effective potential.

Chapter 10

Finite-Temperature Corrections to the Effective Potential

10.1 Corrections to Scalar Field Theory

In the zero-temperature case, the one-loop contribution to the effective potential came from the sum

$$i \sum_{n=1}^{\infty} \int \frac{d^4 k}{(2\pi)^4} \frac{1}{2n} \left[\frac{\frac{1}{2} \lambda \bar{\varphi}^2}{k^2 + i\varepsilon} \right]^n = -\frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \ln \left[1 - \frac{\frac{1}{2} \lambda \bar{\varphi}^2}{k^2 + i\varepsilon} \right]. \quad (10.1)$$

At finite temperature, the replacement of the continuous energy integral by a discrete Matsubara sum gives

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = U(\bar{\varphi}) - \frac{i}{2} \cdot \frac{1}{-i\beta} \sum_{n=-\infty}^{\infty} \int \frac{d^3 k}{(2\pi)^3} \ln \left[1 - \frac{\lambda \bar{\varphi}^2}{2(\omega_n^2 - k^2)} \right], \quad (10.2)$$

$$= U(\bar{\varphi}) + \frac{1}{2\beta} \sum_{n=-\infty}^{\infty} \int \frac{d^3 k}{(2\pi)^3} \ln \left[1 - \frac{\lambda \bar{\varphi}^2}{2(\omega_n^2 - k^2)} \right]. \quad (10.3)$$

For Bose fields, the Matsubara frequencies satisfy $-i\beta\omega_n = 2\pi n$, giving $\omega_n = 2\pi i n/\beta$ and $\omega_n^2 = -4\pi^2 n^2/\beta^2$. Substituting,

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = U(\bar{\varphi}) + \frac{1}{2\beta} \sum_{n=-\infty}^{\infty} \int \frac{d^3 k}{(2\pi)^3} \ln \left[1 + \frac{\lambda \bar{\varphi}^2}{2 \left(\frac{4\pi^2}{\beta^2} n^2 + k^2 \right)} \right]. \quad (10.4)$$

Writing the argument of the logarithm as a ratio,

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = U(\bar{\varphi}) + \frac{1}{2\beta} \sum_{n=-\infty}^{\infty} \int \frac{d^3 k}{(2\pi)^3} \times \left(\ln \left[\frac{4\pi^2}{\beta^2} n^2 + k^2 + \frac{\lambda \bar{\varphi}^2}{2} \right] - \ln \left[\frac{4\pi^2}{\beta^2} n^2 + k^2 \right] \right), \quad (10.5)$$

where the second term contributes only a field-independent constant and is dropped. Denoting $E_k^2 = k^2 + \frac{\lambda \bar{\varphi}^2}{2}$,

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = U(\bar{\varphi}) + \frac{1}{2\beta} \int \frac{d^3 k}{(2\pi)^3} \sum_{n=-\infty}^{\infty} \ln \left[\frac{4\pi^2}{\beta^2} n^2 + E_k^2 \right] + \text{const.} \quad (10.6)$$

Evaluation of the Matsubara Sum

We evaluate the sum

$$X = \sum_{n=-\infty}^{\infty} \ln \left[\frac{4\pi^2}{\beta^2} n^2 + E_k^2 \right] \quad (10.7)$$

by differentiating with respect to E_k . Setting $a^2 = 4\pi^2/\beta^2$ and $y = \beta E_k/2\pi$,

$$\begin{aligned} \frac{dX}{dE_k} &= \sum_{n=-\infty}^{\infty} \frac{2E_k}{a^2 n^2 + E_k^2} \\ &= \frac{2}{E_k} + \frac{4}{a} \sum_{n=1}^{\infty} \frac{y}{y^2 + n^2}, \end{aligned} \quad (10.8)$$

where we separated the $n = 0$ term. Using the identity

$$\sum_{n=1}^{\infty} \frac{y}{y^2 + n^2} = -\frac{1}{2y} + \frac{\pi}{2} \coth \pi y = -\frac{1}{2y} + \frac{\pi}{2} + \frac{\pi e^{-2\pi y}}{1 - e^{-2\pi y}}, \quad (10.9)$$

$$\begin{aligned} \frac{dX}{dE_k} &= \frac{2}{E_k} + \frac{2\beta}{\pi} \left(-\frac{1}{2y} + \frac{\pi}{2} + \frac{\pi e^{-2\pi y}}{1 - e^{-2\pi y}} \right) \\ &= \frac{2}{E_k} - \frac{\beta}{\pi y} + \beta + \frac{2\beta e^{-2\pi y}}{1 - e^{-2\pi y}} \\ &= \beta + \frac{2\beta e^{-\beta E_k}}{1 - e^{-\beta E_k}}, \end{aligned} \quad (10.10)$$

where in the last step we used $y = \beta E_k/2\pi$ so that $\beta/(\pi y) = 2/E_k$, cancelling the first term. Integrating back,

$$X = \int \left(\beta + \frac{2\beta e^{-\beta E_k}}{1 - e^{-\beta E_k}} \right) dE_k = \beta E_k + 2 \ln (1 - e^{-\beta E_k}). \quad (10.11)$$

Substituting back,

$$\begin{aligned} V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= U(\bar{\varphi}) + \frac{1}{2\beta} \int \frac{d^3 k}{(2\pi)^3} (\beta E_k + 2 \ln (1 - e^{-\beta E_k})) \\ &= U(\bar{\varphi}) + \int \frac{d^3 k}{(2\pi)^3} \left[\frac{E_k}{2} + \frac{1}{\beta} \ln (1 - e^{-\beta E_k}) \right]. \end{aligned} \quad (10.12)$$

The first term inside the integral is the zero-point energy. Using the identity

$$i \int_{-\infty}^{\infty} \frac{dx}{2\pi} \ln(-x^2 + y^2 - i\varepsilon) = y + \text{const.}, \quad (10.13)$$

and rotating to Euclidean space via $k^0 \rightarrow ik_E^0$, $-k_0^2 + E_k^2 = k_E^2 + \frac{\lambda \bar{\varphi}^2}{2}$, the zero-temperature part is recovered and we can write

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = \underbrace{U(\bar{\varphi}) + \frac{1}{2} \int \frac{d^4 k_E}{(2\pi)^4} \ln \left[1 + \frac{\lambda \bar{\varphi}^2}{2k_E^2} \right]}_{\text{zero-temperature result}} + \underbrace{\frac{1}{2\pi^2 \beta} \int_0^{\infty} dk k^2 \ln(1 - e^{-\beta E_k})}_{\text{finite-temperature contribution}}. \quad (10.14)$$

Substituting $x = k\beta$, $dk = dx/\beta$, $E_k = (k^2 + \lambda\bar{\varphi}^2/2)^{1/2}$,

$$\Delta V_{\text{eff}}^{(1)}(\bar{\varphi}) = \frac{1}{2\pi^2\beta^4} \int_0^\infty dx x^2 \ln \left[1 - \exp \left[- \left(x^2 + \frac{\lambda\bar{\varphi}^2\beta^2}{2} \right)^{1/2} \right] \right]. \quad (10.15)$$

Defining $y^2 = \lambda\bar{\varphi}^2\beta^2/2$, the finite-temperature correction can be written compactly as

$$V_{\text{eff}}^{(1)}(\bar{\varphi}) = V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) + \frac{1}{2\pi^2\beta^4} I(y), \quad (10.16)$$

where

$$I(y) = \int_0^\infty dx x^2 \ln \left[1 - e^{-(x^2+y^2)^{1/2}} \right]. \quad (10.17)$$

The zero-temperature renormalization counterterms also render this finite-temperature contribution finite, so no additional renormalization is needed.

High-Temperature Expansion

In the high-temperature limit $\bar{\varphi}\beta \ll 1$, we expand $I(y)$ in powers of y^2 . Setting $f(y) = \ln(1 - e^{-(x^2+y^2)^{1/2}})$,

$$\begin{aligned} f(y) &= f(0) + y^2 f'(0) + \mathcal{O}(y^4) \\ &= \ln(1 - e^{-x}) + y^2 \left[\frac{e^{-(x^2+y^2)^{1/2}}}{1 - e^{-(x^2+y^2)^{1/2}}} \cdot \frac{1}{2(x^2+y^2)^{1/2}} \right]_{y=0} + \mathcal{O}(y^4) \\ &= \ln(1 - e^{-x}) + \frac{y^2}{2x} \cdot \frac{e^{-x}}{1 - e^{-x}} + \mathcal{O}(y^4). \end{aligned} \quad (10.18)$$

Substituting into the integral,

$$I(y) = \int_0^\infty dx x^2 \ln(1 - e^{-x}) + \frac{y^2}{2} \int_0^\infty dx \frac{x e^{-x}}{1 - e^{-x}} + \mathcal{O}(y^4). \quad (10.19)$$

For the first term, using $\ln(1 - e^{-x}) = -\sum_{n=1}^\infty e^{-nx}/n$,

$$\int_0^\infty dx x^2 \ln(1 - e^{-x}) = -\sum_{n=1}^\infty \frac{1}{n} \int_0^\infty dx x^2 e^{-nx}. \quad (10.20)$$

Substituting $y = nx$,

$$\int_0^\infty dx x^2 e^{-nx} = \frac{1}{n^3} \int_0^\infty dy y^2 e^{-y} = \frac{\Gamma(3)}{n^3} = \frac{2}{n^3}, \quad (10.21)$$

so the first term gives

$$-\sum_{n=1}^\infty \frac{1}{n} \cdot \frac{2}{n^3} = -2 \sum_{n=1}^\infty \frac{1}{n^4} = -2\zeta(4) = -2 \cdot \frac{\pi^4}{90} = -\frac{\pi^4}{45}. \quad (10.22)$$

For the second term, using $e^{-x}/(1 - e^{-x}) = \sum_{n=0}^\infty e^{-(n+1)x}$,

$$\int_0^\infty dx \frac{x e^{-x}}{1 - e^{-x}} = \sum_{n=1}^\infty \int_0^\infty dx x e^{-nx}. \quad (10.23)$$

Substituting $y = nx$,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \int_0^{\infty} dy y e^{-y} = \sum_{n=1}^{\infty} \frac{\Gamma(2)}{n^2} = \zeta(2) = \frac{\pi^2}{6}. \quad (10.24)$$

Combining,

$$I(y) = -\frac{\pi^4}{45} + \frac{\pi^2 y^2}{12} + \mathcal{O}(y^4). \quad (10.25)$$

Substituting back with $y^2 = \lambda \bar{\varphi}^2 \beta^2 / 2$,

$$\begin{aligned} V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) + \frac{1}{2\pi^2 \beta^4} \left(-\frac{\pi^4}{45} + \frac{\pi^2}{12} \cdot \frac{\lambda \bar{\varphi}^2 \beta^2}{2} + \mathcal{O}(\bar{\varphi}^4) \right) \\ &= \left[V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) - \frac{\pi^2}{90\beta^4} + \frac{\lambda \bar{\varphi}^2}{24\beta^2} + \mathcal{O}(\bar{\varphi}^4) \right]. \end{aligned} \quad (10.26)$$

The first correction $-\pi^2/(90\beta^4) = -\pi^2 T^4/90$ is the familiar Stefan-Boltzmann term, the free energy density of a massless scalar gas. The second correction $\lambda \bar{\varphi}^2/(24\beta^2) = \lambda T^2 \bar{\varphi}^2/24$ is a temperature-dependent mass term, positive and proportional to T^2 . This term is what drives symmetry restoration at high temperature: it adds a positive contribution to the effective mass of the field, pushing the minimum of the effective potential back toward $\bar{\varphi} = 0$ as T increases.

10.2 Corrections to the GSW Model

The finite-temperature correction to the effective potential in the GSW model receives contributions from the two W bosons and the Z boson, each with three polarizations. Using the field-dependent masses $m(W_{1,2}) = \frac{1}{2}g\bar{\varphi}$ and $m(Z) = \frac{1}{2}(g^2 + g'^2)^{1/2}\bar{\varphi}$, the correction is

$$\Delta V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = \frac{3}{2\pi^2 \beta^4} \left\{ 2I\left(\frac{1}{2}g\bar{\varphi}\beta\right) + I\left[\frac{1}{2}(g^2 + g'^2)^{1/2}\bar{\varphi}\beta\right] \right\}. \quad (10.27)$$

Expanding each term using $I(y) = -\pi^4/45 + \pi^2 y^2/12 + \mathcal{O}(y^4)$, the W boson contribution gives

$$\begin{aligned} 2I\left(\frac{1}{2}g\bar{\varphi}\beta\right) &= 2 \left[-\frac{\pi^4}{45} + \frac{\pi^2}{12} \cdot \frac{1}{4} g^2 \bar{\varphi}^2 \beta^2 + \mathcal{O}(\bar{\varphi}^4) \right] \\ &= -\frac{2\pi^4}{45} + \frac{\pi^2}{24} g^2 \beta^2 \bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4), \end{aligned} \quad (10.28)$$

and the Z boson contribution gives

$$\begin{aligned} I\left[\frac{1}{2}(g^2 + g'^2)^{1/2}\bar{\varphi}\beta\right] &= -\frac{\pi^4}{45} + \frac{\pi^2}{12} \cdot \frac{1}{4} (g^2 + g'^2) \bar{\varphi}^2 \beta^2 + \mathcal{O}(\bar{\varphi}^4) \\ &= -\frac{\pi^4}{45} + \frac{\pi^2}{48} (g^2 + g'^2) \bar{\varphi}^2 \beta^2 + \mathcal{O}(\bar{\varphi}^4). \end{aligned} \quad (10.29)$$

Summing these,

$$\begin{aligned}
\Delta V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= \frac{3}{2\pi^2\beta^4} \left[-\frac{2\pi^4}{45} + \frac{\pi^2}{24}g^2\beta^2\bar{\varphi}^2 - \frac{\pi^4}{45} + \frac{\pi^2}{48}(g^2 + g'^2)\bar{\varphi}^2\beta^2 + \mathcal{O}(\bar{\varphi}^4) \right] \\
&= \frac{3}{2\pi^2\beta^4} \left[-\frac{3\pi^4}{45} + \frac{\pi^2\beta^2\bar{\varphi}^2}{24} \left(g^2 + \frac{g^2}{2} + \frac{g'^2}{2} \right) + \mathcal{O}(\bar{\varphi}^4) \right] \\
&= \frac{3}{2\pi^2\beta^4} \left[-\frac{\pi^4}{15} + \frac{\pi^2}{48}\beta^2(3g^2 + g'^2)\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4) \right] \\
&= -\frac{\pi^2}{10\beta^4} + \frac{1}{32\beta^2}(3g^2 + g'^2)\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4).
\end{aligned} \tag{10.30}$$

The full finite-temperature effective potential in the GSW model is therefore

$$\boxed{V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) - \frac{\pi^2}{10\beta^4} + \frac{1}{32\beta^2}(3g^2 + g'^2)\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4).} \tag{10.31}$$

As in the scalar case, the $-\pi^2/(10\beta^4)$ term is the Stefan-Boltzmann free energy of the gauge bosons, and the $(3g^2 + g'^2)\bar{\varphi}^2/(32\beta^2)$ term is a positive temperature-dependent mass correction that drives symmetry restoration at high temperature.

10.3 Corrections to the Minimal SU(5) Model

For the minimal SU(5) GUT, the gauge bosons acquire a field-dependent mass $m = \sqrt{25/8}g\bar{\varphi}$ from the Higgs mechanism. The finite-temperature correction to the effective potential is

$$\begin{aligned}
V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) + \frac{18}{\pi^2\beta^4} I\left(\sqrt{\frac{25}{8}}g\bar{\varphi}\beta\right) \\
&= V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) + \frac{18}{\pi^2\beta^4} \left[-\frac{\pi^4}{45} + \frac{\pi^2}{12} \cdot \frac{25}{8} g^2\bar{\varphi}^2\beta^2 + \mathcal{O}(\bar{\varphi}^4) \right] \\
&= V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) - \frac{2\pi^2}{5\beta^4} + \frac{18}{\pi^2\beta^4} \cdot \frac{25}{8} g^2\bar{\varphi}^2\beta^2 \cdot \frac{\pi^2}{12} + \mathcal{O}(\bar{\varphi}^4),
\end{aligned} \tag{10.32}$$

where the factor of 18 counts the degrees of freedom of the heavy gauge bosons in the SU(5) model. Collecting terms,

$$\boxed{V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = V_{\text{eff}}^{(1)T=0}(\bar{\varphi}) - \frac{2\pi^2}{5\beta^4} + \frac{75}{16\beta^2} g^2\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4).} \tag{10.33}$$

The structure is the same as in the scalar and GSW cases: a Stefan-Boltzmann term $-2\pi^2/(5\beta^4)$ from the free energy of the gauge bosons, and a positive temperature-dependent mass term $75g^2\bar{\varphi}^2/(16\beta^2)$ that drives symmetry restoration. The larger coefficient relative to the GSW case reflects the larger gauge group and the correspondingly greater number of degrees of freedom.

Part III

Cosmological Phase Transitions

Chapter 11

A Brief History of the Early Universe and Cosmological Phase Transitions

11.1 Thermal History of the Early Universe

The universe we observe today emerged from an early hot dense state that has been expanding and cooling ever since. Working backwards in time, the thermal history is punctuated by a sequence of phase transitions, each corresponding to the breaking of a symmetry as the temperature fell below a critical value.

The earliest epoch accessible to our theoretical framework is inflation, a period of accelerated expansion during which the universe expanded by a factor of order 10^{26} over a time of order 10^{-36} to 10^{-32} seconds. At the end of inflation began the reheating phase, during which the universe returned to a thermal state at the reheating temperature T_R through the coupling of the inflaton field ϕ to the Standard Model particles.

Subsequent to reheating, the thermal history proceeds through a series of symmetry breaking transitions. At $T \sim 10^{29} \text{ K} \sim 10^{16} \text{ GeV}$, there may have been a GUT phase transition,

$$\text{GUT} \longrightarrow \text{SU}(2)_L \times \text{U}(1), \quad (11.1)$$

though whether this transition occurred and at what scale depends on the specific GUT model. At $T \sim 10^{15} \text{ K} \sim 100 \text{ GeV}$, the electroweak symmetry was broken,

$$\text{SU}(2)_L \times \text{U}(1) \longrightarrow \text{U}_{\text{em}}(1), \quad (11.2)$$

and at $T \sim 2 \cdot 10^{12} \text{ K} \sim 200 \text{ MeV}$ the QCD confinement transition took place. Each of these transitions is described by a scalar field acting as an order parameter that acquires a vacuum expectation value as the temperature falls through T_c .

11.2 Phase Transitions and the Effective Potential

The nature of a cosmological phase transition is determined by the structure of the finite-temperature effective potential $V_{\text{eff}}(\bar{\varphi}, T)$. We illustrate this with the electroweak phase transition, where the order parameter is the Higgs field h . The tree-level Higgs potential at $T = 0$ is

$$V(h) = -m^2 h^2 + \lambda h^4, \quad m^2 > 0, \quad \lambda > 0, \quad m^2 \sim (100 \text{ GeV})^2. \quad (11.3)$$

At $T \neq 0$, finite-temperature corrections generate a temperature-dependent mass term,

$$\Delta V(h, T) = cT^2 h^2 + \cdots, \quad c > 0, \quad (11.4)$$

so that

$$V(h, T) = V(h) + \Delta V(h, T). \quad (11.5)$$

The positive $T^2 h^2$ term competes with the negative $-m^2 h^2$ term and tends to push the minimum back toward $h = 0$ at high temperature, restoring the symmetry.

The character of the transition depends on how the minimum of $V(h, T)$ evolves with temperature. If the minimum moves smoothly and continuously from $h \neq 0$ at low T to $h = 0$ at high T , the transition is *second order* (or a crossover).

If instead a potential barrier develops between the two minima at some temperature, so that the two phases coexist and the field must tunnel through the barrier to reach the new minimum, the transition is *first order*.

The potential barrier in a first-order transition can arise from higher-order terms in the potential. For example, a cubic term $\sim h^3$ generates a barrier, though for the Standard Model Higgs potential alone this is not sufficient to produce a first-order transition. There is also the possibility of *symmetry non-restoration*, which occurs when $c < 0$, so that the finite-temperature correction deepens the broken-symmetry minimum rather than restoring it. This situation has attracted considerable attention in recent years.

11.3 Bubble Nucleation

In a first-order phase transition, the universe finds itself temporarily trapped in the false vacuum, the metastable minimum of $V_{\text{eff}}(\bar{\varphi}, T)$. The field cannot roll classically to the true vacuum because of the potential barrier separating the two phases. Instead, the transition proceeds through the nucleation of bubbles of the true vacuum within the surrounding false vacuum. Initially, regions of space where the field has already tunnelled to the true vacuum appear as isolated bubbles,

which then expand, driven by the pressure difference between the two phases. New bubbles nucleate, and eventually the expanding bubbles overlap and collide, filling all of space with the true vacuum and completing the transition.

This process has two important phenomenological consequences that have attracted significant attention. The first is electroweak baryogenesis, in which the out-of-equilibrium dynamics at the bubble wall provides the conditions necessary to generate the observed baryon asymmetry of the universe. The second is the production of a stochastic gravitational wave background from bubble collisions and their interaction with the primordial plasma. The detailed analysis of both of these consequences requires a quantitative understanding of the bubble nucleation rate and bubble wall dynamics, which we develop in Part ??.

Chapter 12

The Effective Potential in Cosmology

In Part ?? we developed the effective action formalism and computed the effective potential at zero temperature, using a momentum cutoff to regulate the divergent integrals. Here we revisit this computation in the cosmological context, with two differences. First, we use dimensional regularization and the $\overline{\text{MS}}$ renormalization scheme, which is more systematic and avoids the ambiguities associated with a hard cutoff. Second, we keep the physical application front and center: the question is not just what the quantum-corrected vacuum is, but how it changes as the universe cools, and what determines whether the resulting phase transition is first or second order.

12.1 The Effective Action and Effective Potential

We work with a real scalar field ϕ in ϕ^4 theory,

$$S[\phi] = \int d^4x \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V_0(\phi) \right], \quad V_0(\phi) = \frac{m^2}{2} \phi^2 + \frac{\lambda}{4!} \phi^4, \quad (12.1)$$

where the $\phi \rightarrow -\phi$ symmetry forbids a cubic term. The basic objects of interest are the correlators

$$\langle \Omega | T \{ \phi(x_1) \cdots \phi(x_n) \} | \Omega \rangle = \frac{\int \mathcal{D}\phi \, \phi(x_1) \cdots \phi(x_n) e^{iS[\phi]}}{\int \mathcal{D}\phi \, e^{iS[\phi]}}, \quad (12.2)$$

where $|\Omega\rangle$ is the vacuum of the interacting theory. Our goal is to find a potential for the expectation value $\langle \Omega | \phi(x) | \Omega \rangle$.

Introducing a source $J(x)$, the generating functional is

$$Z[J] = \int \mathcal{D}\phi \, e^{iS[\phi] + i \int d^4x J(x) \phi(x)}, \quad (12.3)$$

from which all correlators are obtained by functional differentiation. The connected generating functional $W[J] = -i \ln Z[J]$ generates the connected Green's functions. Its first derivative defines the classical field,

$$\frac{\delta W[J]}{\delta J(x)} = \langle \Omega | \phi(x) | \Omega \rangle_J \equiv \bar{\phi}(x). \quad (12.4)$$

The effective action is defined by the Legendre transform,

$$\Gamma[\bar{\phi}] \equiv W[J] - \int d^4x J(x) \bar{\phi}(x), \quad (12.5)$$

and satisfies $\delta\Gamma/\delta\bar{\phi}(y) = -J(y)$. The vacuum of the theory in the absence of sources is therefore determined by

$$\frac{\delta\Gamma[\bar{\phi}]}{\delta\bar{\phi}} = 0. \quad (12.6)$$

For a spatially uniform, time-independent configuration $\bar{\phi}(x) = \bar{\phi}_c$, the effective action reduces to

$$\Gamma[\bar{\phi}_c] = - \int d^4x V_{\text{eff}}(\bar{\phi}_c), \quad (12.7)$$

where $V_{\text{eff}}(\bar{\phi}_c)$ is the effective potential, the quantum-corrected potential whose minimum determines the true ground state of the theory. Without proof, $\Gamma[\bar{\phi}]$ is the generating functional of 1PI correlators,

$$V_{\text{eff}}(\bar{\phi}_c) = - \sum_{n=0}^{\infty} \frac{1}{n!} \bar{\phi}_c^n \Gamma^{(n)}(p_i = 0). \quad (12.8)$$

12.2 The Coleman-Weinberg Potential via Dimensional Regularization

The one-loop correction to the effective potential is

$$V_1(\phi) = \frac{1}{2} \int \frac{d^4p_E}{(2\pi)^4} \ln[p_E^2 + \tilde{m}^2(\phi)], \quad (12.9)$$

where

$$\tilde{m}^2(\phi) = \frac{d^2V_0(\phi)}{d\phi^2} = m^2 + \frac{\lambda}{2}\phi^2 \quad (12.10)$$

is the field-dependent mass. This integral is UV divergent and must be regulated. We use dimensional regularization, replacing $d = 4$ by $d = 4 - 2\varepsilon$ and introducing a mass scale μ to keep the action dimensionless,

$$V_1(\phi) = \frac{1}{2} \mu^{4-d} \int \frac{d^dk_E}{(2\pi)^d} \ln(p_E^2 + \tilde{m}^2(\phi)). \quad (12.11)$$

To evaluate this, we differentiate with respect to $\tilde{m}^2(\phi)$ and use the master formula of dimensional regularization,

$$\int \frac{d^dp_E}{(2\pi)^d} \frac{1}{p_E^2 + m^2} = \frac{1}{(4\pi)^{d/2}} (m^2)^{d/2-1} \Gamma\left(1 - \frac{d}{2}\right), \quad (12.12)$$

to obtain

$$\frac{dV_1}{d\tilde{m}^2} = \frac{1}{2} \mu^{4-d} \cdot \frac{1}{(4\pi)^{d/2}} (\tilde{m}^2)^{d/2-1} \Gamma\left(1 - \frac{d}{2}\right). \quad (12.13)$$

Integrating back and substituting $d = 4 - 2\varepsilon$,

$$V_1(\phi) = \frac{1}{64\pi^2} \tilde{m}^4(\phi) \left[-\frac{1}{\varepsilon} + \gamma_E - \ln 4\pi + \ln \frac{\tilde{m}^2(\phi)}{\mu^2} - \frac{3}{2} \right]. \quad (12.14)$$

In the $\overline{\text{MS}}$ scheme, we absorb the divergent combination $\mathcal{C}_{\text{uv}} = 1/\varepsilon - \gamma_E + \ln 4\pi$ into counterterms, leaving the renormalized one-loop potential

$$V_1(\phi) = \frac{1}{64\pi^2} \tilde{m}^4(\phi) \left[\ln \frac{\tilde{m}^2(\phi)}{\mu^2} - \frac{3}{2} \right]. \quad (12.15)$$

This result generalizes immediately to other fields. For a U(1) gauge field with field-dependent mass $\tilde{m}_A^2(\phi) = 2g^2|\phi|^2$ arising from spontaneous symmetry breaking,

$$V_1^{(A)}(\phi) = \frac{3}{64\pi^2} \tilde{m}_A^4(\phi) \left[\ln \frac{\tilde{m}_A^2(\phi)}{\mu^2} - \frac{5}{6} \right], \quad (12.16)$$

where the factor of 3 counts the physical polarizations and the $-5/6$ (rather than $-3/2$) arises from the vector nature of the field, computed in Landau gauge. For Dirac fermions with Yukawa coupling y and field-dependent mass $m_\psi^2 = y^2\phi^2$,

$$V_1^{(\psi)}(\phi) = -\frac{4}{64\pi^2} \tilde{m}_\psi^4(\phi) \left[\ln \frac{\tilde{m}_\psi^2(\phi)}{\mu^2} - \frac{3}{2} \right], \quad (12.17)$$

where the overall minus sign comes from the fermion loop and the factor of 4 counts the spinor degrees of freedom. The general formula, summing over all particles coupling to ϕ , is

$$V_{\text{eff}}(\phi) = V_0(\phi) + \frac{1}{64\pi^2} \sum_i n_i \tilde{m}_i^4(\phi) \left[\ln \frac{\tilde{m}_i^2(\phi)}{\mu^2} - c_i \right], \quad (12.18)$$

where $n_i = (-1)^F \cdot \text{d.o.f.}$ counts degrees of freedom with a minus sign for fermions, and

$$c_i = \begin{cases} \frac{3}{2} & \text{scalars and fermions,} \\ \frac{5}{6} & \text{gauge bosons.} \end{cases} \quad (12.19)$$

This is the Coleman-Weinberg potential in the $\overline{\text{MS}}$ scheme. It supersedes the cutoff-regulated result of Chapter 4 and will be our working formula for the remainder of this part.

12.3 The Coleman-Weinberg Model

We now apply (12.18) to the Coleman-Weinberg model of massless scalar electrodynamics,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + |D_\mu\phi|^2 - \frac{\lambda}{4!}|\phi|^4, \quad D_\mu = \partial_\mu + igA_\mu. \quad (12.20)$$

There is no mass term in the Lagrangian, and the classical action is scale invariant: under $x \rightarrow \lambda x$, $A_\mu \rightarrow \lambda^{-1}A_\mu$, $\phi \rightarrow \lambda^{-1}\phi$, each term in $\mathcal{L}$ scales as λ^{-4} , leaving $S = \int d^4x \mathcal{L}$ invariant.

At the quantum level, scale invariance is broken by the logarithms in (12.18). Writing $\phi = \phi_c + \chi_1 + i\chi_2$ and expanding around the background ϕ_c , the field-dependent masses are

$$\tilde{m}_{\chi_1}^2(\phi_c) = \frac{\lambda}{2}\phi_c^2, \quad (12.21)$$

$$\tilde{m}_{\chi_2}^2(\phi_c) = \frac{\lambda}{6}\phi_c^2, \quad (12.22)$$

$$m_A^2(\phi_c) = 2g^2\phi_c^2. \quad (12.23)$$

In the regime $\lambda^2 \ll g^4$, the scalar loop contributions are negligible and the effective potential is dominated by the gauge boson loop,

$$V_{\text{eff}}(\phi_c) \cong \frac{\lambda}{4!} \phi_c^4 + \frac{3}{16\pi^2} g^4 \phi_c^4 \left[\ln \frac{2g^2 \phi_c^2}{\mu^2} - \frac{5}{6} \right]. \quad (12.24)$$

Redefining the renormalization scale as $\tilde{\mu}^2 = e^{5/6} \mu^2 / (2g^2)$, this simplifies to

$$V_{\text{eff}}(\phi_c) = \frac{\lambda}{4!} \phi_c^4 + \frac{3g^4}{16\pi^2} \phi_c^4 \ln \frac{\phi_c^2}{\tilde{\mu}^2}. \quad (12.25)$$

Defining an effective quartic coupling

$$\lambda_{\text{eff}}(\phi_c) \equiv \lambda + \frac{g^4}{2\pi^2} \ln \frac{\phi_c^2}{\tilde{\mu}^2}, \quad (12.26)$$

the effective potential takes the suggestive form

$$V_{\text{eff}}(\phi_c) = \frac{\lambda_{\text{eff}}(\phi_c)}{4!} \phi_c^4. \quad (12.27)$$

This is precisely analogous to the running coupling in scattering experiments, where the appropriate scale at which to evaluate the coupling is the energy scale of the process. Here the relevant scale is the field value ϕ_c itself. At $\phi_c = \tilde{\mu}$, the logarithm vanishes and $\lambda_{\text{eff}}(\tilde{\mu}) = \lambda$, so $\tilde{\mu}$ plays the role of an RG scale. Quantum corrections have therefore broken the classical scale invariance and generated a preferred scale, producing a nontrivial minimum of the effective potential.

This is the phenomenon of dimensional transmutation already encountered in Chapter 5: a theory with no classical mass scale develops a preferred scale through radiative corrections. The scale $\tilde{\mu}$ at which λ_{eff} is evaluated plays the same role as the renormalization point M in the cutoff treatment of Chapter 4, and the two approaches give the same physical result in different renormalization schemes.

For the Standard Model Higgs, an analogous but more involved running occurs. The tree-level wine-bottle potential has a minimum at ~ 100 GeV with positive λ . However, running λ to higher scales using the Standard Model renormalization group equations, the coupling turns negative at energy scales of order 10^{10} to 10^{12} GeV, implying that the electroweak vacuum is metastable rather than absolutely stable.

The lifetime of this metastable vacuum is many orders of magnitude larger than the age of the universe, so this poses no immediate phenomenological problem. Nevertheless it raises the deep question of why the measured values of the top quark mass and Higgs mass place us precisely in the metastable region of parameter space. For a detailed analysis, see Buttazzo et al., arXiv:1307.3536.

Chapter 13

Finite-Temperature Effective Potential in Cosmology

We now extend the analysis of the previous chapter to finite temperature. The imaginary-time formalism developed in Chapter 9 provides the framework: at temperature $T = 1/\beta$, the partition function is a path integral over fields periodic in Euclidean time with period β , and the loop integrals are replaced by Matsubara sums. Here we apply this machinery to derive the finite-temperature effective potential in the cosmological context, and use it to analyze the temperature dependence of the phase structure.

13.1 The Imaginary-Time Formalism in Brief

The canonical ensemble in quantum mechanics is described by the density matrix $\rho = Z^{-1}e^{-\beta H}$, with partition function $Z = \text{tr}(e^{-\beta H})$ and ensemble averages

$$\langle \mathcal{O} \rangle = \text{tr}(\rho \mathcal{O}) = \frac{1}{Z} \sum_n e^{-\beta E_n} \langle \psi_n | \mathcal{O} | \psi_n \rangle. \quad (13.1)$$

In quantum field theory the trace becomes a path integral over field configurations,

$$Z = \text{tr}(e^{-\beta H}) = \int \mathcal{D}\phi \langle \phi | e^{-\beta H} | \phi \rangle. \quad (13.2)$$

To cast this in path integral form, we use Feynman's formula for the transition amplitude,

$$\langle \phi_b | e^{-iH(t_b - t_a)} | \phi_a \rangle = \int_{\phi(t_a) = \phi_a}^{\phi(t_b) = \phi_b} \mathcal{D}\phi e^{i \int_{t_a}^{t_b} dt \int d^3x \mathcal{L}}, \quad (13.3)$$

and analytically continue to imaginary time $\tau = it$. Setting $t_a = 0$ and $t_b = -i\beta$, identifying $\phi_a = \phi_b$ and integrating over field configurations, the partition function becomes

$$Z = \int_{\phi(0, \mathbf{x}) = \phi(\beta, \mathbf{x})} \mathcal{D}\phi e^{-\int_0^\beta d\tau \int d^3x \mathcal{L}_E}, \quad (13.4)$$

where the Euclidean Lagrangian is

$$\mathcal{L}_E = (\partial_\tau \phi)^2 + (\partial_i \phi)^2 + V(\phi). \quad (13.5)$$

The fields satisfy periodic boundary conditions in Euclidean time, $\phi(0, \mathbf{x}) = \phi(\beta, \mathbf{x})$ for bosons and antiperiodic for fermions. The generating functional at finite temperature is

$$Z^\beta[J] = \int \mathcal{D}\phi \, e^{-\int_0^\beta d\tau \int d^3x [\mathcal{L}_E - J\phi]}, \quad (13.6)$$

and the chain from $Z^\beta[J]$ to the finite-temperature effective potential $V_{\text{eff}}^\beta(\bar{\phi}_c)$ follows exactly the same Legendre transform logic as at zero temperature,

$$Z^\beta[J] \xrightarrow{\text{Legendre}} \Gamma^\beta[\bar{\phi}] \xrightarrow{\bar{\phi}=\bar{\phi}_c} V_{\text{eff}}^\beta(\bar{\phi}_c). \quad (13.7)$$

The key difference from zero temperature lies in the Feynman rules. The propagator becomes $-i/(\omega_k^2 + \mathbf{p}^2 + m^2)$ in Euclidean space, the vertex is unchanged at $-i\lambda$, and the loop integral is replaced by a Matsubara sum,

$$\int \frac{d^4p}{(2\pi)^4} \longrightarrow \frac{i}{\beta} \sum_{k=-\infty}^{\infty} \int \frac{d^3p}{(2\pi)^3}, \quad (13.8)$$

with bosonic Matsubara frequencies $\omega_k = 2\pi k\beta^{-1}$ and fermionic frequencies $\omega_k = (2k + 1)\pi\beta^{-1}$.

13.2 The Thermal Functions J_B and J_F

Following the same steps as in Chapter 9, the one-loop finite-temperature contribution to the effective potential for a scalar field with field-dependent mass $m^2(\phi)$ is

$$V_{1,\beta}(\phi) = \int \frac{d^3p}{(2\pi)^3} \left[\frac{\omega}{2} + \frac{1}{\beta} \ln(1 - e^{-\beta\omega}) \right], \quad \omega = \sqrt{\mathbf{p}^2 + m^2(\phi)}. \quad (13.9)$$

The first term is the zero-point energy, which upon Wick rotation reproduces the zero-temperature Coleman-Weinberg result. The second term is the genuinely thermal contribution. After the substitution $x = |\mathbf{p}|\beta$, it takes the form

$$V_{1,\beta}(\phi) = \frac{T^4}{2\pi^2} J_B\left(\frac{m^2(\phi)}{T^2}\right), \quad (13.10)$$

where the bosonic thermal function is defined as

$$J_B(y^2) = \int_0^\infty dx \, x^2 \ln\left(1 - e^{-\sqrt{x^2+y^2}}\right). \quad (13.11)$$

For a U(1) gauge field with field-dependent mass $m_A^2(\phi) = 2g^2|\phi|^2$, the three physical polarizations contribute

$$V_{1,\beta}^{(A_\mu)}(\phi) = \frac{3T^4}{2\pi^2} J_B\left(\frac{m_A^2(\phi)}{T^2}\right). \quad (13.12)$$

For Dirac fermions with field-dependent mass $m_f^2 = y^2\phi^2$, the antiperiodic boundary conditions give fermionic Matsubara frequencies $\omega_k = (2k + 1)\pi\beta^{-1}$, and the thermal contribution is

$$V_{1,\beta}^{(\psi)}(\phi) = -\frac{2T^4}{\pi^2} J_F\left(\frac{m_f^2(\phi)}{T^2}\right), \quad (13.13)$$

where the fermionic thermal function is

$$J_F(y^2) = \int_0^\infty dx x^2 \ln\left(1 + e^{-\sqrt{x^2+y^2}}\right), \quad (13.14)$$

with a plus sign in the exponential reflecting the antiperiodic boundary conditions. The high-temperature expansions of these functions, valid for $y^2 = m^2(\phi)/T^2 \ll 1$, are

$$J_B(y^2) \cong -\frac{\pi^4}{45} + \frac{\pi^2}{12}y^2 - \frac{\pi}{6}|y|^3 - \frac{1}{32}y^4 \ln \frac{y^2}{a_B} + \dots, \quad (13.15)$$

$$J_F(y^2) \cong \frac{7\pi^4}{360} - \frac{\pi^2}{24}y^2 - \frac{1}{32}y^4 \ln \frac{y^2}{a_F} + \dots, \quad (13.16)$$

where $a_B = \pi^2 e^{3/2-2\gamma_E}$ and $a_F = 16\pi^2 e^{3/2-2\gamma_E}$. The leading terms $\mp\pi^4/45$ and $7\pi^4/360$ are the free energy densities of massless bosons and fermions respectively, the Stefan-Boltzmann contributions. The $\pm\pi^2 y^2/12$ terms are the temperature-dependent mass corrections already computed in Chapter 9. The cubic term $-\pi|y|^3/6$ in J_B is new and is absent for fermions: it arises from the bosonic Matsubara zero mode and, as we shall see, plays a crucial role in generating a potential barrier and hence a first-order phase transition.

13.3 The Coleman-Weinberg Model at Finite Temperature

We now bring together the zero-temperature and finite-temperature contributions to obtain the full effective potential of the Coleman-Weinberg model at temperature T . The total effective potential is

$$V_{\text{eff}}(\phi, T) = V_0(\phi) + \underbrace{V_{1,\text{CW}}(\phi)}_{\lambda_{\text{eff}}(\phi)\phi^4/4!} + V_{1,\beta}(\phi), \quad (13.17)$$

where the zero-temperature part was computed in the previous chapter. In the regime $\lambda^2 \ll g^4$, the thermal contribution is dominated by the gauge boson loop,

$$V_{1,\beta}(\phi) \cong \frac{3T^4}{2\pi^2} J_B\left(\frac{2g^2\phi^2}{T^2}\right). \quad (13.18)$$

In the high-temperature approximation $T^2 \gg m_A^2(\phi) = 2g^2\phi^2$, expanding J_B ,

$$V_{1,\beta}(\phi) \cong \frac{g^2}{4\pi^2} T^2 \phi^2 + \mathcal{O}(\phi^3) + \text{const.}, \quad (13.19)$$

so that the full effective potential becomes

$$V_{\text{eff}}(\phi, T) = \frac{\lambda_{\text{eff}}(\phi)}{4!} \phi^4 + \frac{g^2}{4\pi^2} T^2 \phi^2 + \dots \quad (13.20)$$

The positive $T^2\phi^2$ term drives symmetry restoration at high temperature: it adds a positive mass-squared to the field, pushing the minimum back toward $\phi = 0$. At sufficiently high T the symmetric phase is restored. As the universe cools, the coefficient of ϕ^2 decreases and eventually the broken-symmetry minimum at $\phi \neq 0$ becomes energetically favored.

The cubic term $\sim T|\phi|^3$ from J_B is particularly significant. It generates a potential barrier between the symmetric and broken-symmetry phases that persists down to temperatures below the naive critical temperature, making the transition first order. The height and width of this barrier determine the nucleation rate of bubbles of the broken phase within the symmetric phase, which we analyze in detail in Part ??.

The temperature at which the two minima become degenerate is the critical temperature T_c , derived in Chapter 9 for the purely quartic case. Below T_c the broken-symmetry phase is the global minimum, and the universe undergoes the phase transition. The order and character of this transition, and its cosmological consequences, depend sensitively on the relative magnitudes of the gauge and scalar couplings, and on the full structure of the effective potential including the cubic term from the thermal bosonic loop.

Chapter 14

False Vacuum Decay and Bubble Nucleation

Having established that first-order phase transitions proceed via the nucleation of bubbles of the true vacuum within the false vacuum, we now develop the quantitative framework for computing the nucleation rate. The central object is the bounce solution, a saddle-point of the Euclidean action that dominates the path integral for the false vacuum decay rate. We begin with quantum mechanics, where the essential physics is transparent, and then generalize to quantum field theory.

14.1 Decay Rate in Quantum Mechanics

Consider a quantum mechanical system with a metastable ground state localized at x_0 . If the energy eigenvalue is complex,

$$E_0 = \text{Re } E_0 + i \text{Im } E_0, \quad (14.1)$$

the norm of the wave function decays as

$$\begin{aligned} |\psi_0(t)|^2 &= |e^{-iHt}\psi_0(0)|^2 \\ &= e^{2\text{Im } E_0 t} |\psi_0(0)|^2 \\ &= e^{-\Gamma t} |\psi_0(0)|^2, \end{aligned} \quad (14.2)$$

where $\Gamma = -2\text{Im } E_0$ is the decay rate. To compute E_0 , we use the path integral representation of the transition amplitude,

$$\langle x_b | e^{-i(t_b - t_a)H} | x_a \rangle = \int_{x(t_a)=x_a}^{x(t_b)=x_b} \mathcal{D}x e^{iS(x)}, \quad (14.3)$$

with

$$S(x) = \int_{t_a}^{t_b} dt \left[\frac{1}{2} \left(\frac{dx}{dt} \right)^2 - V(x) \right]. \quad (14.4)$$

We proceed in three steps. First, set $|x_b\rangle = |x_a\rangle = |x_0\rangle$. Second, Wick rotate $t = -i\tau$ with τ real. Third, set $\tau_a = -\hat{\tau}/2$ and $\tau_b = \hat{\tau}/2$.

The left-hand side of (14.3) becomes

$$\langle x_0 | e^{-\hat{H}\hat{\tau}} | x_0 \rangle = \sum_n e^{-E_n \hat{\tau}} |\langle x_0 | \psi_n \rangle|^2 \stackrel{\hat{\tau} \rightarrow \infty}{=} e^{-E_0 \hat{\tau}} |\langle x_0 | \psi_0 \rangle|^2, \quad (14.5)$$

where all excited states are exponentially suppressed. The ground state energy is therefore

$$E_0 = - \lim_{\hat{\tau} \rightarrow \infty} \frac{1}{\hat{\tau}} \log \langle x_0 | e^{-\hat{H}\hat{\tau}} | x_0 \rangle. \quad (14.6)$$

The right-hand side of (14.3) after Wick rotation becomes

$$\int \mathcal{D}x e^{-S_E}, \quad (14.7)$$

where the Euclidean action is

$$S_E = \int_{-\hat{\tau}/2}^{\hat{\tau}/2} d\tau \left[\frac{1}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x) \right]. \quad (14.8)$$

We evaluate (14.7) by expanding around a stationary point $\bar{x}(\tau)$ of S_E . The Euler-Lagrange equation gives

$$\frac{d^2 x(\tau)}{d\tau^2} = V'(x), \quad (14.9)$$

with boundary condition $x(-\hat{\tau}/2) = x(\hat{\tau}/2) = x_0$. Crucially, (14.9) describes a classical particle moving in the inverted potential $-V(x)$.

Setting $x(\tau) = \bar{x}(\tau) + z(\tau)$ and expanding S_E ,

$$\begin{aligned} S_E(x) &= S_E(\bar{x}) - \int d\tau \left(\frac{d^2 \bar{x}}{d\tau^2} - V'(\bar{x}) \right) z \\ &\quad + \frac{1}{2} \int d\tau z(\tau) \left[-\frac{d^2}{d\tau^2} + V''(\bar{x}) \right] z(\tau) + \mathcal{O}(z^3), \end{aligned} \quad (14.10)$$

where integration by parts was used and boundary terms dropped. The linear term vanishes by the equation of motion, leaving

$$\int \mathcal{D}x e^{-S_E} = e^{-S_E(\bar{x})} \int \mathcal{D}z \exp \left(-\frac{1}{2} \int d\tau z \left[-\frac{d^2}{d\tau^2} + V''(\bar{x}) \right] z \right). \quad (14.11)$$

The Gaussian integral over z gives a functional determinant,

$$= N \left[\det \left(-\frac{d^2}{d\tau^2} + V''(\bar{x}) \right) \right]^{-1/2} \stackrel{\hat{\tau} \rightarrow \infty}{=} \left(\frac{\omega}{\pi} \right)^{1/2} e^{-\omega \hat{\tau}/2}, \quad (14.12)$$

where $\omega = \sqrt{V''(x_0)}$. For the trivial stationary path $\bar{x}(\tau) = x_0$, we have $S_E = 0$ and the transition amplitude gives

$$\langle x_0 | e^{-H\hat{\tau}} | x_0 \rangle = \left(\frac{\omega}{\pi} \right)^{1/2} e^{-\omega \hat{\tau}/2}, \quad (14.13)$$

from which $E_0 = \omega/2$.

14.2 The Bounce Solution

The non-trivial stationary path is the bounce: a solution to (14.9) in the inverted potential $-V(x)$ that starts at x_0 at $\tau = -\infty$, rolls to the turning point x_* , and returns to x_0 at $\tau = +\infty$.

Let $\bar{x}(\tau)$ be a bounce solution with $S_E(\bar{x}) = B$. Its contribution to the path integral involves the correction factor

$$K = \left[\frac{\det'(-\partial_\tau^2 + V''(\bar{x}))}{\det(-\partial_\tau^2 + \omega^2)} \right]^{-1/2}, \quad (14.14)$$

where the prime denotes exclusion of the zero eigenvalue.

14.3 The Dilute Gas Approximation

For n widely separated bounces with centers $\tau_1 \ll \tau_2 \ll \cdots \ll \tau_n$, integrating over all center positions gives

$$\int_{-\hat{\tau}/2}^{\hat{\tau}/2} d\tau_1 \int_{\tau_1}^{\hat{\tau}/2} d\tau_2 \cdots \int_{\tau_{n-1}}^{\hat{\tau}/2} d\tau_n = \frac{\hat{\tau}^n}{n!}. \quad (14.15)$$

Summing over all n ,

$$\begin{aligned} \int \mathcal{D}x e^{-S_E} &= \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\hat{\tau}/2} \sum_{n=0}^{\infty} \frac{\hat{\tau}^n}{n!} e^{-Bn} K^n \\ &= \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\hat{\tau}(\omega/2 - Ke^{-B})}, \end{aligned} \quad (14.16)$$

giving

$$\boxed{E_0 = \frac{\omega}{2} - Ke^{-B}}. \quad (14.17)$$

The dilute gas approximation is justified when $B \gg 1$, since the bounce density $n/\hat{\tau} \sim Ke^{-B}$ is exponentially suppressed.

There are two subtleties in computing K . First, the operator $(-\partial_\tau^2 + V''(\bar{x}))$ has a zero eigenvalue from translational invariance of the bounce, which must be excluded from the determinant and treated separately, contributing $(B/2\pi)^{1/2}$. Second, it has a negative eigenvalue requiring analytic continuation. Following Callan and Coleman, the decay rate per unit time is

$$\Gamma = \left(\frac{B}{2\pi}\right)^{1/2} \left[\frac{\det'(-\partial_\tau^2 + V''(\bar{x}))}{\det(-\partial_\tau^2 + \omega^2)} \right]^{-1/2} e^{-B}. \quad (14.18)$$

14.4 Decay Rate in Quantum Field Theory

The generalization to QFT proceeds identically. The decay rate per unit time and volume is

$$\Gamma = \frac{B^2}{4\pi^2} \left[\frac{\det'(-\partial_\tau^2 - \nabla^2 + V''(\bar{\phi}))}{\det(-\partial_\tau^2 - \nabla^2 + \omega^2)} \right]^{-1/2} e^{-B}, \quad (14.19)$$

where the bounce action is

$$B = S_E[\bar{\phi}] = \int d\tau d^3x \left[\frac{1}{2}(\partial_\tau \bar{\phi})^2 + \frac{1}{2}(\nabla \bar{\phi})^2 + V(\bar{\phi}) \right]. \quad (14.20)$$

The functional determinant contributes only a polynomial prefactor; the exponential e^{-B} dominates.

The bounce $\bar{\phi}$ satisfies

$$(\partial_\tau^2 + \nabla^2)\bar{\phi} = V'(\bar{\phi}), \quad (14.21)$$

with boundary conditions $\bar{\phi} \rightarrow \phi_0$ as $\tau \rightarrow \pm\infty$ and $|\mathbf{x}| \rightarrow \infty$. By a theorem of Coleman, Glashow, and Martin (1978), the $O(4)$ -symmetric solution has the lowest action. Writing $\bar{\phi} = \phi(\rho)$ with $\rho = \sqrt{\tau^2 + |\mathbf{x}|^2}$, the equation reduces to

$$\frac{d^2\bar{\phi}}{d\rho^2} + \frac{3}{\rho} \frac{d\bar{\phi}}{d\rho} = V'(\bar{\phi}), \quad (14.22)$$

with $\bar{\phi}'(0) = 0$ and $\bar{\phi} \rightarrow \phi_0$ as $\rho \rightarrow \infty$.

14.5 The Overshoot-Undershoot Algorithm

Equation (14.22) is solved numerically by starting from $\bar{\phi}(0) = \phi_i \in [\phi_0, \phi_1]$ with $\bar{\phi}'(0) = 0$ and evolving to large ρ . If ϕ_i is too large the field overshoots past ϕ_0 ; if too small it undershoots and never reaches ϕ_0 . The correct bounce is found by binary search. The two limiting cases are the thin-wall approximation, where the potential barrier is narrow compared to the bubble radius, and the thick-wall approximation, where it is wide.

14.6 Bubble Nucleation and Dynamics

The bounce has a dual role: it dominates the false vacuum decay rate, and it describes the field configuration after tunneling. The initial bubble is

$$\phi_{\text{bubble}}(t = 0, \mathbf{x}) = \bar{\phi}(\tau = 0, \mathbf{x}), \quad (14.23)$$

$$\partial_t \phi_{\text{bubble}}|_{t=0} = 0. \quad (14.24)$$

After nucleation the bubble evolves by analytic continuation from Euclidean space,

$$\phi_{\text{bubble}}(t, \mathbf{x}) = \bar{\phi}\left(\sqrt{|\mathbf{x}|^2 - t^2}\right), \quad (14.25)$$

and the bubble wall position satisfies $|\mathbf{x}| = \sqrt{t^2 + C^2}$, approaching the speed of light at late times.

14.7 Finite Temperature and the $O(3)$ Bounce

At finite temperature the Euclidean time direction is compactified to a circle of size $\beta = T^{-1}$. At high temperature the $O(4)$ -symmetric bounce transitions to an $O(3)$ -symmetric, time-independent bounce (Linde, 1983). Setting $\partial_\tau \bar{\phi} = 0$, the Euclidean action reduces to

$$S_E = \frac{1}{T} \int d^3x \left[\frac{1}{2} (\nabla \bar{\phi})^2 + V(\bar{\phi}) \right] \equiv \frac{S_3}{T}. \quad (14.26)$$

The two cases are unified as

$$\frac{d^2\phi}{d\rho^2} + \frac{c_i}{\rho} \frac{d\phi}{d\rho} = V'(\phi), \quad c_i = \begin{cases} 3 & O(4) \text{ symmetry,} \\ 2 & O(3) \text{ symmetry,} \end{cases} \quad (14.27)$$

with bounce actions

$$B = \begin{cases} S_E[\bar{\phi}] & O(4), \\ S_3/T & O(3). \end{cases} \quad (14.28)$$

At finite temperature, $V(\bar{\phi})$ is replaced by $V_{\text{eff}}^{\beta}(\bar{\phi})$ and the nucleation rate is $\Gamma \sim e^{-S_3/T}$. The nucleation temperature T_n and percolation temperature T_p , together with the latent heat α and transition rate $\tilde{\beta}$, characterize the transition and determine the gravitational wave spectrum produced during bubble collisions.

Chapter 15

Parameters Characterizing a First-Order Phase Transition

A first-order cosmological phase transition is characterized by several parameters that determine both its dynamics and its observable signatures. Before defining them, we recall the relevant cosmological background. The Robertson-Walker metric is

$$ds^2 = dt^2 - a^2(t) d\mathbf{x}^2, \quad (15.1)$$

with Hubble rate $H = \dot{a}/a$ satisfying the Friedmann equation

$$H^2 = \frac{8\pi G}{3} \rho_{\text{total}}, \quad \rho_{\text{total}} = \frac{\pi^2}{30} g_* T^4 + \Delta\rho, \quad (15.2)$$

where g_* counts the relativistic degrees of freedom and $\Delta\rho$ is the contribution from vacuum energy. The temperature evolves as

$$\frac{dT}{dt} = -HT, \quad (15.3)$$

which follows from entropy conservation in the expanding universe.

15.1 Critical Temperature

The critical temperature T_c is the temperature at which the two phases have equal free energy density, so the effective potential has two degenerate minima. Above T_c the symmetric phase is the global minimum; below T_c the broken phase becomes energetically preferred. The universe does not transition instantaneously at T_c , however, because the transition requires nucleation of bubbles of the true vacuum, which is a tunneling process suppressed by $e^{-S_3/T}$.

15.2 Nucleation Temperature

The number of bubbles nucleated per Hubble volume up to time t is

$$N(t) = \int_{-\infty}^t d\tilde{t} \Gamma(\tilde{t}) H^{-3}(\tilde{t}). \quad (15.4)$$

Using (15.3) to change variables from t to T ,

$$N(T) = \int_T^\infty \frac{d\tilde{T}}{\tilde{T}} \Gamma(\tilde{T}) H^{-4}(\tilde{T}). \quad (15.5)$$

The nucleation temperature T_n is defined by $N(T_n) = 1$, the condition that on average one bubble has nucleated per Hubble volume,

$$\Gamma(T_n) H^{-4}(T_n) \sim 1. \quad (15.6)$$

For $O(3)$ -symmetric instantons at finite temperature, $\Gamma(T) \sim T^4 e^{-S_3/T}$, which gives the approximate condition

$$\frac{S_3(T_n)}{T_n} \approx 145 - 4 \ln \left(\frac{T_n}{100 \text{ GeV}} \right). \quad (15.7)$$

This determines T_n once $S_3(T)$ is known from the bounce calculation.

15.3 The Parameter $\tilde{\beta}$

The parameter $\tilde{\beta}$ characterizes how fast the nucleation rate grows near T_n ,

$$\tilde{\beta} = \frac{1}{\Gamma} \frac{d\Gamma}{dt} \Big|_{N=1}, \quad (15.8)$$

so that $\tilde{\beta}^{-1}$ gives the characteristic duration of the phase transition. Using (15.3),

$$\tilde{\beta} = -HT \frac{d \ln \Gamma}{dT} \Big|_{T=T_n}. \quad (15.9)$$

Writing $\Gamma = A(T) e^{-S_E(T)}$ and neglecting the slowly varying prefactor $A(T)$,

$$\boxed{\frac{\tilde{\beta}}{H} \Big|_{T_n} = T_n \frac{dS_E}{dT} \Big|_{T=T_n}}. \quad (15.10)$$

This is a dimensionless number. When $\tilde{\beta}/H \gg 1$ the phase transition completes in a time much shorter than a Hubble time, so the universe can be treated as approximately static during the transition and the temperature does not change appreciably.

15.4 The Parameter α

The latent heat released during the transition is the energy density difference between the two phases,

$$\Delta\rho = \left(\Delta V - T \frac{\partial \Delta V}{\partial T} \right) \Big|_{T=T_n} \approx \Delta V \Big|_{T=T_n}, \quad (15.11)$$

where ΔV is the free energy difference and the derivative term is subleading. The parameter α is defined as the ratio of this energy density to the radiation energy density,

$$\alpha = \frac{\Delta\rho}{\rho_{\text{rad}}} \Big|_{T=T_n}, \quad (15.12)$$

where $\rho_{\text{rad}} = \pi^2 g_* T_n^4/30$. The parameter α measures the strength of the transition: $\alpha \ll 1$ corresponds to a weak transition in which the latent heat is a small perturbation on the radiation bath, while $\alpha \gg 1$ corresponds to a strongly supercooled transition.

15.5 Reheating Temperature

In the supercooled case $\alpha \gg 1$, the latent heat released by the transition dominates over the radiation energy density. The plasma is reheated to a temperature $T_{\text{rh}} \gg T_n$ determined by energy conservation,

$$(\Delta\rho + \rho_{\text{rad}})_{T=T_n} = \frac{\pi^2}{30} g_* T_{\text{rh}}^4. \quad (15.13)$$

This reheating can have important consequences for baryogenesis and other processes that are sensitive to the thermal history of the universe.

15.6 Bubble Wall Velocity

The bubble wall velocity v_w is determined by the balance between the pressure difference driving the wall outward and the friction from particles in the plasma acquiring mass as they cross the wall. It is intrinsically a non-equilibrium quantity and difficult to calculate from first principles. Three regimes are distinguished.

In the non-relativistic deflagration regime, $v_w \ll 1$, the wall moves slowly and the plasma in front of the wall has time to equilibrate. In the relativistic but non-runaway regime, $v_w \rightarrow 1$ but the Lorentz factor $\gamma = (1 - v_w^2)^{-1/2}$ approaches a finite constant, as the friction force grows to balance the driving pressure. In the runaway regime, $v_w \approx 1$ and γ grows without bound throughout the transition.

In theories with a Higgs boson, the Higgs acquires a mass as it crosses the bubble wall, going from zero mass in the false vacuum to a finite mass in the true vacuum. The resulting friction force is generically large enough to prevent the runaway regime from being reached, so the bubbles at most attain relativistic but finite velocities (Bodeker and Moore, arXiv:1703.08215).

The parameters α , $\tilde{\beta}$, and v_w together determine the spectrum of gravitational waves produced by the phase transition, through the three production mechanisms of bubble wall collisions, sound waves in the plasma, and magnetohydrodynamic turbulence.

Chapter 16

Gravitational Wave Signatures from a First-Order Phase Transition

A first-order cosmological phase transition produces a stochastic background of gravitational waves, whose spectrum encodes information about the transition that is inaccessible to any other observational channel. The parameters α , $\tilde{\beta}$, and v_w introduced in the previous chapter determine both the amplitude and the peak frequency of this spectrum. For an electroweak-scale transition, the predicted signal falls precisely in the millihertz band targeted by the LISA space-based detector, making first-order phase transitions one of the primary science cases for next-generation gravitational wave astronomy.

16.1 Production Mechanisms

There are three main sources of gravitational waves in a first-order phase transition.

The first is the collision of bubble walls. As bubbles of the true vacuum expand, their walls carry kinetic energy proportional to the wall surface area. When two walls collide, this energy is converted into gravitational radiation. In the envelope approximation, only the uncollided portions of the walls contribute, and the signal is concentrated at frequencies $f \sim \tilde{\beta}$.

The second and dominant mechanism in the non-runaway regime is the generation of sound waves in the primordial plasma. As the bubble wall expands, it pushes the plasma ahead of it, creating acoustic perturbations that persist long after the bubble walls have collided and dissipated. Since the energy in sound waves grows with the volume of the bubble rather than its surface area, this mechanism dominates over bubble wall collisions when $v_w < 1$.

The third mechanism is magnetohydrodynamic turbulence, generated when the acoustic motion in the plasma becomes turbulent. Its contribution is generally subdominant, at the ten percent level or less, and is also the least well understood theoretically. We focus on the first two mechanisms here.

16.2 The Linearized Einstein Equations and the Quadrupole Formula

We work in the weak-field limit, treating the metric as a small perturbation about flat spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + \tilde{h}_{\mu\nu}, \quad |\tilde{h}_{\mu\nu}| \ll 1. \quad (16.1)$$

Neglecting the expansion of the universe is justified because the phase transition completes in a time $\tilde{\beta}^{-1} \ll H^{-1}$ much shorter than a Hubble time. Defining the trace-reversed perturbation

$$h_{\mu\nu} = \tilde{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\tilde{h}_{\rho\sigma}\eta^{\rho\sigma}, \quad (16.2)$$

the linearized Einstein equations in Lorenz gauge $\partial_\mu h^{\mu\nu} = 0$ become

$$\partial_\alpha \partial^\alpha h_{\mu\nu} = -16\pi G T_{\mu\nu}. \quad (16.3)$$

This is a wave equation sourced by the stress-energy of the plasma and the scalar field. It is solved by the retarded Green's function,

$$G(x^\mu - y^\mu) = \frac{1}{4\pi} \frac{\delta(x^0 - y^0 - |\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|}, \quad (16.4)$$

giving the retarded solution

$$h_{\mu\nu}(t, \mathbf{x}) = -4G \int d^3y \frac{T_{\mu\nu}(t - |\mathbf{x} - \mathbf{y}|, \mathbf{y})}{|\mathbf{x} - \mathbf{y}|}. \quad (16.5)$$

In the far-field limit $|\mathbf{x} - \mathbf{y}| \approx r$, using stress-energy conservation $\partial^\mu T_{\mu\nu} = 0$ and defining the quadrupole moment tensor

$$I_{ij}(t) = \int d^3y T_{00}(t, \mathbf{y}) y^i y^j, \quad (16.6)$$

one can show that the gravitational wave metric perturbation is

$$\boxed{h_{ij}(t, \mathbf{x}) = -\frac{2G}{r} \frac{d^2 I_{ij}}{dt^2}(t - r).} \quad (16.7)$$

This is the gravitational analogue of the electromagnetic result that a time-varying dipole moment sources radiation: here it is a time-varying quadrupole moment that sources gravitational waves, which is why spherically symmetric processes such as a star collapsing symmetrically produce no gravitational radiation. The power radiated is given by the quadrupole formula,

$$\boxed{\frac{dE_{GW}}{dt} = \frac{G}{5} \frac{d^3 I_{ij}^T}{dt^3} \frac{d^3 I_{ij}^T}{dt^3}}, \quad (16.8)$$

where I_{ij}^T is the traceless part of I_{ij} . The bubble collisions and sound waves both produce anisotropic stress-energy that sources a non-vanishing quadrupole moment, and hence gravitational radiation.

16.3 Order of Magnitude Estimate

To estimate the gravitational wave signal, consider a region of size $\sim v_w \tilde{\beta}^{-1}$ containing a small number of colliding bubbles. The characteristic timescale is $\tilde{\beta}^{-1}$ and the length scale is $v_w \tilde{\beta}^{-1}$. The kinetic energy in the region is set by the fraction of the latent heat converted into bulk motion,

$$E_{\text{kin}} \sim \Delta\rho (v_w \tilde{\beta}^{-1})^3 = \alpha \rho_{\text{rad}} (v_w \tilde{\beta}^{-1})^3. \quad (16.9)$$

The third time derivative of the quadrupole scales as

$$\frac{d^3 I_{ij}^T}{dt^3} \sim \frac{E_{\text{kin}}}{\tilde{\beta}^{-1}} = \alpha \rho_{\text{rad}} v_w^3 \tilde{\beta}^{-2}, \quad (16.10)$$

and substituting into (16.8),

$$\frac{dE_{\text{GW}}}{dt} \sim G \alpha^2 \rho_{\text{rad}}^2 v_w^6 \tilde{\beta}^{-4}. \quad (16.11)$$

Using $G \sim H^2(T_n)/\rho_{\text{rad}}(T_n)$ from the Friedmann equation and converting to the fractional energy density in gravitational waves $\mathcal{S}_{\text{GW}} = \rho_{\text{GW}}/\rho_{\text{tot}}$ with $\rho_{\text{tot}} = (1 + \alpha)\rho_{\text{rad}}$,

$$\mathcal{S}_{\text{GW}} \sim k^2 v_w^3 \frac{\alpha^2}{(1 + \alpha)^2} \left(\frac{H(T_n)}{\tilde{\beta}} \right)^2, \quad (16.12)$$

where k is an efficiency factor, of order unity for runaway bubbles and $\sim \alpha/(1 + \alpha)$ for non-runaway cases, encoding the fraction of latent heat that goes into bulk kinetic energy rather than reheating the plasma. The characteristic frequency of the signal at production is $f_{\text{GW}} \sim \tilde{\beta}$, set by the inverse duration of the transition.

Several features of (16.12) deserve comment. The signal is enhanced by large α , a strongly supercooled transition, and suppressed by large $\tilde{\beta}/H$, a short transition. The dependence on v_w reflects the fact that faster walls convert more latent heat into kinetic energy before the walls collide. The prefactor $(H/\tilde{\beta})^2 \ll 1$ generically suppresses the signal, which is why only the strongest transitions produce a detectable signal.

16.4 The Spectrum Today

The gravitational wave spectrum observed today is obtained by redshifting from the time of the transition at temperature T_n to the present. Using entropy conservation,

$$g_{*,n} T_n^3 a_n^3 = g_{*,0} T_0^3 a_0^3, \quad (16.13)$$

together with the Friedmann equation $H^2 = \rho_{\text{tot}}/3M_{\text{pl}}^2$ with $M_{\text{pl}} = 2.435 \times 10^{18}$ GeV, and the present-day values $H_0 = h \cdot 2.13 \times 10^{-42}$ GeV, $T_0 = 2.43 \times 10^{-13}$ GeV, and $g_{*,0} = 3.97$, the present-day fractional gravitational wave energy density is

$$\Omega_{\text{GW},0} h^2 \sim 1.89 \times 10^{-5} k^2 v_w^3 \frac{\alpha^2}{(1 + \alpha)^2} \left(\frac{H(T_n)}{\tilde{\beta}} \right)^2 \left(\frac{g_{*,n}}{100} \right)^{-1/3}, \quad (16.14)$$

and the peak frequency today is

$$f_{\text{GW},0} \sim 1.7 \times 10^{-5} \text{ Hz} \left(\frac{\tilde{\beta}}{H(T_n)} \right) \frac{T_n}{100 \text{ GeV}} \left(\frac{g_{*,n}}{100} \right)^{1/6}. \quad (16.15)$$

For an electroweak-scale transition with $T_n \sim 100 \text{ GeV}$ and $\tilde{\beta}/H \sim 10^2$, equation (16.15) gives a peak frequency of order 10^{-3} Hz , which falls squarely in the sensitivity band of the LISA space interferometer, planned for launch in the 2030s. The amplitude (16.14) for a moderately strong transition with $\alpha \sim 0.1$ and $k \sim 0.1$ is of order $\Omega_{GW,0} h^2 \sim 10^{-12}$, at the edge of LISA's sensitivity. Stronger transitions with larger α or smaller $\tilde{\beta}/H$ produce correspondingly larger signals.

These results illustrate the power of gravitational wave astronomy as a probe of the early universe. The spectrum (16.14) is determined entirely by the thermodynamic parameters of the phase transition, which in turn are calculable from the finite-temperature effective potential developed in the preceding chapters. A detection by LISA would not only confirm the occurrence of a first-order electroweak phase transition but would also constrain the parameters of the underlying particle physics model, providing information about physics at the electroweak scale that is complementary to and in some respects beyond what is accessible at the LHC.

Chapter 17

Fate of the False Vacuum: Semiclassical Theory

17.1 Introduction

Consider a classical field theory of a single scalar field ϕ in four-dimensional spacetime, with nonderivative self-interactions, described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - U(\phi). \quad (17.1)$$

Suppose the potential $U(\phi)$ has two relative minima, ϕ_+ and ϕ_- , with $U(\phi_-) < U(\phi_+)$, so that ϕ_- is the absolute minimum and ϕ_+ is only a local minimum.

Classically, the field configuration $\phi \equiv \phi_+$, uniform over all space, is a perfectly stable equilibrium: it sits at a local minimum of the potential, and nothing in the classical equations of motion drives it away. Perturbatively, the same is true in the quantum theory: small fluctuations around ϕ_+ are well described by ordinary perturbation theory, and ϕ_+ looks like a perfectly good vacuum state to any finite order in perturbation theory.

Nonperturbatively, however, this is false. Quantum mechanically, the state built on ϕ_+ is unstable: it can decay via barrier penetration, exactly as a particle in a metastable potential well can tunnel out. The state ϕ_+ is therefore called a *false vacuum*, and ϕ_- the *true vacuum*.

The qualitative picture of this decay is the field-theoretic analogue of nucleation phenomena in statistical mechanics: the crystallization of a supersaturated solution, or the boiling of a superheated fluid. In the fluid analogy, $U(\phi)$ plays the role of a free energy as a function of density; the false vacuum is the superheated liquid phase, and the true vacuum is the vapor phase. Thermodynamic fluctuations continually nucleate small bubbles of vapor inside the liquid. A bubble below some critical size shrinks back to nothing, because the surface energy cost of maintaining its boundary outweighs the volume energy gained by converting to the lower-energy phase. A bubble above the critical size, however, finds it energetically favourable to keep growing, and once such a bubble forms, it expands and converts the surrounding liquid to vapor.

The decay of the false vacuum proceeds by exactly the same mechanism, with *quantum* fluctuations replacing thermodynamic ones. Occasionally a bubble of true vacuum nucleates, of just the right size that it is classically favourable for it to grow; once nucleated, the bubble wall expands outward, converting false vacuum to true vacuum as it goes.

The object to compute is therefore the decay probability of the false vacuum per unit time per unit volume, Γ/V . This quantity is directly relevant to cosmology: an

infinitely old universe sitting in a false vacuum must eventually have decayed, no matter how small Γ/V is, simply because an infinite amount of time has elapsed. But the universe is not infinitely old, and the question of whether (and when) a false vacuum decays on cosmological timescales is a genuine, falsifiable question about the history and future of the universe. This point is taken up again in Section ??.

The remainder of this chapter develops the semiclassical formalism for computing Γ/V , following [?]. The exposition proceeds in stages: first reviewing ordinary quantum-mechanical barrier penetration in one dimension (Section 17.2), then generalizing to many degrees of freedom (Section ??), then specializing to field theory (Section ??), working out the thin-wall approximation in closed form (Section ??), and finally following the classical evolution of a bubble after it nucleates (Section ??).

It should be emphasized at the outset what this chapter does *not* cover. The 1977 paper computes only the exponential suppression factor in Γ/V ; the one-loop prefactor multiplying it is the subject of the companion paper by Callan and Coleman, and is not treated here. The present chapter is concerned entirely with the leading semiclassical exponent.

17.2 Barrier Penetration in One Dimension

Before attacking the field theory problem, it is worth carefully rederiving the much simpler quantum-mechanical analogue: a single particle of unit mass tunnelling out of a metastable potential well. This fixes the language and the key result, $\Gamma \sim e^{-B/\hbar}$, that will be generalized step by step.

17.2.1 Setting up the problem

Consider a particle of unit mass moving in one dimension with Lagrangian

$$L = \frac{1}{2}\dot{q}^2 - V(q), \quad (17.2)$$

where $V(q)$ has a local minimum at q_0 , with the zero of energy chosen so that $V(q_0) = 0$. Let σ denote the other zero of V , i.e. the point where the particle, released from rest at q_0 with zero total energy, would classically just reach the far side of the barrier.

Classically there is no instability: a particle placed at rest at q_0 stays there forever, since q_0 is a genuine local minimum. Quantum mechanically, however, there is no exactly stable bound state corresponding to this classical equilibrium, because the particle can tunnel through the barrier separating q_0 from the region beyond σ , where it is free to propagate off to infinity (or to some other, lower-energy region of configuration space).

17.2.2 The WKB exponent

The decay rate associated with this barrier penetration takes the semiclassical form

$$\Gamma = A e^{-B/\hbar} [1 + O(\hbar)], \quad (17.3)$$

where B is computed below and A is a prefactor arising from quantum fluctuations around the dominant tunnelling trajectory. The goal of this section is to derive the exponent B and to show that it is naturally interpreted as a classical action.

The cleanest derivation proceeds by an $\hbar$ -expansion of the Schrödinger equation directly. Write the (energy eigenstate) wavefunction as

$$\psi(q) = e^{if(q)/\hbar}, \quad (17.4)$$

where f is in general complex; this is no loss of generality, since any nonvanishing function can be written this way. Substituting into the time-independent Schrödinger equation

$$-\frac{\hbar^2}{2}\psi''(q) + V(q)\psi(q) = E\psi(q) \quad (17.5)$$

and using

$$\psi' = \frac{if'}{\hbar}\psi, \quad \psi'' = \left[\frac{if''}{\hbar} - \frac{(f')^2}{\hbar^2} \right] \psi, \quad (17.6)$$

gives, after multiplying through by $-2/\psi$ and rearranging,

$$i\hbar f'' - (f')^2 + p(q)^2 = 0, \quad p(q) \equiv \sqrt{2[E - V(q)]}. \quad (17.7)$$

Now expand f in powers of $\hbar$,

$$f(q) = f_0(q) + \hbar f_1(q) + \hbar^2 f_2(q) + \cdots, \quad (17.8)$$

and collect (17.7) order by order. Writing out $(f')^2$ and f'' to the needed order,

$$(f')^2 = (f'_0)^2 + 2\hbar f'_0 f'_1 + O(\hbar^2), \quad f'' = f''_0 + \hbar f''_1 + O(\hbar^2), \quad (17.9)$$

equation (17.7) becomes

$$i\hbar f''_0 + i\hbar^2 f''_1 - (f'_0)^2 - 2\hbar f'_0 f'_1 + p^2 + O(\hbar^2) = 0. \quad (17.10)$$

Matching powers of $\hbar$:

$$O(\hbar^0) : \quad (f'_0)^2 = p^2, \quad (17.11)$$

$$O(\hbar^1) : \quad i f''_0 = 2 f'_0 f'_1. \quad (17.12)$$

Equation (17.11) gives

$$f'_0(q) = \pm p(q) \implies f_0(q) = \pm \int^q p(q') dq'. \quad (17.13)$$

Equation (17.12) then gives

$$f'_1 = \frac{if''_0}{2f'_0} = \frac{i}{2} \frac{d}{dq} \ln(f'_0), \quad (17.14)$$

which integrates to

$$f_1(q) = \frac{i}{2} \ln [\pm p(q)] + \text{const.} \quad (17.15)$$

Assembling $\psi = \exp[i(f_0 + \hbar f_1)/\hbar]$ to this order,

$$\exp(if_1) = \exp\left[-\frac{1}{2} \ln(\pm p)\right] = \frac{1}{\sqrt{\pm p(q)}}, \quad (17.16)$$

so that

$$\psi(q) \approx \frac{C_{\pm}}{\sqrt{p(q)}} \exp\left[\pm \frac{i}{\hbar} \int^q p(q') dq'\right] \quad (17.17)$$

in a classically allowed region, where $p(q)$ is real. This is the standard WKB wavefunction.

17.2.3 The forbidden region and the tunnelling exponent

In the region under the barrier, $E < V(q)$, so $p(q)^2 = 2[E - V(q)] < 0$ and p is pure imaginary. Write

$$p(q) = i\kappa(q), \quad \kappa(q) \equiv \sqrt{2[V(q) - E]} \quad (\text{real, positive}). \quad (17.18)$$

Equation (17.17) then becomes, after the same bookkeeping with $\ln$ of an imaginary argument,

$$\psi(q) \approx \frac{C_{\pm}}{\sqrt{\kappa(q)}} \exp\left[\pm \frac{1}{\hbar} \int^q \kappa(q') dq'\right], \quad (17.19)$$

i.e. a real, exponentially growing or decaying solution under the barrier, rather than an oscillating one. This is the expected behaviour: in a classically forbidden region the wavefunction cannot oscillate, since there is no real classical momentum, but it need not vanish either.

The decaying branch of (17.19), continued from just inside q_0 to the far edge of the barrier at σ , picks up a suppression factor

$$\left| \frac{\psi(\sigma)}{\psi(q_0)} \right| \sim \exp\left[-\frac{1}{\hbar} \int_{q_0}^{\sigma} \kappa(q) dq\right] = \exp\left[-\frac{1}{\hbar} \int_{q_0}^{\sigma} \sqrt{2V(q)} dq\right], \quad (17.20)$$

having set $E = 0$ so that the energy is measured from the bottom of the well. The transmission probability is the square of this amplitude ratio, giving an exponential suppression $e^{-B/\hbar}$ with

$$B = 2 \int_{q_0}^{\sigma} dq [2V(q)]^{1/2}. \quad (17.21)$$

The decay rate Γ inherits the same exponential suppression, since the rate-limiting step in the decay process is precisely this barrier-penetration probability; any remaining, non-exponential dependence on the parameters of the potential is absorbed into the prefactor A , recovering (17.3).

17.2.4 B as a Euclidean action

The exponent B in (17.21) is not merely a convenient integral; it has a direct interpretation as a classical action, and this interpretation is what makes the generalization to many degrees of freedom and to field theory tractable.

Consider the analytic continuation of time to imaginary values, $t \rightarrow -i\tau$ (so τ is real). Under this continuation, $\dot{q} = dq/dt \rightarrow i dq/d\tau$, and the Lagrangian becomes

$$L = \frac{1}{2}\dot{q}^2 - V(q) \longrightarrow -\frac{1}{2}\left(\frac{dq}{d\tau}\right)^2 - V(q) \equiv -L_E, \quad (17.22)$$

where

$$L_E = \frac{1}{2}\left(\frac{dq}{d\tau}\right)^2 + V(q) \quad (17.23)$$

is the *Euclidean* Lagrangian: a particle moving in imaginary time experiences the *inverted* potential $-V(q)$ relative to the original problem, since the sign of the potential term flips relative to the kinetic term. Equivalently, motion under the original barrier in real time becomes motion in the (now classically allowed) inverted-potential problem in imaginary time.

The Euclidean equations of motion following from (17.23), via Hamilton's principle $\delta \int d\tau L_E = 0$, are

$$\frac{d^2 q}{d\tau^2} = \frac{\partial V}{\partial q}, \quad (17.24)$$

with first integral (Euclidean energy conservation)

$$\frac{1}{2} \left(\frac{dq}{d\tau} \right)^2 - V(q) = 0 \quad (17.25)$$

for the trajectory of interest, namely the one with zero total energy, released from rest. From (17.25),

$$\frac{dq}{d\tau} = \sqrt{2V(q)} \implies d\tau = \frac{dq}{\sqrt{2V(q)}}, \quad (17.26)$$

and the Euclidean action evaluated on this classical trajectory is

$$S_E = \int d\tau L_E = \int d\tau \left[\frac{1}{2} \left(\frac{dq}{d\tau} \right)^2 + V(q) \right] = \int d\tau \left(\frac{dq}{d\tau} \right)^2 = \int dq \frac{dq}{d\tau} = \int_{q_0}^{\sigma} dq \sqrt{2V(q)}, \quad (17.27)$$

using (17.25) twice (once to replace $\frac{1}{2}(dq/d\tau)^2$ by V in L_E , giving $L_E = (dq/d\tau)^2$, and once to replace $dq/d\tau$ itself). Comparing (17.27) with (17.21),

$$B = 2 \int_{q_0}^{\sigma} dq \sqrt{2V(q)} = 2 S_E. \quad (17.28)$$

This trajectory — starting at rest at q_0 at $\tau \rightarrow -\infty$, moving under the inverted potential out to σ at some finite τ (taken to be $\tau = 0$ by time-translation invariance), and by time-reversal symmetry of (17.24)–(17.25) bouncing back and returning to q_0 as $\tau \rightarrow +\infty$ — is called *the bounce*. The full round trip out to σ and back doubles the one-way action, which is exactly the origin of the factor of 2 in (17.21): B is the *total* Euclidean action of the complete bounce trajectory,

$$B = S_E[\text{bounce}] = \int_{-\infty}^{\infty} d\tau L_E. \quad (17.29)$$

This reformulation, $\Gamma \sim e^{-S_E[\text{bounce}]/\hbar}$, is the form in which the result generalizes. Sections ?? and ?? promote this one-dimensional bounce to, respectively, a trajectory in many-dimensional configuration space and a classical solution of the Euclidean field equations.

17.3 Barrier Penetration in Field Theory

Having established the semiclassical formalism for barrier penetration in quantum mechanics, we now translate the prescription to the field-theory problem of false vacuum decay. The central object is the *bounce*, the field configuration that dominates the path integral in the classically forbidden region.

The starting point is the Euclidean (imaginary-time) equation of motion, obtained from the Minkowskian field equation by the formal substitution $t \rightarrow -i\tau$,

$$\left(\frac{\partial^2}{\partial \tau^2} + \nabla^2 \right) \phi = U'(\phi), \quad (17.30)$$

where a prime denotes differentiation with respect to ϕ . The passage from real to imaginary time takes us from Minkowski space to Euclidean space, and it is in this Euclidean setting that barrier penetration processes are naturally described.

The bounce solution $\phi(\tau, \mathbf{x})$ must satisfy two boundary conditions. The first requires that the field returns to the false vacuum ϕ_+ as imaginary time runs to $\pm\infty$,

$$\lim_{\tau \rightarrow \pm\infty} \phi(\tau, \mathbf{x}) = \phi_+. \quad (17.31)$$

This is the field-theoretic counterpart of the particle starting and ending at the classical equilibrium point q_0 as $\tau \rightarrow \pm\infty$, as established in the previous section. Physically, it expresses the fact that far in imaginary time the field sits undisturbed in the false vacuum. The second boundary condition is

$$\frac{\partial \phi}{\partial \tau}(0, \mathbf{x}) = 0, \quad (17.32)$$

which is the field-theoretic analogue of the vanishing of the particle's velocity at the turning point $\tau = 0$. It reflects the time-reversal symmetry of the bounce: the field configuration for positive τ is simply the mirror image of that for negative τ .

The tunneling amplitude is controlled by the coefficient B , which is the Euclidean action evaluated on the bounce solution,

$$B = S_E = \int d\tau d^3x \left[\frac{1}{2} \left(\frac{\partial \phi}{\partial \tau} \right)^2 + \frac{1}{2} (\nabla \phi)^2 + U(\phi) \right]. \quad (17.33)$$

The decay rate per unit volume then takes the semiclassical form $\Gamma/V \sim Ae^{-B/\hbar}$, so a larger B corresponds to a more suppressed tunneling rate.

For the expression in (17.33) to yield a finite value of B , the integrand must fall off sufficiently fast in all directions. Consider the behaviour of the field at large spatial separation from the bubble, $|\mathbf{x}| \rightarrow \infty$. There the field approaches some constant value ϕ_∞ , so the gradient terms $(\partial \phi / \partial \tau)^2$ and $(\nabla \phi)^2$ both vanish. The integrand then reduces to $U(\phi_\infty)$, and the spatial integral becomes

$$\int d^3x U(\phi_\infty) \sim U(\phi_\infty) \cdot |\mathbf{x}|^3 \rightarrow \infty, \quad (17.34)$$

unless $U(\phi_\infty) = 0$. The potential U has been normalised so that $U(\phi_+) = 0$ at the false vacuum; the true vacuum satisfies $U(\phi_-) < 0$ and so does not give a zero. Therefore finiteness of B forces

$$\lim_{|\mathbf{x}| \rightarrow \infty} \phi(\tau, \mathbf{x}) = \phi_+. \quad (17.35)$$

This is not an independent physical assumption but a derived condition: it follows directly from requiring that the tunneling exponent be finite, which is the minimal consistency requirement for the semiclassical approximation. It also has a clear physical meaning: far from the nucleated bubble, the field remains undisturbed in the false vacuum.

The three boundary conditions (17.31), (17.32), and (17.35) are all consistent with the assumption that the bounce possesses full $O(4)$ invariance, that is, that ϕ depends on τ and $\mathbf{x}$ only through the $O(4)$ -invariant combination

$$\rho = (\tau^2 + |\mathbf{x}|^2)^{1/2}. \quad (17.36)$$

If $O(4)$ -non-invariant bounces exist, they carry a higher Euclidean action than the $O(4)$ -invariant one and may therefore be safely ignored. We thus restrict attention to solutions of the form $\phi = \phi(\rho)$.

Under the $O(4)$ symmetry assumption, the field equation (17.30) reduces to a single nonlinear ordinary differential equation, which we now derive explicitly. Since ϕ depends on $(\tau, \mathbf{x})$ only through $\rho = (\tau^2 + |\mathbf{x}|^2)^{1/2}$, we compute each term in the wave operator separately. For the imaginary-time part, the chain rule gives

$$\frac{\partial \phi}{\partial \tau} = \phi'(\rho) \frac{\partial \rho}{\partial \tau} = \phi'(\rho) \frac{\tau}{\rho}, \quad (17.37)$$

where a prime now denotes $d/d\rho$. Differentiating once more,

$$\frac{\partial^2 \phi}{\partial \tau^2} = \phi''(\rho) \frac{\tau^2}{\rho^2} + \phi'(\rho) \frac{\partial}{\partial \tau} \left(\frac{\tau}{\rho} \right). \quad (17.38)$$

The remaining derivative is

$$\frac{\partial}{\partial \tau} \left(\frac{\tau}{\rho} \right) = \frac{1}{\rho} - \frac{\tau^2}{\rho^3} = \frac{|\mathbf{x}|^2}{\rho^3}, \quad (17.39)$$

so that

$$\frac{\partial^2 \phi}{\partial \tau^2} = \phi'' \frac{\tau^2}{\rho^2} + \phi' \frac{|\mathbf{x}|^2}{\rho^3}. \quad (17.40)$$

For the spatial part, by the same reasoning applied to each Cartesian component x^i ,

$$\frac{\partial^2 \phi}{\partial (x^i)^2} = \phi'' \frac{(x^i)^2}{\rho^2} + \phi' \frac{\rho^2 - (x^i)^2}{\rho^3}. \quad (17.41)$$

Summing over $i = 1, 2, 3$,

$$\nabla^2 \phi = \phi'' \frac{|\mathbf{x}|^2}{\rho^2} + \phi' \frac{3\rho^2 - |\mathbf{x}|^2}{\rho^3}. \quad (17.42)$$

Adding (17.40) and (17.42),

$$\frac{\partial^2 \phi}{\partial \tau^2} + \nabla^2 \phi = \phi'' \left(\frac{\tau^2 + |\mathbf{x}|^2}{\rho^2} \right) + \phi' \left(\frac{|\mathbf{x}|^2 + 3\rho^2 - |\mathbf{x}|^2}{\rho^3} \right). \quad (17.43)$$

Using $\tau^2 + |\mathbf{x}|^2 = \rho^2$ in the first bracket and cancelling $|\mathbf{x}|^2$ in the second,

$$\frac{\partial^2 \phi}{\partial \tau^2} + \nabla^2 \phi = \phi'' + \frac{3}{\rho} \phi'. \quad (17.44)$$

Substituting into (17.30) yields the reduced equation,

$$\frac{d^2 \phi}{d\rho^2} + \frac{3}{\rho} \frac{d\phi}{d\rho} = U'(\phi). \quad (17.45)$$

The coefficient $3/\rho$ in (17.45) is the contribution of the angular degrees of freedom of four-dimensional Euclidean space to the radial Laplacian; more generally, for an $O(n)$ -invariant field in n dimensions it would be $(n-1)/\rho$. Here $n = 4$, one imaginary-time direction and three spatial directions, giving the factor of 3. There is a useful mechanical

analogy: if one interprets ϕ as the position of a particle and ρ as time, equation (17.45) describes a particle moving in the inverted potential $-U(\phi)$ and subject to a ρ -dependent viscous damping force with coefficient $3/\rho$, which weakens as ρ increases. This analogy will be used subsequently to establish the existence of the bounce and to understand its qualitative properties.

Finally, the boundary conditions (17.31) and (17.35) both collapse under $O(4)$ symmetry to the single condition

$$\lim_{\rho \rightarrow \infty} \phi(\rho) = \phi_+, \quad (17.46)$$

and condition (17.32) becomes

$$\left. \frac{d\phi}{d\rho} \right|_{\rho=0} = 0, \quad (17.47)$$

which is also required for regularity of the solution at the origin. Together, (17.46) and (17.47) furnish the two boundary conditions for the ordinary differential equation (17.45).

Part IV

Quantum Field Theory in Curved Spacetime

The preceding parts of this thesis developed a systematic framework for understanding the vacuum structure of quantum field theories and its evolution with temperature. Beginning with the effective action formalism in flat spacetime, we computed the one-loop effective potential and saw how quantum corrections can qualitatively change the vacuum structure of a theory, generating symmetry breaking where none existed classically. Extending this to finite temperature, we found that thermal corrections restore symmetry at high temperatures, driving cosmological phase transitions as the early universe cooled. Throughout all of this, the background spacetime was treated as fixed and flat, or at most as a classical FRW geometry.

This assumption is an approximation, and in many physically important situations it breaks down. In the very early universe, near the Planck scale, the energy densities involved are large enough that the geometry itself is dynamical and cannot be treated as a fixed background. Near a collapsing star, the curvature of spacetime grows without bound as the horizon forms. In the inflationary epoch, the exponential expansion of the universe produces a de Sitter geometry whose causal structure is fundamentally different from that of Minkowski space. In all of these situations, the interplay between quantum matter and curved geometry becomes essential.

The appropriate framework for addressing these questions is *semiclassical gravity*, in which the spacetime geometry is treated classically via the Einstein equations, but the matter fields are promoted to quantum operators. The backreaction of quantum matter on the geometry is captured by the semiclassical Einstein equation,

$$G_{\mu\nu} = 8\pi\langle\hat{T}_{\mu\nu}\rangle, \quad (17.48)$$

where the right-hand side is the expectation value of the quantum stress-energy tensor in some state. This is the natural generalization of the classical field equations to include quantum effects, and it is valid as long as the curvature remains well below the Planck scale so that quantum gravitational effects can be ignored.

The necessity of this framework can be seen directly from the effective action. In a curved background, the field equation acquires a non-minimal coupling to the curvature, $(\square - m^2 + \xi R)\phi = 0$, which shifts the effective mass of the field and can qualitatively change the phase structure. More fundamentally, the very notion of a vacuum state becomes observer-dependent in curved spacetime. In flat spacetime, the Poincaré invariance of the Minkowski metric singles out a unique vacuum state agreed upon by all inertial observers, and the effective potential is a well-defined quantity. In a general curved spacetime there is no such preferred frame, and different observers, or the same observer at different times, may disagree on the particle content of a given state. The Bogoliubov transformation formalism, developed in Chapter 21, makes this precise.

The physical consequences of this observer-dependence are profound. The most celebrated is the Hawking effect: a black hole formed by gravitational collapse emits a steady thermal flux of radiation at a temperature $T_H = \kappa/2\pi$, where κ is the surface gravity of the horizon. This connects back to the cosmological context in a direct way. The de Sitter spacetime produced during inflation also possesses a horizon, the cosmological horizon, and quantum fields in this background experience a Gibbons-Hawking temperature $T = H/2\pi$, where H is the Hubble rate. Quantum fluctuations in this thermal de Sitter background seed the primordial density perturbations that eventually grow into the large-scale structure of the universe we observe today.

The chapters of this part develop the necessary tools in a logical sequence. We begin with the causal structure of the Schwarzschild spacetime, introducing the Kruskal and

Penrose diagrams that make the global geometry transparent. We then develop black hole thermodynamics, establishing the Bekenstein-Hawking entropy and the generalized second law. The general formalism of quantum field theory in curved spacetime follows, including the Klein-Gordon inner product, the Bogoliubov transformation, and the ambiguity in the notion of particles. We then study two concrete examples of particle production, the moving mirror and the uniformly accelerating detector, which isolate the essential physics of the Hawking effect in a simpler setting. Finally we derive Hawking radiation itself, both in the 2D model where the calculation can be done exactly, and in the full 4D case where greybody factors modify the spectrum.

Chapter 18

Spacetime Causal Structure

The effective action formalism developed in Part ?? was built on the assumption that the background spacetime is flat Minkowski space. Before we can extend it to curved spacetime, we need to understand the global causal structure of the spacetimes we are working with. This chapter introduces the Schwarzschild geometry and its maximal extension, develops the Kruskal and Penrose diagram technology that makes the causal structure transparent, and identifies the key features, the horizon and the singularity, that will drive the physics of particle production in later chapters.

We work in units $\hbar = c = G = 1$ and use the mostly plus metric signature $g_{\mu\nu} = (-, +, +, +)$ throughout this part.

18.1 The Schwarzschild Metric and the Tortoise Coordinate

The Schwarzschild metric outside a spherically symmetric mass distribution is

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2, \quad r > 2M. \quad (18.1)$$

There is an apparent singularity at $r = 2M$, but this is a coordinate artifact rather than a true curvature singularity. To see this, we introduce the tortoise coordinate r_* , defined by

$$r_* = \int \frac{dr}{1 - \frac{2M}{r}} = r + 2M \ln\left(\frac{r}{2M} - 1\right), \quad 2M < r < \infty, \quad -\infty < r_* < +\infty. \quad (18.2)$$

The name comes from the fact that an infinite coordinate interval $r_* \rightarrow -\infty$ corresponds to approaching the horizon $r \rightarrow 2M$, so the horizon recedes to minus infinity in this coordinate. In terms of r_* , the t - r part of the metric becomes

$$- \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} = \left(1 - \frac{2M}{r}\right) (-dt^2 + dr_*^2), \quad (18.3)$$

which is conformally flat, with an overall factor that vanishes at the horizon but leaves the conformal structure intact. Introducing null coordinates

$$u = t - r_*, \quad v = t + r_*, \quad (18.4)$$

the metric takes the form

$$ds^2 = - \left(1 - \frac{2M}{r}\right) du dv + r^2 d\Omega^2. \quad (18.5)$$

18.2 Kruskal Coordinates and the Maximal Extension

The singularity at $r = 2M$ in the original coordinates is removed by a further change of coordinates. Any transformation of the form $\tilde{u} = f(u)$, $\tilde{v} = g(v)$ preserves the null structure of the metric. We choose

$$\tilde{u} = -4Me^{-u/4M}, \quad \tilde{v} = 4Me^{v/4M}, \quad (18.6)$$

which maps $u \in (-\infty, +\infty)$ to $\tilde{u} \in (-\infty, 0)$ and $v \in (-\infty, +\infty)$ to $\tilde{v} \in (0, +\infty)$. In these Kruskal coordinates the metric becomes

$$ds^2 = -\frac{2M}{r(\tilde{u}, \tilde{v})} e^{-r(\tilde{u}, \tilde{v})/2M} d\tilde{u} d\tilde{v} + r^2(\tilde{u}, \tilde{v}) d\Omega^2, \quad (18.7)$$

where $r(\tilde{u}, \tilde{v})$ is determined implicitly by

$$\tilde{u} \tilde{v} = 4M^2 \left(1 - \frac{r}{2M}\right) e^{-r/2M}. \quad (18.8)$$

The metric is now manifestly regular at $r = 2M$: when $r = 2M$ either $\tilde{u}$ or $\tilde{v}$ vanishes, but the prefactor $e^{-r/2M}$ is finite. The true singularity at $r = 0$ corresponds to $\tilde{u} \tilde{v} = 4M^2$, a hyperbola in the $(\tilde{u}, \tilde{v})$ plane. The singularity is genuine because the curvature invariants diverge there, and geodesics reach it in finite affine parameter.

The Kruskal extension reveals four distinct regions of the maximally extended Schwarzschild spacetime. Region I is the exterior of the black hole, $r > 2M$, where $t = \text{const}$ lines converge at spacelike infinity and $r = \text{const}$ lines are timelike surfaces. Region II is the black hole interior: once inside, the roles of r and t are exchanged, with $r = \text{const}$ becoming spacelike surfaces and an observer being unable to remain at constant r . Regions III and IV are the white hole interior and the second exterior region respectively; they are physically irrelevant for realistic gravitational collapse but are part of the maximal extension.

18.3 The Penrose Diagram

Compactifying the Kruskal coordinates via

$$\bar{u} = \tan^{-1} \frac{\tilde{u}}{4M}, \quad \bar{v} = \tan^{-1} \frac{\tilde{v}}{4M}, \quad (18.9)$$

maps the entire spacetime into a finite region. This choice is not unique; any monotonic functions would preserve the causal relations. The resulting Penrose diagram represents the full causal structure of the spacetime in a finite picture, with null rays propagating at 45 throughout.

The causal structure encoded in the diagram is physically important. An observer in region I can receive signals from region I and from region IV, but cannot communicate with the interior (region II) or with another exterior observer in region III. The singularity at $r = 0$ is a spacelike surface in regions II, meaning it lies in the future of every observer who crosses the horizon, and cannot be avoided.

The Penrose diagram for 2D Minkowski spacetime is obtained by a similar construction. In null coordinates $u = t - x$, $v = t + x$, the metric is $ds^2 = -du dv$. The compactification

$$u' = 2 \tan^{-1} u, \quad v' = 2 \tan^{-1} v, \quad -\pi \leq u', v' \leq \pi, \quad (18.10)$$

maps the full Minkowski space to a finite diamond. The conformal factor is

$$\Omega^2 = \frac{1}{4} \sec^2 \frac{u'}{2} \sec^2 \frac{v'}{2}, \quad (18.11)$$

and after a conformal rescaling $g_{\mu\nu} \rightarrow \Omega^{-2} g_{\mu\nu}$ the metric takes the same form $d\tilde{s}^2 = du' dv'$ over the compact region. The boundaries of the diagram correspond to future and past null infinity $\mathcal{I}^\pm$, future and past timelike infinity $i^\pm$, and spacelike infinity i^0 . Null rays terminate on $\mathcal{I}^\pm$ because $u = \tan(u'/2) \rightarrow \pm\infty$ as $u' \rightarrow \pm\pi$. In 4D the same construction applies, with each interior point representing a 2-sphere, except on the symmetry axis and at i^0 .

Chapter 19

Spacetime Causal Structure

The effective action formalism developed in Part ?? was built on the assumption that the background spacetime is flat Minkowski space. Before we can extend it to curved spacetime, we need to understand the global causal structure of the spacetimes we are working with. This chapter introduces the Schwarzschild geometry and its maximal extension, develops the Kruskal and Penrose diagram technology that makes the causal structure transparent, and identifies the key features, the horizon and the singularity, that will drive the physics of particle production in later chapters.

We work in units $\hbar = c = G = 1$ and use the mostly plus metric signature $g_{\mu\nu} = (-, +, +, +)$ throughout this part.

19.1 The Schwarzschild Metric and the Tortoise Coordinate

The Schwarzschild metric outside a spherically symmetric mass distribution is

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2, \quad r > 2M. \quad (19.1)$$

There is an apparent singularity at $r = 2M$, but this is a coordinate artifact rather than a true curvature singularity. To see this, we introduce the tortoise coordinate r_* , defined by

$$r_* = \int \frac{dr}{1 - \frac{2M}{r}} = r + 2M \ln\left(\frac{r}{2M} - 1\right), \quad 2M < r < \infty, \quad -\infty < r_* < +\infty. \quad (19.2)$$

The name comes from the fact that an infinite coordinate interval $r_* \rightarrow -\infty$ corresponds to approaching the horizon $r \rightarrow 2M$, so the horizon recedes to minus infinity in this coordinate. In terms of r_* , the t - r part of the metric becomes

$$- \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} = \left(1 - \frac{2M}{r}\right) (-dt^2 + dr_*^2), \quad (19.3)$$

which is conformally flat, with an overall factor that vanishes at the horizon but leaves the conformal structure intact. Introducing null coordinates

$$u = t - r_*, \quad v = t + r_*, \quad (19.4)$$

the metric takes the form

$$ds^2 = - \left(1 - \frac{2M}{r}\right) du dv + r^2 d\Omega^2. \quad (19.5)$$

19.2 Kruskal Coordinates and the Maximal Extension

The singularity at $r = 2M$ in the original coordinates is removed by a further change of coordinates. Any transformation of the form $\tilde{u} = f(u)$, $\tilde{v} = g(v)$ preserves the null structure of the metric. We choose

$$\tilde{u} = -4Me^{-u/4M}, \quad \tilde{v} = 4Me^{v/4M}, \quad (19.6)$$

which maps $u \in (-\infty, +\infty)$ to $\tilde{u} \in (-\infty, 0)$ and $v \in (-\infty, +\infty)$ to $\tilde{v} \in (0, +\infty)$. In these Kruskal coordinates the metric becomes

$$ds^2 = -\frac{2M}{r(\tilde{u}, \tilde{v})} e^{-r(\tilde{u}, \tilde{v})/2M} d\tilde{u} d\tilde{v} + r^2(\tilde{u}, \tilde{v}) d\Omega^2, \quad (19.7)$$

where $r(\tilde{u}, \tilde{v})$ is determined implicitly by

$$\tilde{u} \tilde{v} = 4M^2 \left(1 - \frac{r}{2M}\right) e^{-r/2M}. \quad (19.8)$$

The metric is now manifestly regular at $r = 2M$: when $r = 2M$ either $\tilde{u}$ or $\tilde{v}$ vanishes, but the prefactor $e^{-r/2M}$ is finite. The true singularity at $r = 0$ corresponds to $\tilde{u} \tilde{v} = 4M^2$, a hyperbola in the $(\tilde{u}, \tilde{v})$ plane. The singularity is genuine because the curvature invariants diverge there, and geodesics reach it in finite affine parameter.

The Kruskal extension reveals four distinct regions of the maximally extended Schwarzschild spacetime. Region I is the exterior of the black hole, $r > 2M$, where $t = \text{const}$ lines converge at spacelike infinity and $r = \text{const}$ lines are timelike surfaces. Region II is the black hole interior: once inside, the roles of r and t are exchanged, with $r = \text{const}$ becoming spacelike surfaces and an observer being unable to remain at constant r . Regions III and IV are the white hole interior and the second exterior region respectively; they are physically irrelevant for realistic gravitational collapse but are part of the maximal extension.

19.3 The Penrose Diagram

Compactifying the Kruskal coordinates via

$$\bar{u} = \tan^{-1} \frac{\tilde{u}}{4M}, \quad \bar{v} = \tan^{-1} \frac{\tilde{v}}{4M}, \quad (19.9)$$

maps the entire spacetime into a finite region. This choice is not unique; any monotonic functions would preserve the causal relations. The resulting Penrose diagram represents the full causal structure of the spacetime in a finite picture, with null rays propagating at 45 throughout.

The causal structure encoded in the diagram is physically important. An observer in region I can receive signals from region I and from region IV, but cannot communicate with the interior (region II) or with another exterior observer in region III. The singularity at $r = 0$ is a spacelike surface in regions II, meaning it lies in the future of every observer who crosses the horizon, and cannot be avoided.

The Penrose diagram for 2D Minkowski spacetime is obtained by a similar construction. In null coordinates $u = t - x$, $v = t + x$, the metric is $ds^2 = -du dv$. The compactification

$$u' = 2 \tan^{-1} u, \quad v' = 2 \tan^{-1} v, \quad -\pi \leq u', v' \leq \pi, \quad (19.10)$$

maps the full Minkowski space to a finite diamond. The conformal factor is

$$\Omega^2 = \frac{1}{4} \sec^2 \frac{u'}{2} \sec^2 \frac{v'}{2}, \quad (19.11)$$

and after a conformal rescaling $g_{\mu\nu} \rightarrow \Omega^{-2} g_{\mu\nu}$ the metric takes the same form $d\tilde{s}^2 = du' dv'$ over the compact region. The boundaries of the diagram correspond to future and past null infinity $\mathscr{I}^\pm$, future and past timelike infinity $i^\pm$, and spacelike infinity i^0 . Null rays terminate on $\mathscr{I}^\pm$ because $u = \tan(u'/2) \rightarrow \pm\infty$ as $u' \rightarrow \pm\pi$. In 4D the same construction applies, with each interior point representing a 2-sphere, except on the symmetry axis and at i^0 .

Chapter 20

Black Hole Thermodynamics

Before developing the full quantum field theory in curved spacetime, we establish that black holes behave as thermodynamic objects even at the semiclassical level. This was historically the first hint that black holes and thermodynamics are deeply connected, and it sets up the physical question that the Hawking calculation answers: if a black hole has entropy proportional to its horizon area, what is the temperature associated with that entropy, and is there a corresponding thermal radiation? The gedanken experiments of this chapter fix the Bekenstein-Hawking entropy $S_{BH} = A/4$ and the generalized second law, providing the thermodynamic framework within which Hawking radiation will later be understood.

The idea that black holes carry entropy and behave as thermodynamic objects was proposed by Bekenstein on the basis of a simple but compelling analogy. We develop this analogy here and subject it to quantitative tests.

20.1 The Bekenstein Entropy Conjecture

Consider throwing a box of gas with entropy S_p into a black hole. The gas disappears behind the horizon and its entropy is lost to an outside observer. But the second law of thermodynamics forbids the total entropy of the universe from decreasing. This suggests that the black hole itself must carry entropy to compensate.

Two facts from general relativity support this. First, vacuum solutions to the Einstein equations with regular horizons, that is, black holes, are determined by just two parameters: mass M and angular momentum J (plus charge Q for non-vacuum solutions). This is the no-hair theorem. Second, the area of a black hole horizon never decreases; it only grows. For a Schwarzschild black hole,

$$A = 4\pi r_h^2 = 4\pi(2M)^2 = 16\pi M^2. \quad (20.1)$$

The area has the same monotonicity property we expect of entropy. Bekenstein therefore proposed that the black hole entropy is proportional to the horizon area,

$$S_{BH} = \alpha \cdot A_{\text{horizon}} \stackrel{\text{Schwarzschild}}{=} 16\pi\alpha M^2, \quad (20.2)$$

where α is a coefficient to be determined. The area A here is the proper area, coming from the angular part of the metric $r^2(\tilde{u}, \tilde{v}) d\Omega^2$, and is therefore coordinate-independent.

20.2 The Qubit Gedanken Experiment

To test the conjecture and fix α , consider lowering a qubit inside a box into a Schwarzschild black hole using a string. The qubit has entropy $S_p = \ln 2$ (it can be spin up or spin down in a mixed state) and mass μ . The question is whether the total entropy $S_{\text{tot}} = S_{BH} + S_{\text{outside}}$ can decrease in this process.

The qubit has four-momentum p_μ with

$$\mu^2 = -g^{\mu\nu} p_\mu p_\nu. \quad (20.3)$$

Since we lower the qubit radially, $p_r = 0$, and the energy measured at infinity is

$$E = \mu \sqrt{1 - \frac{2M}{r}}, \quad (20.4)$$

which is simply gravitational redshift. By energy conservation, the change in black hole mass when the qubit falls in at radius r is $\delta M = E$. It appears that by bringing the qubit arbitrarily close to the horizon we can make δM , and hence δS_{BH} , arbitrarily small, in apparent contradiction with the second law.

The resolution comes from the uncertainty principle. To control the number of particles in the box we need the uncertainty in energy to satisfy $\delta E < \mu$, which means the particle can only be localized to a region larger than its Compton wavelength,

$$\delta l \geq \frac{1}{\mu}. \quad (20.5)$$

The proper distance δl and the coordinate distance δr near the horizon are related by

$$dl^2 = \frac{dr^2}{1 - \frac{2M}{r}} \implies l = \int_{r_h}^{r_h + \delta r} \frac{dr}{\sqrt{1 - \frac{2M}{r}}} \simeq 2\sqrt{2M\delta r}, \quad (20.6)$$

so the constraint $\delta l \geq 1/\mu$ gives $\delta r \geq 1/(8M\mu^2)$. Near the horizon at $r = r_h + \delta r$,

$$E^2 = \mu^2 \left(1 - \frac{2M}{r_h + \delta r}\right) \simeq \mu^2 \frac{\delta r}{2M}, \quad (20.7)$$

and so $E = \mu \sqrt{\delta r/2M}$. Using the minimum δr ,

$$E \geq \mu \cdot \frac{1}{\sqrt{2M}} \cdot \frac{1}{\sqrt{8M\mu^2}} = \frac{1}{4M}, \quad (20.8)$$

giving the fundamental result

$$\boxed{\delta M \geq \frac{1}{4M}}. \quad (20.9)$$

The change in horizon area is then

$$\delta A_h = \delta(16\pi M^2) = 32\pi M \delta M \geq 8\pi > 0, \quad (20.10)$$

in Planck units. This is finite, not zero, so the entropy of the black hole does increase. Dropping one qubit of entropy $\ln 2$ corresponds to a minimum area increase $\delta A_h \geq 8\pi$, so the minimum entropy increase satisfies

$$\frac{\delta S_{BH}}{\delta A_h} = \alpha \implies \alpha = \frac{\ln 2}{8\pi}. \quad (20.11)$$

The correct coefficient, fixed by Hawking's calculation, is $\alpha = 1/4$, giving the Bekenstein-Hawking entropy

$$S_{BH} = \frac{A_h}{4}. \quad (20.12)$$

Our estimate gives the right order of magnitude and confirms the positivity of δS_{tot} .

20.3 The Generalized Second Law

We now verify the generalized second law $\delta S_{\text{tot}} \geq 0$ for two further scenarios.

Geodesic Infall with Angular Momentum

Consider the same setup but now the qubit falls along a geodesic with angular momentum $\tilde{L}$. For a massive particle in the Schwarzschild geometry, the geodesic equation gives

$$\dot{r}^2 = \tilde{E}^2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{\tilde{L}^2}{r^2}\right) \equiv \tilde{E}^2 - V_{\text{eff}}(r), \quad (20.13)$$

where $\tilde{E} = E/\mu$ and $\tilde{L} = L/\mu$ are the specific energy and angular momentum. The energy deposited into the black hole is

$$E = \mu \left[\dot{r}^2 + f(r) \left(1 + \frac{\tilde{L}^2}{r^2}\right) \right]^{1/2}, \quad (20.14)$$

and the changes in mass and angular momentum are $\delta M = E$, $\delta J = \tilde{L}$. The change in black hole entropy is

$$\delta S_{BH} = 32\pi\alpha M^2 \mu \cdot E = 32\pi\alpha M^2 \mu^2 \left[\dot{r}^2 + f(r) \left(1 + \frac{\tilde{L}^2}{r^2}\right) \right]^{1/2}. \quad (20.15)$$

Near the horizon, $f(r_h + \delta r) = \delta r/2M$ and $\dot{r}^2 \sim 1/(8M\mu^2\delta r)$, so

$$E \sim \frac{1}{\sqrt{8M\delta r}} \geq \frac{1}{\sqrt{8M \cdot \frac{1}{8M\mu^2}}} = \mu, \quad (20.16)$$

giving $\delta S_{BH} \geq 32\pi\alpha M^2 \mu^2$. With $\alpha = 1/4$ this gives $\delta S_{BH} \geq 8\pi M^2 \mu^2 \gg \ln 2$, so

$$\delta S_{\text{tot}} = \delta S_{BH} - \ln 2 \geq 8\pi M^2 \mu^2 - \ln 2 > 0, \quad (20.17)$$

regardless of the angular momentum of the infalling particle.

Box of Radiation

Consider a black hole and a box of radiation at temperature T with energy E_{box} and entropy $S_{\text{box}} = E_{\text{box}}/T$. A machine lowers the box to near the horizon, opens it, lets the

radiation fall in, then raises the empty box back. The energy deposited into the black hole at $r = r_h + \delta r$ is

$$\delta M = E_{\text{box}} \sqrt{f(r_h + \delta r)} \simeq E_{\text{box}} \sqrt{\frac{\delta r}{2M}}. \quad (20.18)$$

The change in black hole entropy is

$$\delta S_{BH} = 32\pi\alpha M^2 \delta M = 32\pi\alpha M^2 E_{\text{box}} \sqrt{\frac{\delta r}{2M}}, \quad (20.19)$$

and the entropy lost from outside is $\delta S_{\text{outside}} = -S_{\text{box}} = -E_{\text{box}}/T$. The box must be at least as large as the thermal wavelength of the radiation, $\delta r \geq 1/T$, so

$$\sqrt{\frac{\delta r}{2M}} \geq \frac{1}{\sqrt{2MT}}, \quad (20.20)$$

and therefore

$$\delta S_{BH} \geq \frac{32\pi\alpha M^2 E_{\text{box}}}{\sqrt{2MT}}. \quad (20.21)$$

With $\alpha = 1/4$ and $T_{BH} = 1/(8\pi M)$,

$$\delta S_{BH} \geq \frac{8\pi M^2 E_{\text{box}}}{\sqrt{2MT}} \gg \frac{E_{\text{box}}}{T} = S_{\text{box}}, \quad (20.22)$$

so $\delta S_{\text{tot}} = \delta S_{BH} - S_{\text{box}} \geq 0$. The machine cannot violate the second law.

Charged Black Hole

For a black hole with charge Q , the metric is the Reissner-Nordström metric,

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}, \quad (20.23)$$

with electric potential $A_0 = Q/r$. The horizon is at

$$r_h = M + \sqrt{M^2 - Q^2}, \quad (20.24)$$

and the horizon area is

$$A_h = 4\pi r_h^2 = 4\pi \left(M + \sqrt{M^2 - Q^2} \right)^2. \quad (20.25)$$

Now consider lowering a charged particle with mass μ and charge q to near the horizon. The conserved energy for a charged particle in this background is $E = \mu\sqrt{f(r)} + \dot{r}^2 + qA_0$, so the energy and charge deposited into the black hole are $\delta M = E_\infty$ and $\delta Q = q$. The change in horizon area follows from differentiating $f(r_h) = 0$,

$$\frac{\partial r_h}{\partial M} = \frac{r_h^2}{r_h^2 - Q^2}, \quad \frac{\partial r_h}{\partial Q} = \frac{-Qr_h}{r_h^2 - Q^2}, \quad (20.26)$$

giving

$$\delta A_h = \frac{8\pi r_h^2}{r_h^2 - Q^2} (\delta M - \Phi_h \delta Q), \quad (20.27)$$

where $\Phi_h = Q/r_h$ is the horizon electric potential. The condition $\delta M - \Phi_h \delta Q > 0$ ensures the particle actually falls in and is not repelled. Using the uncertainty principle as before, $\delta r \geq 1/(2\mu)$, and combining with the entropy lost outside,

$$\delta S_{\text{tot}} = \delta S_{BH} - S_{\text{box}} = \frac{2\pi r_h^2}{r_h^2 - Q^2} (\delta M - \Phi_h \delta Q) - \frac{E_{\text{box}}}{T} \geq 0, \quad (20.28)$$

so the generalized second law holds for the Reissner-Nordström black hole as well.

Chapter 21

Quantum Fields in Curved Spacetime

We now develop the general formalism of quantum field theory in curved spacetime. The central lesson is that the notion of a particle, and hence of a vacuum state, is not an intrinsic property of a quantum field but depends on the choice of mode decomposition, which in turn depends on the observer and the geometry. This is a fundamental departure from flat spacetime, where Poincaré invariance singles out a unique vacuum. The Bogoliubov transformation formalism developed here is the technical tool that quantifies this observer-dependence, and it will be applied directly in the following chapters to compute particle production in specific spacetimes. It also provides the conceptual foundation for understanding why the effective potential, which we computed in Parts ?? and ?? assuming a unique vacuum, requires generalization in curved spacetime.

21.1 The General Action and Equation of Motion

We consider a free scalar field on a curved spacetime background, described by the action

$$S = \int d^d x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} (m^2 - \xi R) \phi^2 \right], \quad (21.1)$$

where R is the Ricci scalar and ξ is a non-minimal coupling to the curvature. The special value

$$\xi = \frac{d-2}{4(d-1)} \quad (21.2)$$

is the conformal coupling, for which the massless action acquires an additional symmetry: it becomes invariant under a simultaneous scaling of the metric and the field. The equation of motion is

$$(\square - m^2 + \xi R) \phi = 0, \quad \square = g^{\mu\nu} \nabla_\mu \nabla_\nu. \quad (21.3)$$

In flat spacetime this reduces to the Klein-Gordon equation, which we review first before treating the curved case.

21.2 Quantization in Flat Spacetime

In 4D Minkowski spacetime the field is expanded in plane wave modes,

$$\phi(t, \mathbf{x}) = \int d^3 k \left(u_k(t, \mathbf{x}) a_k + u_k^*(t, \mathbf{x}) a_k^\dagger \right), \quad (21.4)$$

where the positive and negative frequency modes are

$$u_k = \frac{1}{(2\pi)^{3/2}\sqrt{2\omega}} e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}, \quad (21.5)$$

$$u_k^* = \frac{1}{(2\pi)^{3/2}\sqrt{2\omega}} e^{+i\omega t - i\mathbf{k}\cdot\mathbf{x}}, \quad (21.6)$$

with dispersion relation $\omega = +\sqrt{\mathbf{k}^2 + m^2}$. The canonical conjugate momentum is

$$\Pi(t, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}(t, \mathbf{x}), \quad (21.7)$$

and the equal-time commutation relations are

$$[\Pi(t, \mathbf{x}), \phi(t, \mathbf{x}')] = -i \delta(\mathbf{x} - \mathbf{x}'), \quad (21.8)$$

$$[\Pi(t, \mathbf{x}), \Pi(t, \mathbf{x}')] = [\phi(t, \mathbf{x}), \phi(t, \mathbf{x}')] = 0. \quad (21.9)$$

The vacuum $|0\rangle$ is defined by $a_k|0\rangle = 0$ for all k , and n -particle states are built by acting with creation operators $a_k^\dagger$.

The modes are organized by the Klein-Gordon inner product, defined for two solutions φ_1, φ_2 of the field equation as

$$\langle \varphi_1, \varphi_2 \rangle \equiv -i \int_{\text{fixed } t} d^3x (\varphi_1 \partial_t \varphi_2^* - \partial_t \varphi_1 \cdot \varphi_2^*). \quad (21.10)$$

This product is conserved in time, as follows from the Klein-Gordon equation. It is not positive definite on the space of all solutions, but it is positive definite on the space of positive-frequency solutions and negative definite on negative-frequency solutions, with the two spaces orthogonal to each other. The orthonormality relations for the plane wave modes are

$$\langle u_k, u_{k'} \rangle = \delta(\mathbf{k} - \mathbf{k}'), \quad (21.11)$$

$$\langle u_k^*, u_{k'}^* \rangle = -\delta(\mathbf{k} - \mathbf{k}'), \quad (21.12)$$

$$\langle u_k, u_{k'}^* \rangle = 0. \quad (21.13)$$

We verify the first relation explicitly. Computing the time derivatives $\partial_t u_{k'}^* = +i\omega' u_{k'}^*$ and $\partial_t u_k = -i\omega u_k$,

$$\begin{aligned} \langle u_k, u_{k'} \rangle &= -i \int d^3x u_k (+i\omega') u_{k'}^* - u_{k'}^* (-i\omega) u_k \\ &= (\omega + \omega') \int d^3x u_k u_{k'}^* \\ &= \frac{\omega + \omega'}{(2\pi)^3 \cdot 2\sqrt{\omega\omega'}} \int d^3x e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} \\ &= \frac{\omega + \omega'}{2\sqrt{\omega\omega'}} \delta(\mathbf{k} - \mathbf{k}'). \end{aligned} \quad (21.14)$$

Since $\delta(\mathbf{k} - \mathbf{k}') \neq 0$ only when $\mathbf{k} = \mathbf{k}'$, i.e. $\omega = \omega'$, the prefactor evaluates to $2\omega/(2\omega) = 1$, confirming the first relation. The second follows from antisymmetry of the inner product, and the third from the fact that $\langle u_k, u_{k'}^* \rangle \propto (\omega - \omega')\delta(\mathbf{k} + \mathbf{k}')$, which vanishes when $\mathbf{k}' = -\mathbf{k}$ forces $\omega = \omega'$. These relations allow the creation and annihilation operators to be extracted directly from the field,

$$a_k = \langle \phi, u_k \rangle, \quad a_k^\dagger = -\langle \phi, u_k^* \rangle, \quad (21.15)$$

with commutation relations $[a_k, a_{k'}^\dagger] = \langle u_k, u_{k'} \rangle = \delta(\mathbf{k} - \mathbf{k}')$.

21.3 Quantization in Curved Spacetime

In a curved spacetime we choose a spacelike foliation: a family of spacelike hypersurfaces Σ parameterized by a time function $t(x^\mu)$.

The unit future-pointing normal to Σ is

$$n_\mu = \frac{\partial_\mu t}{|\partial_\mu t|}, \quad n_\mu n_\nu g^{\mu\nu} = -1, \quad (21.16)$$

and the induced metric on Σ is

$$(g_\Sigma)_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad (21.17)$$

which satisfies $(g_\Sigma)_{\mu\nu} n^\mu n^\nu = 0$, confirming that it projects vectors onto the surface. The Klein-Gordon inner product generalizes to

$$\langle \varphi_1, \varphi_2 \rangle = -i \int_\Sigma d^{d-1} \sigma \sqrt{g_\Sigma} n^\mu (\varphi_1 \partial_\mu \varphi_2^* - \partial_\mu \varphi_1 \cdot \varphi_2^*). \quad (21.18)$$

This inner product is independent of the choice of spacelike surface Σ , as follows from the Gauss divergence theorem. If Σ_1 and Σ_2 are two spacelike slices bounding a four-volume Ω , then

$$\langle \varphi_1, \varphi_2 \rangle_{\Sigma_1} - \langle \varphi_1, \varphi_2 \rangle_{\Sigma_2} = \int_\Omega d^4 x \sqrt{g} \nabla^\mu (\varphi_1^* \nabla_\mu \varphi_2 - \nabla_\mu \varphi_1^* \cdot \varphi_2) = 0, \quad (21.19)$$

where the last equality follows because both φ_1 and φ_2 satisfy the Klein-Gordon equation. The inner product therefore has the same properties as in flat space, provided the contribution from spatial infinity is negligible.

The field is expanded in a complete set of modes $\{u_i, u_i^*\}$ satisfying the same orthonormality relations as before,

$$\phi(x) = \sum_i \left(u_i(x) a_i + u_i^*(x) a_i^\dagger \right), \quad (21.20)$$

$$\langle u_i, u_j \rangle = \delta_{ij}, \quad \langle u_i^*, u_j^* \rangle = -\delta_{ij}, \quad \langle u_i, u_j^* \rangle = 0. \quad (21.21)$$

These are no longer plane waves; their form depends on the spacetime geometry. The canonical momentum is

$$\Pi = \frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \sqrt{g_\Sigma} n^\mu \partial_\mu \phi, \quad (21.22)$$

and the equal-time commutation relations take the same form as in flat space. The creation and annihilation operators are extracted as

$$a_i = \langle \phi, u_i \rangle = -i \int d\sigma (\sqrt{g_\Sigma} n^\mu \partial_\mu u_i^* \cdot \phi - u_i^* \Pi), \quad (21.23)$$

$$a_i^\dagger = -\langle \phi, u_i^* \rangle = i \int d\sigma (\sqrt{g_\Sigma} n^\mu \partial_\mu u_i \cdot \phi - u_i \Pi), \quad (21.24)$$

satisfying $[a_i, a_j^\dagger] = \delta_{ij}$. The vacuum $|0\rangle$ is defined by $a_i|0\rangle = 0$.

21.4 Ambiguity in the Notion of Particles and Bogoliubov Transformations

In Minkowski spacetime the construction of a vacuum state and Fock space is unambiguous because there is a natural set of modes, the plane waves (??), singled out by the Poincaré symmetry of the background. The vector $\partial/\partial t$ is a Killing vector of Minkowski space, orthogonal to the spacelike hypersurfaces $t = \text{const}$, and the plane wave modes are its eigenfunctions with eigenvalues $-i\omega$ for $\omega > 0$. The vacuum is then the unique state invariant under the Poincaré group.

In curved spacetime the Poincaré group is no longer a symmetry, and in general there are no Killing vectors at all with which to define positive-frequency modes. In special cases, such as de Sitter space or the Schwarzschild exterior, there may be Killing vectors associated with particular symmetries, and one can define natural coordinates adapted to them. But even when such coordinates exist, they do not exhibit the same privileged status as the inertial coordinates of Minkowski space. The spirit of general covariance is that no coordinate system is physically preferred, and so no mode decomposition is intrinsically more natural than any other. The ambiguity in the choice of modes is therefore not a technical inconvenience but a fundamental feature of quantum field theory in curved spacetime.

The mode decomposition above is not unique. Consider a second complete set of modes $\{\tilde{u}_j, \tilde{u}_j^*\}$ with the same orthonormality properties, giving a second quantization with operators $\hat{a}_j$ and vacuum $|\tilde{0}\rangle$ defined by $\hat{a}_j|\tilde{0}\rangle = 0$. Since both sets are complete, we can expand one in terms of the other,

$$\tilde{u}_j(x) = \sum_i (\alpha_{ji} u_i(x) + \beta_{ji} u_i^*(x)), \quad (21.25)$$

$$\tilde{u}_j^*(x) = \sum_i (\alpha_{ji}^* u_i^* + \beta_{ji}^* u_i), \quad (21.26)$$

where the Bogoliubov coefficients are

$$\alpha_{ji} = \langle \tilde{u}_j, u_i \rangle, \quad \beta_{ji} = -\langle \tilde{u}_j, u_i^* \rangle. \quad (21.27)$$

The orthonormality of both sets implies the relations

$$\sum_j (\alpha_{ji}^* \alpha_{jk} - \beta_{ji} \beta_{jk}^*) = \delta_{ik}, \quad (21.28)$$

$$\sum_j (\alpha_{ji}^* \beta_{jk} - \beta_{ji} \alpha_{jk}^*) = 0, \quad (21.29)$$

and their transposes. These can be used to invert the expansion,

$$u_i = \sum_j (\alpha_{ji} \tilde{u}_j - \beta_{ji} \tilde{u}_j^*). \quad (21.30)$$

The two sets of operators are related by the Bogoliubov transformation,

$$a_i = \sum_j (\alpha_{ji} \hat{a}_j + \beta_{ji}^* \hat{a}_j^\dagger), \quad (21.31)$$

$$\hat{a}_j = \sum_i (\alpha_{ji}^* a_i - \beta_{ji}^* a_i^\dagger). \quad (21.32)$$

The two quantizations are inequivalent precisely when the β coefficients are nonzero. To see what this means physically, compute the expectation value of the number operator $\hat{N}_j = \hat{a}_j^\dagger \hat{a}_j$ in the vacuum $|0\rangle$ of the first quantization,

$$\begin{aligned} \langle 0 | \hat{a}_j^\dagger \hat{a}_j | 0 \rangle &= \langle 0 | \sum_i \left(\alpha_{ji} a_i^\dagger - \beta_{ji} a_i \right) \sum_k \left(\alpha_{jk}^* a_k - \beta_{jk}^* a_k^\dagger \right) | 0 \rangle \\ &= \sum_{i,k} \beta_{ji} \beta_{jk}^* \underbrace{\langle 0 | a_i a_k^\dagger | 0 \rangle}_{\delta_{ik}} \\ &= \sum_i |\beta_{ji}|^2. \end{aligned} \tag{21.33}$$

The state $|0\rangle$, which contains no particles of type a , contains $\sum_i |\beta_{ji}|^2$ particles of type $\hat{a}_j$. This is the key result: different choices of modes, corresponding to different observers or different regions of spacetime, lead to genuinely inequivalent notions of particle and vacuum. The particle content of a state is observer-dependent in curved spacetime, in sharp contrast with the situation in flat spacetime where all inertial observers agree.

In summary, the field equation $(\square - m^2 + \xi R)\phi = 0$ in curved spacetime admits multiple inequivalent quantizations related by Bogoliubov transformations,

$$\phi = \sum_i \left(u_i a_i + u_i^* a_i^\dagger \right) = \sum_j \left(\tilde{u}_j \hat{a}_j + \tilde{u}_j^* \hat{a}_j^\dagger \right), \tag{21.34}$$

and the vacuum of one observer contains particles as seen by another whenever $\beta_{ji} \neq 0$. The physical consequences of this ambiguity are explored in the following chapters.

Chapter 22

Particle Production and the Unruh Effect

With the general formalism in place, we study two concrete examples that isolate the essential physics of particle production in the simplest possible settings, before tackling the full black hole problem. The moving mirror model shows that particle production can occur even in flat spacetime, driven purely by a time-dependent boundary condition. The Unruh effect shows that particle production is observer-dependent: the Minkowski vacuum contains a thermal distribution of particles as seen by a uniformly accelerating observer. Both of these results are more than analogies. The logarithmic relation between the in and out coordinates in the mirror problem is precisely the same mathematical structure that appears near a black hole horizon, and the Unruh temperature $T = a/2\pi$ is the local limit of the Hawking temperature experienced by a static observer near the horizon. Understanding these simpler cases in detail makes the derivation of Hawking radiation in the next chapter both natural and transparent.

22.1 The Moving Mirror

We now consider a concrete example of particle production in flat spacetime, where the non-trivial physics comes not from curvature but from boundary conditions. Consider a massless scalar field in 2D flat spacetime with metric

$$ds^2 = -dt^2 + dx^2 = -du dv, \quad u = t - x, \quad v = t + x. \quad (22.1)$$

Half the space is bounded by a mirror with trajectory $x = z(t)$. The field satisfies a Dirichlet boundary condition at the mirror,

$$\phi|_{x=z(t)} = 0, \quad (22.2)$$

and the field equation $\partial^2 \phi / \partial u \partial v = 0$ has the general solution $\phi = f(u) + g(v)$, a sum of left-moving and right-moving waves. The boundary condition enforces a relation between f and g , so that the motion of the mirror mixes the two.

We choose in-modes that in the past were standing waves,

$$\phi_{\omega}^{\text{in}} = \frac{1}{\sqrt{4\pi\omega}} (e^{-i\omega v} - e^{-i\omega u}), \quad t < 0, \quad (22.3)$$

where $e^{-i\omega v}$ is the incoming wave and $e^{-i\omega u}$ is the reflected wave. On the boundary $u = v$ and the two terms cancel, satisfying the Dirichlet condition. For $t > 0$, the mirror has moved and the reflected wave acquires a phase,

$$\phi_\omega^{\text{in}} = \frac{1}{\sqrt{4\pi\omega}} (e^{-i\omega v} - e^{-i\omega v_0(u)}), \quad t > 0, \quad (22.4)$$

where $v_0(u)$ is determined by the mirror trajectory. If τ_u is the proper time at which the mirror is at position $u = \tau_u - z(\tau_u)$, then $v_0 = \tau_u + z(\tau_u) = 2\tau_u - u$, giving

$$\phi_\omega^{\text{in}} = \frac{1}{\sqrt{4\pi\omega}} (e^{-i\omega v} - e^{-i\omega(2\tau_u - u)}). \quad (22.5)$$

This mode is natural for an observer in the past (simple $e^{-i\omega v}$ dependence) but complicated for an observer in the future. We therefore also define out-modes natural for the future observer,

$$\phi_{\tilde{\omega}}^{\text{out}} = \frac{1}{\sqrt{4\pi\tilde{\omega}}} (e^{-i\tilde{\omega}u} - e^{-i\tilde{\omega}(2\tau_v - v)}), \quad (22.6)$$

where $\tau_v + z(\tau_v) = v$. The Bogoliubov coefficient β is computed on the surface $t = 0$ where the in-modes are still simple,

$$\beta_{\tilde{\omega}\omega} = -\langle \phi_{\tilde{\omega}}^{\text{out}}, (\phi_\omega^{\text{in}})^* \rangle = -i \int_0^\infty dx \left(\phi_{\tilde{\omega}}^{\text{out}} \dot{\phi}_\omega^{\text{in}*} - \dot{\phi}_{\tilde{\omega}}^{\text{out}} \phi_\omega^{\text{in}*} \right) \Big|_{t=0}. \quad (22.7)$$

On this surface $u = -x$ and $v = x$, giving

$$\beta_{\tilde{\omega}\omega} = - \int_0^\infty \frac{dx}{2\pi} \sqrt{\frac{\omega}{\tilde{\omega}}} (e^{i(\tilde{\omega}+\omega)x} + e^{i(\tilde{\omega}-\omega)x-2i\tilde{\omega}\tau_x}), \quad (22.8)$$

where $\tau_x + z(\tau_x) = x$. The number of produced particles is

$$\frac{dN}{d\tilde{\omega}} = \int_0^\infty |\beta_{\tilde{\omega}\omega}|^2 d\omega. \quad (22.9)$$

Constant Velocity Mirror

When the mirror moves with constant velocity $z = -\alpha t$, $|\alpha| < 1$, we have $\tau_v = v/(1 - \alpha)$ and so $\tau_x = x/(1 - \alpha)$. Substituting into (22.8) and regularizing the oscillating integrals by $e^{-\epsilon x}$,

$$\beta_{\tilde{\omega}\omega} = \frac{\sqrt{\tilde{\omega}\omega}}{2\pi} \left(\frac{2\alpha}{1 - \alpha} \right) \frac{1}{(\tilde{\omega} + \omega)} \frac{1}{\left(\frac{1+\alpha}{1-\alpha} \tilde{\omega} + \omega \right)}, \quad (22.10)$$

where we used $\int_0^\infty dx e^{iAx - \epsilon x} = i/A$. Integrating $|\beta_{\tilde{\omega}\omega}|^2$ over ω ,

$$\boxed{\frac{dN}{d\tilde{\omega}} = \frac{1}{\pi^2 \tilde{\omega}} \left(\frac{1}{4\alpha} \ln \frac{1+\alpha}{1-\alpha} - \frac{1}{2} \right)}. \quad (22.11)$$

This result is finite but non-thermal, reflecting the fact that a uniformly moving mirror produces particles but not a thermal spectrum.

Accelerating Mirror

Consider now a mirror that starts at rest, accelerates, and asymptotically approaches a null ray, with trajectory

$$z = B - t - Ae^{-2Kt}, \quad t \rightarrow \infty. \quad (22.12)$$

Tracing out-modes backward in time, the reflected waves never go beyond the point $v = B$. The relation $\tau_v = -(1/2K) \ln((B-v)/A)$ for $v < B$ gives

$$\beta_{\tilde{\omega}\omega} \simeq -\frac{1}{2\pi} \sqrt{\frac{\omega}{\tilde{\omega}}} \int_0^\infty dy e^{i(\tilde{\omega}-\omega)(B-y)} \left(\frac{y}{A}\right)^{i\tilde{\omega}/K}, \quad (22.13)$$

where we substituted $y = B - x$ and extended the upper limit to infinity since only the behavior near $y = 0$ matters. Using the Gamma function integral $\Gamma(z) = \int_0^\infty dy y^{z-1} e^{-y}$,

$$|\beta_{\tilde{\omega}\omega}|^2 = \frac{1}{(K\pi)^2} \frac{\omega}{\tilde{\omega}} e^{-\pi\tilde{\omega}/2K} \Gamma\left(1 + \frac{i\tilde{\omega}}{K}\right) \Gamma\left(1 - \frac{i\tilde{\omega}}{K}\right). \quad (22.14)$$

Using $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$ and $(-i)^{-i\tilde{\omega}/K} = e^{-\pi\tilde{\omega}/2K}$,

$$|\beta_{\tilde{\omega}\omega}|^2 = \frac{1}{2\pi K} \frac{\omega}{(\omega - \tilde{\omega})^2} \frac{1}{e^{2\pi\tilde{\omega}/K} - 1}. \quad (22.15)$$

Integrating over ω and isolating the divergence at high frequency,

$$\frac{dN}{d\tilde{\omega}} = \frac{1}{2\pi K} \frac{1}{e^{2\pi\tilde{\omega}/K} - 1} \int \frac{d\omega}{\omega}. \quad (22.16)$$

The prefactor is a Bose-Einstein distribution with temperature

$$T = \frac{K}{2\pi}. \quad (22.17)$$

The integral diverges, but this represents a continuous flux rather than a divergent total number. If the mirror stops after retarded time u_f , the integral cuts off at $\omega_{\max} \sim e^{Ku_f}/A$ and

$$\frac{dN}{d\tilde{\omega} du_f} = \frac{1}{2\pi} \cdot \frac{1}{e^{2\pi\tilde{\omega}/K} - 1}, \quad (22.18)$$

a steady thermal flux at temperature $T = K/2\pi$. This is the prototype of Hawking radiation, as we shall see in the next chapter.

22.2 The Unruh-DeWitt Detector

We now set up a general framework for understanding what a moving observer actually measures. Consider a detector with internal Hamiltonian H_D , following a worldline $x^\mu(\tau)$ and coupled to a scalar field $\varphi(x)$ via the interaction

$$S_{\text{int}} = \lambda \int_{-\infty}^{\infty} d\tau \varphi(x(\tau)) \hat{D}(\tau), \quad (22.19)$$

where $\hat{D}(\tau) = e^{iH_D\tau}\hat{D}(0)e^{-iH_D\tau}$ is the detector operator in the interaction picture and λ is the coupling strength. To first order in perturbation theory, the transition amplitude from the ground state $|E_0, 0\rangle$ to an excited state $|E, \psi\rangle$ is

$$\mathcal{A} = i\lambda \int_{-\infty}^{\infty} d\tau \langle E | \hat{D}(\tau) | E_0 \rangle \langle \psi | \varphi(x(\tau)) | 0 \rangle. \quad (22.20)$$

Summing $|\mathcal{A}|^2$ over all final field states $|\psi\rangle$ and using completeness,

$$\sum_{\psi} \langle 0 | \varphi(x(\tau_1)) | \psi \rangle \langle \psi | \varphi(x(\tau_2)) | 0 \rangle = \langle 0 | \varphi(x(\tau_1)) \varphi(x(\tau_2)) | 0 \rangle \equiv G^+(x(\tau_1), x(\tau_2)), \quad (22.21)$$

the total transition probability is

$$P_{EE_0} = \lambda^2 \left| \langle E | \hat{D}(0) | E_0 \rangle \right|^2 \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} d\tau_2 e^{-i\Delta E(\tau_1 - \tau_2)} G^+(x(\tau_1), x(\tau_2)), \quad (22.22)$$

where $\Delta E = E - E_0$. The detector response function is defined as

$$F(\Delta E) = \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} d\tau_2 e^{-i\Delta E(\tau_1 - \tau_2)} G^+(x(\tau_1), x(\tau_2)). \quad (22.23)$$

For stationary trajectories, G^+ depends only on the proper time difference $\tau = \tau_1 - \tau_2$, and changing variables to $\tau = \tau_1 - \tau_2$ and $T = (\tau_1 + \tau_2)/2$ shows that the transition rate per unit proper time is

$$\frac{dP_{EE_0}}{d\tau} = \lambda^2 \left| \langle E | \hat{D}(0) | E_0 \rangle \right|^2 \int_{-\infty}^{\infty} d\tau e^{-i\Delta E\tau} g(\tau), \quad (22.24)$$

where $g(\tau) = G^+(x(\tau_1), x(\tau_2))$ depends only on $\tau_1 - \tau_2$. The response function $F(\Delta E)$ is therefore the Fourier transform of the Wightman function evaluated along the detector's worldline.

22.3 The Accelerating Detector and the Unruh Effect

We now apply the detector formalism to a uniformly accelerating observer in 4D Minkowski spacetime. The trajectory is

$$x = y = 0, \quad z = \sqrt{t^2 + B^2}, \quad (22.25)$$

with parametric form

$$t = B \sinh \frac{\tau}{B}, \quad z = B \cosh \frac{\tau}{B}, \quad (22.26)$$

and constant proper acceleration $a^\mu a_\mu = 1/B^2$.

The Wightman function in the Minkowski vacuum is

$$G^+(x, x') = \langle 0 | \varphi(x) \varphi(x') | 0 \rangle = \frac{1}{4\pi^2} \frac{1}{(t - t' - i\varepsilon)^2 - |\mathbf{x} - \mathbf{x}'|^2}. \quad (22.27)$$

Substituting the accelerating trajectory, one finds

$$G^+(x(\tau), x(\tau')) = -\frac{1}{16\pi^2 B^2} \frac{1}{\sinh^2 \left(\frac{\tau - \tau'}{2B} - i\varepsilon \right)}. \quad (22.28)$$

The detector response function is therefore

$$F(\Delta E) = \int_{-\infty}^{\infty} \left(-\frac{d\tau}{16\pi^2 B^2} \right) \frac{e^{-i\Delta E\tau}}{\sinh^2\left(\frac{\tau}{2B} - i\varepsilon\right)}. \quad (22.29)$$

We evaluate this by contour integration. The poles of $1/\sinh^2(\tau/2B)$ occur at $\tau/2B = i\pi n$, i.e. $\tau = 2\pi i B n$ for integer n . For $\Delta E > 0$ we close the contour in the lower half plane, picking up poles at $\tau = -2\pi i B n$ for $n = 1, 2, \dots$ (the $i\varepsilon$ prescription shifts the poles away from $n = 0$),

$$F(\Delta E) = -\frac{1}{16\pi^2 B^2} \sum_{n=1}^{\infty} (-2\pi i) \operatorname{Res}_{\tau=-2\pi i B n} \frac{e^{-i\Delta E\tau}}{\sinh^2\left(\frac{\tau}{2B}\right)}. \quad (22.30)$$

Computing the residues and summing the geometric series,

$$F(\Delta E) = \frac{\Delta E}{2\pi} \sum_{n=1}^{\infty} e^{-2\pi B \Delta E n} = \boxed{\frac{\Delta E}{2\pi} \cdot \frac{1}{e^{2\pi B \Delta E} - 1}}, \quad (22.31)$$

which is a Planck distribution with temperature

$$T_U = \frac{1}{2\pi B} = \frac{a}{2\pi}, \quad (22.32)$$

where $a = 1/B$ is the proper acceleration. This is the Unruh effect: a uniformly accelerating observer in the Minkowski vacuum measures a thermal bath at temperature $T_U = a/2\pi$. An inertial observer in the same state sees no particles at all; the thermal nature of the state is entirely an effect of the acceleration.

The Unruh effect has a deep connection to the existence of a causal horizon for the accelerating observer. The observer in the right Rindler wedge cannot receive signals from the left wedge, and this causal inaccessibility is responsible for the thermal character of the state, in exact analogy with the Hawking effect for black holes, to which we turn in the next chapter.

Chapter 23

Hawking Radiation

We now bring together all the tools developed in this part to derive the central result: a black hole formed by gravitational collapse emits a steady thermal flux of radiation at the Hawking temperature $T_H = 1/8\pi M$. The derivation proceeds by tracing the relation between modes defined on past null infinity and modes defined on future null infinity, through the collapsing geometry. The logarithmic relationship between the in and out coordinates near the horizon, which we already encountered in the accelerating mirror, is the key ingredient. The result connects back to the thermodynamic picture of Chapter 20: the Hawking temperature is precisely the temperature one would assign to a black hole of entropy $S_{BH} = A/4$ using the thermodynamic relation $T = dM/dS$. It also connects forward to the cosmological context of Part ??: the same mechanism, quantum fluctuations near a causal horizon, is responsible for the Gibbons-Hawking temperature of de Sitter space and ultimately for the primordial density perturbations seeded during inflation.

23.1 The Thermal Spectrum from Bogoliubov Coefficients

The thermal character of Hawking radiation can be derived more directly from the general Bogoliubov transformation formalism of Chapter 21, without going through the detailed mode calculation. We sketch this argument here.

In Minkowski spacetime, the second quantization of a free scalar field is based on a distinguished inertial frame, which determines the spacetime slicing and the decomposition into positive and negative frequency modes. The most general classical solution is

$$\varphi^{\text{cl}}(\mathbf{x}, t) = \sum_{i, \omega} [a_{\omega i} f_{\omega i}(\mathbf{x}, t) + a_{\omega i}^* f_{\omega i}^*(\mathbf{x}, t)], \quad f_{\omega i}(\mathbf{x}, t) = e^{i\omega t} \varphi_{\omega i}(\mathbf{x}), \quad (23.1)$$

and in the quantum theory the coefficients $a_{\omega i}$ are promoted to operators with vacuum $a_{\omega i}|0\rangle = 0$.

In curved spacetime, general coordinate invariance means there is no distinguished Lorentz frame and hence no unique vacuum state. Two observers $\mathcal{O}_1$ and $\mathcal{O}_2$ at different points will in general have different notions of positive frequency, and the vacuum of $\mathcal{O}_1$ will contain particles from the point of view of $\mathcal{O}_2$. This is Parker particle production.

Concretely, the field can be expanded in the modes of either observer,

$$\varphi(\mathbf{x}, t) = \sum_k \left(a_k f_k(\mathbf{x}, t) + a_k^\dagger f_k^*(\mathbf{x}, t) \right), \quad (23.2)$$

$$\varphi(\mathbf{x}, t) = \sum_\ell \left(b_\ell g_\ell(\mathbf{x}, t) + b_\ell^\dagger g_\ell^*(\mathbf{x}, t) \right), \quad (23.3)$$

where the inner product is

$$\langle f, g \rangle = \frac{1}{2i} \int_{t=\text{const}} d^3x \left(\bar{f} \dot{g} - \dot{\bar{f}} g \right). \quad (23.4)$$

The modes g_ℓ are in general non-trivial linear combinations of f_k and f_k^* ,

$$g_\ell = \sum_k (\alpha_{\ell k} f_k + \beta_{\ell k} f_k^*), \quad (23.5)$$

with Bogoliubov coefficients

$$\alpha_{\ell k} = \langle f_k, g_\ell \rangle, \quad \beta_{\ell k} = -\langle f_k^*, g_\ell \rangle. \quad (23.6)$$

The matrix $M = \begin{pmatrix} \alpha & \beta \\ \beta^* & \alpha^* \end{pmatrix}$ is unitary, giving the constraint $\alpha\alpha^* - \beta\beta^* = 1$. Expanding the field in both bases and comparing coefficients,

$$a_k = \sum_\ell \left(b_\ell \alpha_{\ell k} + b_\ell^\dagger \beta_{\ell k}^* \right), \quad (23.7)$$

$$b_\ell = \sum_k \left(a_k \alpha_{\ell k}^* - a_k^\dagger \beta_{\ell k} \right). \quad (23.8)$$

The expectation value of the number operator $N_k = b_k^\dagger b_k$ in the vacuum $|0\rangle$ of $\mathcal{O}_1$ is

$$\begin{aligned} \langle 0|N_k|0\rangle &= \langle 0|b_k^\dagger b_k|0\rangle \\ &= \langle 0|\sum_\ell \left(-a_\ell \beta_{\ell k} + a_\ell^\dagger \alpha_{\ell k} \right) \sum_{\ell'} \left(a_{\ell'} \alpha_{\ell' k}^* - a_{\ell'}^\dagger \beta_{\ell' k}^* \right) |0\rangle \\ &= \sum_{\ell, \ell'} \beta_{\ell k} \beta_{\ell' k}^* \langle 0|a_\ell a_{\ell'}^\dagger|0\rangle \\ &= \sum_{\ell, \ell'} \beta_{\ell k} \beta_{\ell' k}^* \langle 0|\left(\delta_{\ell\ell'} + a_{\ell'}^\dagger a_\ell \right)|0\rangle \\ &= \sum_\ell |\beta_{\ell k}|^2, \end{aligned} \quad (23.9)$$

where only the term involving $[a_\ell, a_{\ell'}^\dagger] = \delta_{\ell\ell'}$ survives. The vacuum of $\mathcal{O}_1$ therefore contains $\sum_\ell |\beta_{\ell k}|^2$ particles of type b_k .

The thermal nature of Hawking radiation follows from a special property of the Bogoliubov coefficients near a black hole horizon. As shown in Hawking's original derivation, the logarithmic relation between the in and out coordinates near the horizon implies

$$|\alpha_{k\ell}|^2 = e^{\omega_k/T_H} |\beta_{k\ell}|^2, \quad \forall k, \ell, \quad (23.10)$$

where ω_k is the energy of mode k and T_H is the Hawking temperature. Using the unitarity constraint $\sum_{\ell} (|\alpha_{k\ell}|^2 - |\beta_{k\ell}|^2) = 1$,

$$\sum_{\ell} |\beta_{k\ell}|^2 (e^{\omega_k/T_H} - 1) = 1, \quad (23.11)$$

and therefore

$$\langle 0|N_k|0\rangle = \sum_{\ell} |\beta_{k\ell}|^2 = \frac{1}{e^{\omega_k/T_H} - 1}, \quad (23.12)$$

a Planck distribution at the Hawking temperature T_H . The black hole vacuum, as seen by an asymptotic observer, contains a thermal distribution of particles at every frequency.

23.2 Particle Production in Gravitational Collapse

We now derive the most important consequence of quantum field theory in curved spacetime: the thermal radiation emitted by a black hole, discovered by Hawking in 1974. We model the gravitational collapse by a thin spherical shell of mass M collapsing to form a black hole. By Birkhoff's theorem, the metric outside the shell is the Schwarzschild metric,

$$ds_{\text{out}}^2 = -f(r) du dv, \quad f(r) = 1 - \frac{2M}{r}, \quad u, v = t \mp r_*, \quad (23.13)$$

and the metric inside the shell is flat,

$$ds_{\text{in}}^2 = -d\tilde{u} d\tilde{v}, \quad \tilde{u}, \tilde{v} = \tilde{t} \mp r. \quad (23.14)$$

The history of the collapsing matter does not affect the particle production; what matters is the relationship between the coordinates on either side of the shell near the moment of horizon crossing.

The 2D Model

We work in the 2D reduction, suppressing the angular coordinates. The shell trajectory is $r = r_s(\tilde{t})$, and matching the metrics across the shell gives

$$\frac{du}{d\tilde{u}} = \frac{\sqrt{(1 - \dot{r}_s^2)f(r_s) + \dot{r}_s^2} - \dot{r}_s}{f(r_s)(1 - \dot{r}_s)}, \quad (23.15)$$

$$\frac{dv}{d\tilde{v}} = \frac{\sqrt{(1 - \dot{r}_s^2)f(r_s) + \dot{r}_s^2} + \dot{r}_s}{f(r_s)(1 + \dot{r}_s)}, \quad (23.16)$$

where $\dot{r}_s = dr_s/d\tilde{t}$. As the shell crosses the horizon, $f(r_s) \rightarrow 0$. Writing $r_s(\tilde{t}) = r_h + A(\tilde{t} - \tilde{t}_h) + \dots$ with $0 > A > -1$ and $f(r) \simeq (r - r_h)/\kappa_h$ near the horizon, one finds that $dv/d\tilde{v}$ remains regular at horizon crossing,

$$v = \text{const} + \alpha \tilde{v} \Big|_{\text{horizon crossing}}, \quad (23.17)$$

so v is a smooth function of $\tilde{v}$ there. By contrast, $du/d\tilde{u}$ diverges logarithmically,

$$u = -\kappa_h \ln |\tilde{u} - \tilde{u}_h| + \text{const}, \quad (23.18)$$

where $\kappa_h = f'(r_h)/2$ is the surface gravity. This logarithmic relation is the key to the thermal spectrum.

The natural in-modes on past null infinity $\mathcal{I}^-$ are

$$\varphi_\omega^{\text{in}} = \frac{1}{\sqrt{4\pi\omega}} (e^{-i\omega v} - e^{-i\omega u}), \quad (23.19)$$

and the natural out-modes for an asymptotic observer on future null infinity $\mathcal{I}^+$ are

$$\varphi_{\tilde{\omega}}^{\text{out}} = \frac{1}{\sqrt{4\pi\tilde{\omega}}} (e^{-i\tilde{\omega}u} - e^{-i\tilde{\omega}u_0(v)}), \quad (23.20)$$

where $u_0(v)$ is the image of the out-ray under reflection from the center. Using the logarithmic relation,

$$u_0 = -\kappa_h \ln |v - v_h| + \text{const}, \quad (23.21)$$

so the out-modes contain the same logarithmic phase as the accelerating mirror. The computation of the Bogoliubov coefficients therefore proceeds identically to the mirror case with the identification $K \rightarrow \kappa_h$, giving a thermal flux at temperature

$$T_H = \frac{\kappa_h}{2\pi} = \frac{f'(r_h)}{4\pi} = \frac{1}{8\pi M}. \quad (23.22)$$

This is the Hawking temperature. The particle production rate per unit retarded time is

$$\frac{dN}{d\tilde{\omega} du} = \frac{1}{2\pi} \cdot \frac{1}{e^{\tilde{\omega}/T_H} - 1}, \quad (23.23)$$

a steady thermal flux that persists as long as the black hole exists. An observer at $\mathcal{I}^+$ will keep receiving this radiation even though no signal can escape from inside the horizon; the radiation originates from quantum fluctuations just outside the horizon, not from inside it.

The 4D Model

In four dimensions the analysis is more involved due to the angular structure. The field modes take the form

$$\varphi_{\omega lm} = \frac{Y_{lm}(\theta, \phi)}{r} R_{\omega l}(r) e^{-i\omega t}, \quad (23.24)$$

where the radial function satisfies a Schrödinger-like equation,

$$-\frac{d^2 R_{\omega l}}{dr_*^2} + V_{\text{eff}}(r_*) R_{\omega l} = \omega^2 R_{\omega l}, \quad V_{\text{eff}}(r_*) = \left(\frac{l(l+1)}{r^2} + \frac{2M}{r^3} \right) \left(1 - \frac{2M}{r} \right). \quad (23.25)$$

The potential vanishes both at the horizon ($r_* \rightarrow -\infty$) and at infinity ($r_* \rightarrow +\infty$), so the modes are free waves in both asymptotic regions. A wave incident from infinity partially reflects off the potential barrier and partially transmits through to the horizon. The transmission coefficient $\Gamma_{\tilde{\omega}l}$, called the greybody factor, modifies the thermal flux,

$$\frac{dN}{d\tilde{\omega} dt} = \frac{\Gamma_{\tilde{\omega}l}}{2\pi} \cdot \frac{1}{e^{\tilde{\omega}/T_H} - 1}. \quad (23.26)$$

The greybody factor satisfies $\Gamma_{\tilde{\omega}l} \rightarrow 0$ as $l \rightarrow \infty$ or as $\tilde{\omega} \rightarrow 0$ at fixed l , so low-frequency and high-angular-momentum modes are suppressed. The radiation is therefore not exactly thermal, but it is in detailed balance with a thermal bath at temperature T_H : the emission and absorption coefficients are equal by time-reversal symmetry, so a black hole in a thermal bath at T_H would be in equilibrium.

23.3 The Eternal Black Hole and the Choice of Vacuum

For the eternal (maximally extended) Schwarzschild black hole, the metric in Kruskal coordinates is

$$ds^2 = -\frac{2M}{r} e^{-r/2M} d\bar{u} d\bar{v}, \quad \bar{u} = -4M e^{-u/4M}, \quad \bar{v} = 4M e^{v/4M}. \quad (23.27)$$

The physical content of the theory depends crucially on the choice of quantum vacuum state, and there are three natural choices.

The Boulware Vacuum

The Boulware vacuum $|0\rangle_B$ is defined by the modes

$$\varphi_{\omega,R}^B = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega u}, \quad \varphi_{\omega,L}^B = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega v}, \quad (23.28)$$

which are plane waves for an asymptotically distant observer and correspond to the natural vacuum seen by a static observer far from the black hole. However, these modes are defined only in region I of the Kruskal diagram and are singular at the horizon. The Wightman function in the Boulware vacuum is

$$G_B^+ = -\frac{1}{4\pi} \ln [(u - u' - i\varepsilon)(v - v' - i\varepsilon)], \quad (23.29)$$

which has no poles in the lower half plane. A static detector at fixed r therefore measures zero particle flux in the Boulware vacuum, $F_B(\Delta E) = 0$.

The Hartle-Hawking Vacuum

The Hartle-Hawking vacuum $|0\rangle_H$ is defined by modes that are regular everywhere except at the singularity,

$$\varphi_{\omega,R}^H = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega \bar{u}}, \quad \varphi_{\omega,L}^H = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega \bar{v}}. \quad (23.30)$$

These modes are smooth across the horizon and provide a regular quantization for the full extended spacetime. The Bogoliubov coefficients between the Hartle-Hawking and Boulware modes are

$$\beta_{\tilde{\omega}\omega,RR} = -\langle \varphi_{\tilde{\omega}R}^B, (\varphi_{\omega,R}^H)^* \rangle = \frac{-i}{2\pi\sqrt{\omega\tilde{\omega}}} \left(\frac{i}{4M\omega} \right)^{i4M\tilde{\omega}} \Gamma(1 + i4M\tilde{\omega}), \quad (23.31)$$

which are nonzero and describe a thermal flux between the black hole and its environment. The Wightman function is

$$G_H^+(t, r, t', r') = -\frac{1}{4\pi} \ln [(\bar{u} - \bar{u}' - i\varepsilon)(\bar{v} - \bar{v}' - i\varepsilon)]. \quad (23.32)$$

For a static detector at fixed r , substituting the trajectory gives

$$\begin{aligned} G_H^+(t, r, t', r) &= -\frac{1}{4\pi} \ln \left[\cosh \left(\frac{t - t' - i\varepsilon}{4M} \right) - 1 \right] + \tilde{G}(r) \\ &= -\frac{1}{4\pi} \ln \left[\cosh \left(\frac{\tau - \tau' - i\varepsilon}{4M\sqrt{1 - \frac{2M}{r}}} \right) - 1 \right] + \tilde{G}(r), \end{aligned} \quad (23.33)$$

where we used $d\tau = \sqrt{1 - 2M/r} dt$. The Wightman function has periodic poles in the imaginary time direction with period

$$\Delta\tau = i \cdot 8\pi M \sqrt{1 - \frac{2M}{r}}, \quad (23.34)$$

so the detector measures a temperature

$$T = \frac{1}{8\pi M \sqrt{1 - \frac{2M}{r}}}. \quad (23.35)$$

An observer closer to the horizon measures a higher temperature due to the blueshift factor $\sqrt{1 - 2M/r}^{-1}$, diverging at the horizon itself. This is consistent with the Unruh effect: a static observer in Schwarzschild spacetime is accelerated with respect to a freely falling frame, and the acceleration grows without bound as the observer approaches the horizon.

For a freely falling observer crossing the horizon at fixed Kruskal coordinate $\bar{r} = (\bar{v} - \bar{u})/2$, the Wightman function is

$$G_H^+(\bar{t}, \bar{r}, \bar{t}', \bar{r}) = -\frac{1}{2\pi} \ln(\bar{t} - \bar{t}' - i\varepsilon), \quad (23.36)$$

which has no poles in the lower half plane, so the freely falling observer's detector measures nothing. The Hartle-Hawking vacuum is regular from the perspective of a freely falling observer crossing the horizon. This state physically describes a black hole in thermal equilibrium with its own radiation.

The Unruh Vacuum

For a realistic gravitational collapse, there is no incoming radiation from past null infinity, only outgoing Hawking radiation. The appropriate state is the Unruh vacuum $|0\rangle_U$, defined by

$$\varphi_{\omega,R}^U = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega\bar{u}}, \quad \varphi_{\omega,L}^U = \frac{1}{\sqrt{4\pi\omega}} e^{-i\omega v}. \quad (23.37)$$

This state uses the Kruskal mode for the right-movers (regular at the future horizon) and the Boulware mode for the left-movers (no incoming flux from the past). It is regular at the future horizon but singular at the past horizon, which is not physically relevant for a collapsing star. A static detector at fixed r in the Unruh vacuum sees a one-sided thermal flux of outgoing radiation at the Hawking temperature $T_H = 1/8\pi M$, with no incoming flux from infinity. This is the state that correctly describes the endpoint of gravitational collapse.

23.4 Hawking Radiation in the Bigger Picture

The three results we have obtained, the Unruh effect, the thermal spectrum from the accelerating mirror, and Hawking radiation from a black hole, are all manifestations of the same underlying physics: the observer-dependence of the vacuum state in the presence of a causal horizon. In each case, a state that appears to be the vacuum for one class of observers contains a thermal distribution of particles for another class.

For the black hole, the connection is made precise by the identification of the surface gravity κ_h with the acceleration of a static observer near the horizon. As $r \rightarrow r_h$, the proper acceleration required to remain stationary diverges as $a \sim 1/(4M\sqrt{1-2M/r})$, and the local Unruh temperature $T = a/2\pi$ coincides with the blueshifted Hawking temperature $T_H/\sqrt{1-2M/r}$. The Hawking effect and the Unruh effect are therefore the same phenomenon seen from different perspectives: global versus local.

The Hawking temperature $T_H = 1/8\pi M$ implies that black holes evaporate by radiating energy, causing M to decrease and T_H to increase. This leads to the information paradox: if the radiation is exactly thermal, then no information about the initial state that formed the black hole is encoded in the outgoing radiation, apparently violating unitarity. The resolution of this paradox remains one of the central open problems in theoretical physics, at the intersection of quantum mechanics, general relativity, and thermodynamics.

Conclusion

This thesis has developed a unified framework for understanding the vacuum structure of quantum field theories and its role in the physics of the early universe, proceeding from the abstract machinery of the effective action to the concrete phenomenon of Hawking radiation.

We began in Part I by constructing the effective action formalism from first principles. Starting from the generating functional $Z[J]$ and passing through the connected functional $W[J]$ via a Legendre transform, we arrived at the effective action $\Gamma[\varphi_c]$, whose equation of motion governs the full quantum dynamics of the field. The central object for our purposes was the effective potential $V_{\text{eff}}(\varphi_c)$, whose minima determine the true vacuum of the quantum theory. We gave it two complementary physical interpretations: as the generating functional for 1PI Green's functions at zero external momenta, and as the minimum energy density over all normalized states with a given field expectation value. The one-loop computation in scalar field theory, first using a hard momentum cutoff and then using dimensional regularization and the $\overline{\text{MS}}$ scheme, gave the Coleman-Weinberg potential, and we applied it to gauge theories including the GSW electroweak model and GUT models. The striking phenomenon of dimensional transmutation in massless scalar electrodynamics showed how quantum corrections can generate a mass scale and a nontrivial vacuum structure in a classically scale-invariant theory.

Part II extended the formalism to finite temperature using the imaginary-time formalism of thermal field theory. The key structural change is that the periodicity of fields in Euclidean time with period $\beta = 1/T$, the KMS condition, discretizes the loop momenta into Matsubara frequencies. The finite-temperature effective potential acquires thermal corrections that generate a positive mass term proportional to $T^2\varphi_c^2$, driving symmetry restoration at high temperatures even in theories that are spontaneously broken at $T = 0$. We computed these corrections explicitly for scalar field theory, the GSW model, and minimal $\text{SU}(5)$, and derived the critical temperature T_c at which the two phases become degenerate.

Part III placed this analysis in its cosmological context. The thermal history of the early universe is a sequence of symmetry breaking transitions, each driven by the temperature-dependent effective potential falling below its critical value. We revisited the Coleman-Weinberg potential in the $\overline{\text{MS}}$ scheme and introduced the thermal functions J_B and J_F , whose high-temperature expansions reveal the full structure of the finite-temperature corrections including the cubic term $\sim T|m|^3$ that is responsible for generating a potential barrier and hence a first-order phase transition. The Coleman-Weinberg model at finite temperature provided a concrete example of how a classically scale-invariant theory develops a preferred scale and undergoes a first-order transition as the universe cools.

Part IV turned to the dynamics of false vacuum decay and bubble nucleation, the mechanism by which a first-order phase transition actually proceeds. The universe,

trapped in a metastable false vacuum, tunnels to the true vacuum through the nucleation of bubbles governed by the bounce solution of the Euclidean field equations. We analyzed the $O(4)$ -symmetric bounce at zero temperature and the $O(3)$ -symmetric bounce at finite temperature, computed the nucleation rate, and characterized the transition by the parameters α and $\tilde{\beta}$ that determine the gravitational wave spectrum produced by bubble collisions. This provides a direct observational window into the thermal history of the early universe, potentially accessible to future detectors such as LISA.

Part V addressed the fundamental limitation of everything that preceded it: the assumption that the background spacetime is fixed and flat. In the early universe and near compact objects, the curvature of spacetime is not negligible, and quantum field theory must be formulated on a curved background. We developed the semiclassical gravity framework, in which the geometry is treated classically via the Einstein equations but matter fields are quantized. The central new feature is the observer-dependence of the vacuum state: in the absence of a global Poincaré symmetry, different observers define inequivalent mode decompositions related by Bogoliubov transformations, and the vacuum of one observer contains particles as seen by another. We made this precise through the moving mirror model and the Unruh effect, before deriving the Hawking effect: a black hole formed by gravitational collapse emits a steady thermal flux of radiation at temperature $T_H = \kappa/2\pi$, where κ is the surface gravity. The Bekenstein-Hawking entropy $S_{BH} = A/4$ and the generalized second law complete the thermodynamic picture of black holes.

Several threads connect these results into a unified story. The effective potential, computed perturbatively in flat spacetime, acquires curvature corrections in a curved background through the non-minimal coupling $\xi R\phi^2$, which can shift the phase structure of the theory. The thermal functions J_B and J_F that appear in the finite-temperature effective potential are the same integrals that appear in the Stefan-Boltzmann free energy of the Hawking radiation. The Hawking temperature $T_H = 1/8\pi M$ satisfies $T = dM/dS_{BH}$, confirming the thermodynamic consistency of the Bekenstein-Hawking entropy. And the same mechanism of particle production near a causal horizon that gives rise to Hawking radiation also gives rise to the Gibbons-Hawking temperature $T = H/2\pi$ of de Sitter space, which seeds the primordial density perturbations generated during inflation.

The work of this thesis has been primarily foundational, building the theoretical tools needed to address these questions rather than pursuing any single one of them to its conclusion. The effective action formalism, the imaginary-time thermal field theory, and the Bogoliubov transformation framework are the basic ingredients of any serious treatment of quantum fields in the early universe. Having developed them systematically and seen how they connect to one another, the natural next step is to apply them more carefully to specific physical scenarios, refining the approximations made here and confronting the results with observation.

Appendix A

Proof that $Z(J) = e^{iW(J)}$

The connection between $Z(J)$ and $W(J)$ follows from the combinatorics of Wick diagrams. The S-matrix is

$$S = T \exp \left[-i \int_{-\infty}^{\infty} dt H_I(t) \right], \quad (\text{A.1})$$

where H_I is the interaction picture Hamiltonian, which here includes a source term. Expanding S in a Taylor series and applying Wick's theorem gives a sum over Wick diagrams,

$$S = \sum \mathcal{W}, \quad (\text{A.2})$$

where each $\mathcal{W}$ is a Wick graph. For example, for an interaction of the form $H_I(x) = \lambda \bar{\psi}(x)\psi(x)\varphi(x)$, one of the terms at order λ^2 looks like

Unlike Feynman diagrams, Wick diagrams are labelled. Taking the vacuum expectation value of S selects only diagrams with no uncontracted external legs,

$$Z(J) = \langle 0|S|0 \rangle = \sum \mathcal{W}_u, \quad (\text{A.3})$$

where $\mathcal{W}_u$ denotes vacuum graphs. A vacuum graph is defined as a graph with no external legs, only internal lines forming closed loops,

Any collection of vacuum graphs combined into a single disconnected diagram is still a vacuum graph, since no external legs appear anywhere. Similarly, $iW(J) = \sum \mathcal{W}_{cu}$, where $\mathcal{W}_{cu}$ are the connected vacuum graphs. It then follows from the theorem proved in Appendix B that

$$Z(J) = \sum \mathcal{W}_u = \exp \sum \mathcal{W}_{cu} = e^{iW(J)}, \quad (\text{A.4})$$

which completes the proof. $\square$

Appendix B

The Sum of All Wick Diagrams is the Exponential of the Connected Ones

We prove that

$$\sum \text{all Wick diagrams} = \exp\left(\sum \text{connected Wick diagrams}\right). \quad (\text{B.1})$$

Consider the case where H_I contains a single interaction term. Two diagrams belong to the same pattern P if they differ only by a permutation of vertices. Define $S(P)$ as the number of permutations that leave a diagram of pattern P unchanged, and $n(P)$ as the number of vertices. Then the S-matrix is

$$S = \sum_P \frac{:\mathcal{O}(P):}{S(P)}, \quad (1)$$

where $:\mathcal{O}(P):$ is the normal-ordered operator represented by the Wick diagram. Now consider a pattern consisting of n_i copies of the connected pattern P_i , with $i = 1, 2, \dots$ an ordering of all connected patterns. Then

$$\mathcal{O}(P) = \prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}, \quad (2)$$

$$S(P) = \prod_{i=1}^{\infty} S(P_i)^{n_i}. \quad (3)$$

Substituting into (1), the additional $n_i!$ in the denominator accounts for the fact that permuting graphs of a given connected pattern does not yield a new Wick graph,

$$S = \sum_{\substack{n_i=0 \\ i=0}}^{\infty} \frac{:\prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}:}{\prod_{i=1}^{\infty} [S(P_i)^{n_i} n_i!]}. \quad (4)$$

Factoring the sum over each n_i independently,

$$\begin{aligned}
S &= \prod_{i=1}^{\infty} \sum_{n_i=0}^{\infty} \frac{1}{n_i!} \left[\frac{\mathcal{O}(P_i)}{S(P_i)} \right]^{n_i} \\
&=: \prod_{i=1}^{\infty} \exp\left(\frac{\mathcal{O}(P_i)}{S(P_i)}\right) : \\
&=: \exp\left(\sum_{i=1}^{\infty} \frac{\mathcal{O}(P_i)}{S(P_i)}\right) : \\
&=: \exp \sum \mathcal{W}_c :, \tag{5}
\end{aligned}$$

where $\mathcal{W}_c$ denotes connected Wick graphs and we used (2) and (3) in the last step. This establishes that

$$\sum \text{all Wick diagrams} = e^{\sum \text{connected Wick diagrams}}. \tag{6}$$

□

The proof of $Z(J) = e^{iW(J)}$ in Appendix A then follows immediately by taking the vacuum expectation value of (6).

Appendix C

Equivalence of the Loop Expansion and the $\hbar$ Expansion

We show that an L -loop graph carries a factor of $\hbar^{L-1}$, so that the loop expansion and the $\hbar$ expansion are equivalent. Starting from

$$Z(J) = N \int [\mathcal{D}\varphi] \exp \left[\frac{i}{\hbar} S(\varphi, J) \right], \quad (\text{C.1})$$

the vertex factors are the coefficients of the interaction terms in $\hbar^{-1}S(\varphi, J)$, so each vertex contributes a factor of $\hbar^{-1}$. The propagators are the inverses of the quadratic operators in the exponent, so each internal line contributes a factor of $\hbar$. A graph with V vertices and I internal lines therefore carries an overall factor of

$$\hbar^{I-V}. \quad (\text{C.2})$$

For a connected graph, each vertex fixes one internal momentum except for one overall energy-momentum conservation constraint, so the number of independent loop momenta is

$$L = I - V + 1. \quad (\text{C.3})$$

Hence the $\hbar$ -dependence of a connected graph is

$$\hbar^{I-V} = \hbar^{L-1}, \quad (\text{C.4})$$

which shows that the loop expansion is precisely the expansion in powers of $\hbar$. $\square$

Appendix D

Derivation of $\delta W(J)/\delta J = \varphi$

This appendix supplies the step used in the proof of Theorem 3, following Jackiw and Kerman (1979). The two equivalent expressions for $W(J)$ are

$$e^{\frac{i}{\hbar}W(J)} = \langle -, t | +, t \rangle, \quad (\text{D.1})$$

$$e^{\frac{i}{\hbar}W(J)} = \langle 0 | T \exp \left(\frac{i}{\hbar} \int (dx) J(x) \Phi(x) \right) | 0 \rangle. \quad (\text{D.2})$$

The states $|\pm, t\rangle$ solve the time-dependent Schrödinger equation

$$i\hbar \partial_t |\pm, t\rangle = H |\pm, t\rangle - \int d\mathbf{x} J(t, \mathbf{x}) \Phi(\mathbf{x}) |\pm, t\rangle, \quad (\text{D.3})$$

with boundary conditions $\lim_{t \rightarrow \mp\infty} |\pm, t\rangle = |0\rangle$, where $|0\rangle$ is the ground state of H with $H|0\rangle = 0$. The states $|\psi_{\pm}, t\rangle$ are related to $|\pm, t\rangle$ by

$$|+, t\rangle = \exp \frac{i}{\hbar} \left(\int_{-\infty}^t dt' w(t') \right) |\psi_+, t\rangle, \quad (\text{D.4})$$

$$|-, t\rangle = \exp \frac{i}{\hbar} \left(- \int_t^{\infty} dt' w^*(t') \right) |\psi_-, t\rangle, \quad (\text{D.5})$$

and satisfy $(i\hbar \partial_t - H + \int J \Phi dx) |\pm, t\rangle = 0$ with the same boundary conditions. Their overlap gives

$$\langle -, t | +, t \rangle = e^{\frac{i}{\hbar}W(J)}. \quad (\text{D.6})$$

Expanding this explicitly,

$$\begin{aligned} & \langle \psi_-, t | \left[\exp \frac{i}{\hbar} \left(- \int_t^{\infty} dt' w^*(t') \right) \right]^* \exp \frac{i}{\hbar} \left(\int_{-\infty}^t w(t') dt' \right) | \psi_+, t \rangle \\ &= \langle \psi_-, t | \exp \frac{i}{\hbar} \left(\int_t^{\infty} dt' w(t') + \int_{-\infty}^t dt' w(t') \right) | \psi_+, t \rangle \\ &= \langle \psi_-, t | \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' w(t') | \psi_+, t \rangle \\ &= \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' w(t') \underbrace{\langle \psi_-, t | \psi_+, t \rangle}_1 \\ &= \exp \frac{i}{\hbar} \left(\int_{-\infty}^{\infty} dt' w(t') \right), \end{aligned} \quad (\text{D.7})$$

so that $W = \int_{-\infty}^{\infty} dt w(t)$. Using the eigenvalue equation for $|\psi_+, t\rangle$,

$$\int_{-\infty}^{\infty} dt \langle \psi_-, t | \left(i\hbar \partial_t - H + \int d\mathbf{x} J\Phi \right) |\psi_+, t\rangle = \int_{-\infty}^{\infty} dt w(t) \underbrace{\langle \psi_-, t | \psi_+, t \rangle}_1, \quad (\text{D.8})$$

which gives

$$W(J) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\hbar \partial_t - H + \int d\mathbf{x} J\Phi |\psi_+, t\rangle. \quad (\text{D.9})$$

Taking the functional derivative,

$$\begin{aligned} \frac{\delta W(J)}{\delta J} &= \int_{-\infty}^{\infty} dt \frac{\delta \langle \psi_-, t |}{\delta J} \underbrace{\left(i\hbar \partial_t - H + \int d\mathbf{x} J\Phi \right) |\psi_+, t\rangle}_{w(t)|\psi_+, t\rangle} \\ &\quad + \int_{-\infty}^{\infty} dt \langle \psi_-, t | \frac{\delta}{\delta J} \left(i\hbar \partial_t - H + \int d\mathbf{x} J\Phi \right) |\psi_+, t\rangle \\ &\quad + \int_{-\infty}^{\infty} dt \langle \psi_-, t | \left(i\hbar \partial_t - H + \int d\mathbf{x} J\Phi \right) \frac{\delta |\psi_+, t\rangle}{\delta J}, \end{aligned} \quad (\text{D.10})$$

where the constraint $\langle \psi_-, t | \Phi(\mathbf{x}) | \psi_+, t \rangle = \phi(t, \mathbf{x})$ was used for the second term. Integrating the third term by parts,

$$i\hbar \left\langle \psi_-, t \left| \frac{\delta |\psi_+, t\rangle}{\delta J} \right. \right\rangle \Big|_{-\infty}^{\infty} + \left\langle \psi_-, t \left| \underbrace{\left(-i\hbar \partial_t - H + \int d\mathbf{x} J\Phi \right) \frac{\delta |\psi_+, t\rangle}{\delta J}}_{\langle \psi_-, t | w(t) \rangle} \right. \right\rangle, \quad (\text{D.11})$$

so that

$$\begin{aligned} \frac{\delta W(J)}{\delta J} &= \phi + i\hbar \left\langle \psi_-, t \left| \frac{\delta}{\delta J} |\psi_+, t\rangle \right. \right\rangle \Big|_{-\infty}^{\infty} \\ &\quad + \int_{-\infty}^{\infty} dt w(t) \left[\frac{\delta \langle \psi_-, t |}{\delta J} |\psi_+, t\rangle + \langle \psi_-, t | \frac{\delta}{\delta J} |\psi_+, t\rangle \right], \end{aligned} \quad (\text{D.12})$$

where the expression inside the brackets vanishes upon differentiating the normalization constraint $\langle \psi_-, t | \psi_+, t \rangle = 1$. It remains to show that the boundary term also vanishes. At $t \rightarrow -\infty$, $|\psi_+, t\rangle \rightarrow |0\rangle$ independently of J , so $\delta |\psi_+, t\rangle / \delta J = 0$ at the lower limit. For the upper limit, rewrite it as

$$i\hbar \frac{\delta}{\delta J} \underbrace{\langle \psi_-, t | \psi_+, t \rangle}_1 - i\hbar \left(\frac{\delta \langle \psi_-, t |}{\delta J} \right) |\psi_+, t\rangle, \quad (\text{D.13})$$

where the second term vanishes because at $t \rightarrow \infty$, $\langle \psi_-, t | \rightarrow \langle 0 |$, also independently of J . The boundary term therefore vanishes at both limits, and

$$\therefore \quad \boxed{\frac{\delta W(J)}{\delta J} = \phi.} \quad (\text{D.14})$$

Appendix E

Evaluation of the One-Loop Momentum Integral

We evaluate

$$I = \int \frac{d^4 k_E}{(2\pi)^4} \ln(k_E^2 + V'') = \frac{1}{8\pi^2} \int_0^\Lambda dk_E k_E^3 \ln(k_E^2 + V''), \quad (\text{E.1})$$

where the angular integration in four-dimensional Euclidean space gives a factor $2\pi^2$. Substituting $u = k_E^2$, $du = 2k_E dk_E$,

$$I = \frac{1}{16\pi^2} \int_0^{\Lambda^2} u du \ln(u + V''). \quad (\text{E.2})$$

Integrating by parts with $f = \ln(u + V'')$, $dg = u du$,

$$I = \frac{1}{16\pi^2} \left\{ \left[\frac{u^2}{2} \ln(u + V'') \right]_0^{\Lambda^2} - \int_0^{\Lambda^2} \frac{u^2}{2} \cdot \frac{1}{u + V''} du \right\}, \quad (\text{E.3})$$

where the boundary term evaluates to $\frac{\Lambda^4}{2} \ln(\Lambda^2 + V'')$. Using the partial fraction decomposition $\frac{u^2}{u + V''} = u - V'' + \frac{(V'')^2}{u + V''}$,

$$\begin{aligned} \frac{1}{2} \int_0^{\Lambda^2} \frac{u^2}{u + V''} du &= \frac{1}{2} \int_0^{\Lambda^2} \left(u - V'' + \frac{(V'')^2}{u + V''} \right) du \\ &= \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln(\Lambda^2 + V'') - (V'')^2 \ln V'' \right]. \end{aligned} \quad (\text{E.4})$$

Assembling the pieces,

$$I = \frac{1}{16\pi^2} \left\{ \frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') - \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln \left(\frac{\Lambda^2 + V''}{V''} \right) \right] \right\}. \quad (\text{E.5})$$

Expanding $\ln(\Lambda^2 + V'') = \ln \Lambda^2 + \ln(1 + V''/\Lambda^2) \simeq \ln \Lambda^2 + V''/\Lambda^2 - (V'')^2/\Lambda^4 + \dots$, the leading terms are

$$\frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') \simeq \Lambda^4 \ln \Lambda + \frac{\Lambda^2 V''}{2} - \frac{(V'')^2}{2}, \quad (\text{E.6})$$

$$\frac{(V'')^2}{2} \ln(\Lambda^2 + V'') \simeq \frac{(V'')^2}{2} \ln \Lambda^2 + \mathcal{O}(1/\Lambda^2). \quad (\text{E.7})$$

Collecting all terms,

$$I = \frac{1}{16\pi^2} \left\{ \Lambda^4 \left(\ln \Lambda - \frac{1}{4} \right) + \Lambda^2 V'' + \frac{(V'')^2}{2} \left(\ln \frac{\Lambda^2}{V''} - \frac{1}{2} \right) \right\} + \mathcal{O}\left(\frac{1}{\Lambda^2}\right). \quad (\text{E.8})$$

Dropping the $\Lambda^n \ln \Lambda$ terms, which are field-independent and do not affect the location of the minimum of $U(\rho)$, the one-loop integral contributing to (4.52) is

$$\frac{\hbar}{2} I = \hbar \left[\frac{\Lambda^2}{32\pi^2} V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2} \left(-\frac{1}{2} + \ln \frac{V''(\rho)}{\Lambda^2} \right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right) \right]. \quad (\text{E.9})$$

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